

Steering a warp drive without exotic matter

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Abstract

A useful warp drive must change its velocity, yet known positive-energy constructions are static or constant-velocity, while accelerating ones require exotic matter; as Bobrick and Martire noted, warp drives “do not have any natural way of changing their velocities.” We first settle the question of principle by reading the Bondi–Sachs balance as a propulsion law: any asymptotically flat drive with a confined, dominant-energy material source (and standard peeling) changes its Bondi four-momentum only by radiating to null infinity, so it cannot steer without radiating, the general-relativistic counterpart of the relativistic rocket. We then exhibit a positive-energy spacetime that saturates it. Prescribing the passenger worldtube and accepting positive null radiation as the exterior (the exact Kinnersley photon rocket), we solve for the matching shell: the mechanism is photon-rocket recoil on a tidally protected flat cavity, and the decoupled flat interior, not a warp field, makes it a warp drive. The exterior energy conditions reduce to $n^2 \geq 0$, and steering obeys the closed-form control law $-\dot{m} \geq 3m|a|$, paid by Bondi mass loss. The static matched shell satisfies the surface dominant energy condition for $2m/R < 24/25$, and admissible accelerating shells exist unconditionally at small acceleration about it, below a rigorous closed-form acceleration–compactness ceiling. The least-radiating maneuver is the gravitational-wave–silent Damour dipole, and minimum-radiation steering is a hyperbolic geodesic with the universal Tsiolkovsky constant 3. A slow fixed-axis burn assembles, perturbatively, into a time-evolved spacetime. The drive is causal, subluminal, and a *reaction* drive: steering costs an order-unity fraction of the rest mass for mildly relativistic boosts, but needs no exotic matter. Steering a warp drive is a problem of energy budget, not negative energy.

1 Introduction

Alcubierre’s warp drive [1] showed that general relativity permits a compact region (a “bubble”) to be transported through an ambient spacetime while its interior remains locally flat. The price was exotic matter: the construction violates the null energy condition (NEC), and with it the weak, strong, and dominant conditions. This is not an accident of Alcubierre’s particular ansatz. Santiago, Schuster, and Visser [2] proved that *generic* warp drives, defined through the standard shift-vector ansatz, violate the NEC; Lobo and Visser [3] had earlier quantified the unavoidable exotic content, and Pfenning and Ford [4] showed that the required negative energy is further constrained by quantum inequalities [5].

Two design philosophies dominate the field, and both are overconstrained. *Metric-first* designs [1, 6, 7] choose the geometry and let Einstein’s equations dictate the matter, which then turns out to be exotic. *Source-first* or static designs [8, 9, 10, 11] prescribe a physically reasonable matter profile and tolerate the geometry that results; these reach positive energy or reduced exotic content, but are *static or constant-velocity*. The positive-energy claims of Lentz [12] and Fell–Heisenberg [13] take a different route, but their energy-condition status remains disputed [14] (Table 1). Bobrick and Martire state the gap plainly: warp drives “do not have any natural way of changing their velocities,” and “no metric which describes an accelerating warp drive solution has so far been presented in the literature” [8]. Barzegar, Buchert, and Vigneron [15] (see also [16]) raise a complementary structural requirement: a physical warp drive must possess *covariant* (not merely coordinate) velocity and acceleration, and a genuine mechanism *derived from matter* rather than reverse-engineered from a chosen metric. Buchert and Frackowiak [17] report an instability of warp fields in that metric-first class (Sec. 14).

The most recent positive-energy constructions sharpen this gap rather than close it. Fuchs *et al.* [9] match an explicit positive-energy shell, but at constant velocity, and name radiation- and momentum-transfer steering as the open direction beyond simple photon rockets; Rodal [10] reduces the exotic requirement to a predominantly positive invariant density, and Garattini and Zatrımaylov [11] satisfy the *averaged* conditions in a de Sitter bubble, both at fixed velocity. The discriminator the present construction meets, and none of these do, is the observer-robust *pointwise* dominant energy condition, in bulk *and* shell and over all observers, *together with* a covariant, matter-derived acceleration (Table 1, Sec. 14).

In a companion paper [18] we tested whether the source-first route can be pushed to admissibility. Using the observer-robust energy-condition framework of Ref. [19], an exhaustive scan of compact-support shell configurations found *none* admissible: the dominant-energy failures localize at the smooth source–vacuum transition, persist in the static limit, and appear as Hawking–Ellis [20] Type IV stress-energy in the transition tail. The obstruction is geometric, tied to the transition itself rather than to the warp velocity, and we re-derive this boundary cost in closed form below (Prop. 7). That result reframes the problem: if the obstruction is the transition itself, the design should remove the smooth source–vacuum transition from its center and abandon the requirement that the exterior be vacuum.

This paper. We propose a third philosophy, *worldtube-first*. We prescribe the passenger worldtube (a timelike curve Γ together with a regular cavity) and we *accept positive null radiation* as the exterior matter we are willing to carry; Einstein’s equations then *determine* the shell. The exterior is taken to be the exact Kinnersley–Robinson–Trautman photon rocket [21, 22, 23, 24, 25], a known exact solution whose source is pure outgoing null dust. Its energy conditions reduce to a single positivity, $n^2 \geq 0$ (Sec. 5), and its central worldline is covariantly accelerated, by the recoil of anisotropic radiation, exactly as a photon rocket. We excise the worldline singularity, replace $r < R(u, \Omega)$ by a regular cavity, and match across a timelike worldtube using the standard Lanczos junction together with flux-source balance laws [26, 27]: the *timelike*-shell balance laws with an external crossing flux, mandatory here because the exterior radiation is emitted at the shell, the cavity within being vacuum.

The resulting object, a *radiative momentum warpsell*, is causal, subluminal, self-gravitating, and built entirely from matter that satisfies the energy conditions in the bulk. Its acceleration is real and is paid for by *Bondi mass loss*. The organizing result is not the metric but a *conservation law*: no asymptotically flat, positive-energy drive can change its velocity without radiating four-momentum to null infinity, and the construction below is the explicit positive-energy object that *saturates* it. This makes rigorous, as a Bondi–Sachs conservation law, the momentum-balance obstruction Bobrick and Martire could state only heuristically [8]: because warp spacetimes are asymptotically flat, a confined drive acquires four-momentum only by radiating it, and we exhibit the positive-energy object that pays exactly that price with $n^2 \geq 0$. We summarize the contributions:

1. A **model-independent no-reactionless-steering principle**, the capstone (Sec. 3, Theorem 1): for *any* asymptotically flat, dominant-energy drive with a confined material source (and Bondi–Sachs peeling), the Bondi four-momentum changes only through radiation to null infinity, so steering forces radiation. The balance follows directly from the curvature across three independent exact spacetimes. The contributions below are its faces in an explicit positive-energy instance that *saturates* it.
2. **Bulk energy conditions** for the exterior collapse to the single scalar inequality $n^2 \geq 0$ (Sec. 5).
3. A **control law** $-\dot{m} \geq 3m|a|$, the local dipole face of the principle (Corollary 3), ties covariant proper acceleration to Bondi mass loss, with ma the Bondi momentum flux, the thrust (Sec. 6).
4. An **unconditional small-amplitude existence theorem**: admissible accelerating shells exist about the static Schwarzschild–Minkowski anchor, whose surface dominant energy condition holds strictly for $2m/R < 24/25$. Exact matching reduces to a *single* closed-form scalar $D(x)$ (58), bounded away from zero ($|D| > \frac{4}{3}$) across the window, so an admissible accelerating shell exists for every small $a > 0$ (Theorem 2, App. D), with matched surface stress of Hawking–Ellis Type I and a strictly positive observer-robust surface-DEC margin.

5. A **perturbative time-evolved assembly** for slow fixed-axis sub-horizon burns (Proposition 6): the per-cut snapshots assemble into a dynamical admissible solution, conditional on one constant-cut compatibility identity (64) whose leading $\mathcal{O}(a^2)$ ($\ell = 2$) *obstruction* we show to cancel gauge-completely (the four $\ell = 2$ junction rows independent, the reduced Jacobian nondegenerate); the all-orders identity and the rotating-axis assembly are left for future work (Sec. 8.2).
6. An **acceleration–compactness frontier** $a_{\max} R \leq g(2m/R)$ for a positive-energy shell, with one rigorous closed-form member, the kinematic ceiling $g(x) < \frac{1}{2}(1 - x)$ (Prop. 5); the operative dominant-energy frontier $g(x)$ lies between that ceiling and a numerical lower bound, a shape-response effect binding at the rear pole (Sec. 8, App. C).
7. **Thickening with admissible matter**: the thin shell is the limit of finite-thickness energy-condition-respecting matter: a *tangential-pressure* wall is strictly DEC-admissible across its whole width, recovers the anchor for $2m/R < 4/5$, and stays admissible once made radiative, whereas a generic *smooth* (radial-tension) thickening reproduces the boundary cost (Sec. 9); its $\mathcal{O}(\mu)$ back-reaction is closed at adiabatic linear order (App. F).
8. **Maneuver design as geometric control**: the exterior, being exactly Kinnersley, radiates no gravitational waves at the Damour dipole (Sec. 10), and that news-silence selects the dipole as the minimum-radiation maneuver, *proved* to saturate the control law (Sec. 11). The complementary *free-profile* problem is solved in closed form: bulk positivity and news-silence collapse the cost to a single hyperbolic norm on the velocity space, so the minimum-radiation velocity change is a *hyperbolic geodesic* on $\mathbb{H}^3 = SO^+(3, 1)/SO(3)$ with the universal Tsolkovsky constant 3 (Theorem 5), and multi-burn steering is a state-constrained geodesic against the moving frontier.

The scope is stated plainly in Sec. 15: the construction is subluminal, not reactionless, and astronomically but *positively* costly. Warp propulsion is thus an energy-budget engineering problem, not an exotic-matter impossibility.

Throughout we use $G = c = 1$, signature $(-, +, +, +)$, and $\kappa = 8\pi$, matching Refs. [19, 18].

2 What is a warp drive? A worldtube-first definition

The recurring criticism of warp spacetimes is that their “motion” is a coordinate statement [15, 16]: a shift vector $\beta^i(t, x)$ in an ADM form $ds^2 = -dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt)$ encodes a coordinate velocity that need not correspond to any covariant acceleration of matter. We therefore adopt an operational, gauge-invariant definition.

Definition 1 (Warpshell). A *warpshell* is a spacetime $\mathcal{M} = \mathcal{M}_- \cup_{\Sigma} \mathcal{M}_+$ such that: (i) the interior $\mathcal{M}_-$ is a regular timelike worldtube (the cavity) with bounded tidal tensor along a central worldline Γ ; (ii) Σ is a timelike hypersurface across which the induced metric is continuous (the first junction condition) and the surface stress-energy S_{ab} is well defined; (iii) the exterior $\mathcal{M}_+$ satisfies the energy conditions; and (iv) the central worldline carries a prescribed covariant proper acceleration $a^\mu = u^\nu \nabla_\nu u^\mu$ with gauge-invariant magnitude $\sqrt{a_\mu a^\mu}$. The warpshell is *admissible* if, in addition, S_{ab} satisfies the surface energy conditions.

Definition 1 makes “acceleration” a scalar invariant of a physical worldline, not a coordinate velocity, answering the covariance demand of Refs. [15, 16]. It also inverts the design order demanded there: we will fix Γ and the admissible exterior matter *first*, and let the Einstein equations return S_{ab} . Alcubierre and Natário bubbles are warpshells in this sense only in a degenerate, non-accelerating limit; the constant-velocity positive-energy shells of Refs. [8, 9] realize (i)–(iii) but reach (iv) only in the non-accelerating limit $a^\mu = 0$. Our task is to satisfy all four *with the exterior energy conditions intact*.

In what sense a warp drive. The object we construct is, mechanically, a rocket: it expends mass as radiation, and its passengers feel proper acceleration. What distinguishes it from a bare rocket, and places it in the warp-drive lineage, is the structure of Definition 1: a passenger cavity that is exactly flat, so the interior tidal tensor vanishes and the occupants ride in a tidally flat, accelerated laboratory, joined to an exterior that is positive-energy in every frame. It is the accelerating, positive-energy member of the family whose constant-velocity members are the shells of Refs. [8, 9]: what it shares with the Alcubierre program is the protected flat interior, and what it adds is a covariant, matter-derived change of state of motion. Unlike the Alcubierre bubble it

carries *no* local expansion or contraction of space ahead of or behind the cavity (no York-time “warp field”): the cavity is rigidly Rindler-accelerated and the exterior is the photon rocket, so the warp-drive connection is precisely the decoupled, tidally protected passenger laboratory, not a spatial-expansion mechanism. We claim no superluminal travel and no reactionless propulsion (Sec. 15); the contribution is that the *acceleration* long missing from positive-energy warp drives is achievable at the price of ordinary radiation.

3 The no-reactionless-steering principle

The organizing principle of the paper is a statement stronger than “this particular construction radiates”: *no* confined, positive-energy drive can steer without paying a momentum debt to null infinity. We state it first, as a model-independent theorem with a self-contained proof skeleton, and only then build the saturating object: the explicit worldtube-first construction of Sec. 4 below (whose control law (13) (positivity of the null-dust amplitude, Prop. 3) and momentum balance (11) we invoke as needed) is its local, positive-energy *saturating* face. We state the law, identify the one hypothesis the construction must supply, and record how the Kinnersley photon rocket of Sec. 4 saturates it.

Theorem 1 (No reactionless steering, model-independent). *Let $(\mathcal{M}, g)$ be asymptotically flat at future null infinity $\mathcal{I}^+$, admitting a Bondi–Sachs expansion with the standard Newman–Penrose peeling behavior [28, 29, 30]. Let its material (timelike) source be spatially confined on each retarded cut (of compact support, or decaying faster than the Bondi falloff) while outgoing null radiation may reach $\mathcal{I}^+$ (this is the matter-flux channel n^2 below), and let the total stress-energy satisfy the dominant energy condition, with non-negative Bondi mass $m_B \geq 0$ (used only for the no-loan refinement, Cor. 2; the flux-free no-go itself needs no sign hypothesis, and $m_B \geq 0$ holds unconditionally for the photon-rocket exterior below, where $m_B = m(u) > 0$). Then the Bondi four-momentum $P_B^\mu(u)$ evolves only through the four-momentum radiated to $\mathcal{I}^+$,*

$$\frac{dP_B^\mu}{du} = -\frac{1}{4\pi} \oint \left[|\mathcal{N}(u, \Omega)|^2 + 4\pi n^2(u, \Omega) \right] \hat{\ell}^\mu(\Omega) d\Omega, \quad (1)$$

with $\mathcal{N}$ the Bondi news¹ and $n^2 \geq 0$ the outgoing matter flux. Consequently, on any retarded-time segment with vanishing news and no matter flux, P_B^μ is constant: the confined drive cannot change its total Bondi four-momentum, and hence, between stationary Bondi frames, its velocity (Prop. 1), without radiating.

Proof. The argument composes three classical ingredients; only the framing is new, and three hypotheses do distinct work. The central no-go, constancy of P_B^μ on a flux-free segment (Corollary 1), is the flux balance (1) evaluated at zero flux and needs *no* sign hypothesis; the dominant energy condition enters only to sign the two integrands non-negative, orienting the radiated budget future-directed; and Bondi-mass positivity enters one level deeper, in the no-loan refinement (Corollary 2) alone, the only step conditional for an arbitrary distributional shell. (i) *The four-momentum is well-defined at a cut.* Under peeling the mass aspect $m_B(u, \Omega)$ and the $\ell = 0, 1$ momentum aspect are finite on each cut of $\mathcal{I}^+$, and their lowest two moments define $P_B^\mu(u) = \frac{1}{4\pi} \oint m_B(u, \Omega) \hat{\ell}^\mu(\Omega) d\Omega$, the Dray–Streubel/Geroch–Winicour supermomentum charge, supertranslation-covariant at a fixed cut [31, 32, 33]. (ii) *The flux-balance law.* Equation (1) is the Bondi–van der Burg–Metzner–Sachs four-momentum-loss formula [34, 35], with the gravitational channel $|\mathcal{N}|^2$, whose radiative symplectic flux structure at $\mathcal{I}^+$ is that of Ashtekar–Streubel [36], and the matter channel $4\pi n^2$, with $n^2 := \lim_{r \rightarrow \infty} r^2 T_{uu}$ the coefficient of the outgoing null flux $T_{ab} \sim (n^2/r^2) \ell_a \ell_b$ ($\ell_a = -(du)_a$), non-negative by the dominant energy condition. (The matter reaching $\mathcal{I}^+$ is massless, so its leading outgoing radiative flux $T_{uu} \geq 0$ is along ℓ_a and the single scalar $n^2 \geq 0$ sets both the energy and the momentum moment of the matter flux; a massive tail, decaying faster than the Bondi falloff by hypothesis, does not reach $\mathcal{I}^+$.) Both integrands are non-negative and contracted with the future-directed null $\hat{\ell}^\mu$. (iii) *Positivity bounds the budget.* The no-go itself is already the flux law: with both integrands of (1) non-negative, P_B^μ is constant on any flux-free segment. Positivity enters one level deeper, and here we separate two cases at the point of assertion. For a *regular* (smooth) asymptotically flat/null spacetime, and, unconditionally, for the explicit construction, whose Bondi mass is the exterior value $m_B(u) = m(u) > 0$ (Cor. 4), the dominant energy condition gives $m_B(u) \geq 0$: the positive-energy theorem is established both spinorially, by the Witten–Nester argument [37], and independently by the Schoen–Yau

¹We normalize the news so that the gravitational energy flux is $-\frac{1}{4\pi} \oint |\mathcal{N}|^2 d\Omega$, with $|\mathcal{N}|^2 = \mathcal{N}\bar{\mathcal{N}}$ the modulus of the complex (spin-weight -2) Bondi news; only the matter channel n^2 enters the explicit construction below.

minimal-surface method [38], and it is carried to the Bondi mass at $\mathcal{I}^+$ by the Witten–Nester spinor argument on an asymptotically null slice [39, 40, 41], where the Nester–Witten integrand is a non-negative bulk integral under DEC. The positivity step carries the standard hypotheses of those theorems (a complete asymptotically flat/null slice with no naked inner boundary, and, for the distributional shell, the DEC in the generalized surface-stress sense), which the present construction satisfies (the cavity is regular and the shell is admissible, Theorem 2). For the distributional shell the relevant results are the low-regularity positive-mass theorems of Lee–LeFloch [42] and their Lorentzian, junction-aware spinor extension by Shibuya [43]: the Witten–Nester integrand is then a non-negative bulk integral *plus* a non-negative surface term whenever the shell satisfies the surface DEC (Lemma 6, App. B), exactly the strictly-positive Type-I margin Theorem 2 establishes. Only the sign of that surface term is needed here, and it is fixed by the surface DEC alone; the full distributional positive-Bondi-mass theorem at $\mathcal{I}^+$ (existence and decay of the Witten spinor across the C^0 junction on an asymptotically null slice) is left open (Sec. 15). For the explicit construction it is unnecessary: $m_B(u) = m(u) > 0$ at every cut (Cor. 4). Hence the four-momentum radiable on $[u_0, u_f]$ is bounded by $m_B(u_0)$ (Corollary 2): one cannot *borrow* momentum by driving the mass negative. Positivity thus closes the negative-Bondi-mass bookkeeping loophole; it does *not*, by itself, constrain the redistribution of momentum among matter, shell, and field at fixed P_B^μ , the supertranslation-sensitive content addressed separately by Proposition 1. $\square$

Corollary 1 (Propulsion no-go). *On any segment with $\mathcal{N} \equiv 0$ and $n^2 \equiv 0$, P_B^μ is constant: a confined positive-energy drive cannot change its velocity (the Bondi-frame ratio P_B^i/P_B^0 between stationary cuts, Prop. 1) without radiating four-momentum to $\mathcal{I}^+$.*

Corollary 2 (Monotone mass; no momentum loan). *The energy component of (1) is $\dot{m}_B = -\frac{1}{4\pi} \oint [|\mathcal{N}|^2 + 4\pi n^2] d\Omega \leq 0$; with $m_B \geq 0$ the four-momentum radiable on $[u_0, u_f]$ is bounded by $m_B(u_0)$, so momentum cannot be acquired by driving the mass negative. In particular a mass-preserving ($\dot{m}_B \equiv 0$) drive cannot steer: with both integrands of (1) non-negative, $\dot{m}_B = 0$ forces $\mathcal{N} \equiv 0$ and $n^2 \equiv 0$, hence zero thrust, so the would-be reactionless dipole $n^2 \propto \cos \vartheta$ (zero monopole, nonzero momentum moment) is exactly what the bulk positivity $n^2 \geq 0$ forbids. The Bondi-mass sign $m_B \geq 0$ is unconditional for the Kinnersley rocket, where $m_B(u) = m(u) > 0$ at every cut (Cor. 4); for an arbitrary confined distributional shell it is conditional on the distributional positive-Bondi-mass theorem left open in Sec. 15, the surface DEC fixing only the sign of the junction term (Lemma 6).*

Corollary 3 (Steering cost; reduction to the control law). *Any $\Delta P_B^i \neq 0$ requires a nonzero $\ell = 1$ flux, which strictly costs Bondi mass. For the Kinnersley rocket two facts hold separately. First, the gravitational channel vanishes (the Damour dipole is news-free, $\mathcal{N} \equiv 0$), so the four-momentum loss (1) is carried by the matter channel alone. Second, bulk positivity $n^2(\vartheta) \geq 0$, binding at the forward pole, is exactly the control law $-\dot{m} \geq 3m|a|$ (13), whose saturated integral is the Tsiolkovsky budget (14). The global law (1) thus reduces, in the rocket limit, to the dipole control law.*

Confined sources only. The theorem governs the *total* Bondi four-momentum of a confined source, and the confinement hypothesis is what gives the no-go its force. A device that expels *unconfined massive* reaction mass can change its payload’s velocity while the total Bondi four-momentum stays fixed, the momentum being carried to timelike infinity i^+ by the propellant rather than to $\mathcal{I}^+$; but that is an ordinary rocket with a material exhaust, not a reactionless drive, and its propellant leaves every bounded region, violating the hypothesis. For a positive-energy source that stays confined the only momentum channel to infinity is the null radiation reaching $\mathcal{I}^+$, so steering forces that radiation.

The remaining hypothesis needing care is the identification of P_B^i with the *kinematic* steering of the passenger worldline.

Proposition 1 (Center of momentum at null infinity). *The integrated momentum change of a confined drive between two stationary Bondi frames is the integrated $\ell = 1$ flux ΔP_B^i of (1), unambiguously and supertranslation-covariantly; the corresponding velocity change is read from the Bondi-frame ratio P_B^i/P_B^0 between those frames. The instantaneous identification of a center-of-mass worldline with the $\ell = 1$ mass-aspect moment is, by contrast, supertranslation-ambiguous in general: the relativistic center of mass has no invariant curved-space analog, the same ambiguity that lets a deformable body “swim” through curved spacetime by cyclic shape change with no momentum exchange [44, 45, 46]. We therefore state the instantaneous COM $\leftrightarrow$ scri identification as a hypothesis (a choice of Bondi rest frame); it is discharged exactly by the present construction, whose exterior is the Kinnersley rocket with the closed-form worldline*

$P_B^\mu(u) = m(u)v^\mu(u)$ [23], leaving no center-of-mass ambiguity. The two effects are complementary: the no-reactionless-steering principle forbids reactionless velocity change (ΔP_B^i), whereas swimming produces a finite position displacement at fixed P_B^μ .

Corollary 4 (Unambiguous steering velocity for the Kinnersley rocket). *For the Kinnersley photon-rocket exterior of Sec. 4 the outgoing Robinson–Trautman congruence is shear-free, so its canonical cuts are good cuts ($\sigma^0 = 0$) on which the $\ell = 1$ mass-aspect moment is supertranslation-unambiguous [25] (a choice of this canonical shear-free cut family with the explicit Bonnor four-momentum, not a frame-invariant center of mass), and there the Bondi four-momentum is exactly the Bonnor value $P_B^\mu(u) = m(u)v^\mu(u)$ [23] at every retarded cut (v^μ the unit four-velocity, $m(u) > 0$). This closed-form four-momentum discharges the hypothesis of Proposition 1: the instantaneous steering velocity $v^i(u) = P_B^i(u)/P_B^0(u)$ is read directly off P_B^μ with no center-of-mass choice, so the no-reactionless-steering principle applies to the instantaneous velocity, not merely between stationary Bondi frames. The residual freedom in these good cuts is the four-parameter BMS translation subgroup, under which P_B^μ transforms as a Lorentz four-vector, so the steering velocity $v^i = P_B^i/P_B^0$ is translation-covariant. (We claim the unambiguous velocity, not a fully invariant center-of-mass worldline, which the no-go does not use.)*

Relation to the classical Bondi–Sachs balance. The flux-balance law (1) is classical Bondi–Sachs, and the recoil of a source by anisotropic radiation is older still [47, 48]; we claim no novelty for it. The content of Theorem 1 is (i) reading it as a propulsion *no-go* (reactionless steering is forbidden in general relativity, Corollary 1); (ii) closing the “borrow momentum for free” loophole with Bondi-mass positivity (Corollary 2); (iii) separating it cleanly from the center-of-mass swimming effect (Proposition 1); and (iv) exhibiting an *explicit, fully positive-energy* spacetime that *saturates* it. It is the positive-energy complement of Santiago–Schuster–Visser [2]: they forbid the metric-first warp drive (the prescribed shift-vector ansatz with vacuum exterior, tested over *all* observers); the present no-go prices the source-first one and shows the price is payable with $n^2 \geq 0$. It supplies the covariant accelerator that Refs. [15, 16] demand ($dP_B^i/du \neq 0$ is a geometric Bondi-frame momentum flux at null infinity), and places the constant-velocity positive-energy shells [8, 9, 10] as the $dP_B^i/du = 0$ boundary case. The Tsolkovsky budget (14) is its integrated form, making the construction the general-relativistic counterpart of the relativistic rocket [49]. Geometrically the law is a moment-map/symplectic statement on the Ashtekar–Streubel radiative phase space (Sec. 11.1): the asymptotic-translation Hamiltonian is the radiated four-momentum flux, P_B^μ the cut charge for the same asymptotic translation (linked by the balance law $dP_B^\mu/du = -F^\mu$), and the covariant steering is the frame-invariant projected recoil R^μ (41), which vanishes on the constant-velocity ($C_4 = 0$) face of Prop. 8.

Concretely, the explicit construction realizes the matter channel n^2 of (1) as the outgoing Kinnersley dipole of Sec. 6, whose monopole is the mass loss $-\dot{m}$ and whose $\ell = 1$ moment is the thrust ma ; the following proposition records that this instance saturates the law through the matter channel alone.

Proposition 2 (Photon-rocket instance, saturating the flux law). *Let $\mathcal{M}_+$ be the exact Kinnersley–Robinson–Trautman photon-rocket exterior of Sec. 4, matched to a confined cavity across a timelike Σ . Then the Bondi four-momentum obeys (11), whose monopole and dipole are the moment identities $\oint n^2 d\Omega = -\dot{m}$ and $-\oint n^2 \cos\vartheta d\Omega = ma$ (the thrust, App. B). Consequently (i) the worldline accelerates, with instantaneous-rest-frame thrust $dP_B^i/du = ma \neq 0$ (covariantly $dP_B^\mu/du = \dot{m}v^\mu + ma^\mu$), only if $n^2 \not\equiv 0$, so steering requires radiation; and (ii) if $n^2 \equiv 0$ (the Schwarzschild limit (8)) then $\dot{m} = 0$ and $a = 0$, so P_B^μ is constant and the cavity cannot self-accelerate. The exterior DEC ($n^2 \geq 0$, Prop. 3) makes the radiated momentum future-directed, and the dipole emission has vanishing news, so this instance saturates (1) through the matter channel alone.*

The law can be read directly off the curvature, and its asymptotic hypotheses hold on the explicit spacetimes. The balance (11) closes two independent ways (kinematically from $P_B^\mu = m(u)v^\mu(u)$ and from the integrated radiated flux of the *curvature-derived* n^2) as an identity of the field equations, the bulk dipole (12) delivering the thrust ma (Sec. 6). The law specializes cleanly across three exact spacetimes: on the accelerating Kinnersley rocket the radiated flux equals the kinematic recoil and the news vanishes (the Damour dipole is gravitational-wave silent); on static Schwarzschild the flux vanishes; and on a Vaidya [50] monopole (the $G_0 \rightarrow 0$ straight-worldline limit of the same rocket) the energy flux equals $-\dot{m}$ with *zero* radiated momentum: isotropic mass loss does not steer. The peeling hypothesis holds analytically on the

algebraically special Kinnersley exterior, with the Damour dipole's Ψ_4 silence genuine rather than a coordinate artifact (App. I). On a saturating maneuver the Bondi mass is monotone and strictly positive with $\dot{m}_B = -F^0$. The no-reactionless-steering law is thus universal, with the photon rocket as its positive-energy saturating instance.

4 Canonical construction

4.1 The three regions

The spacetime has the nested structure

$$\mathcal{M} = \mathcal{M}_- \cup_{\Sigma} \mathcal{M}_+, \quad \Sigma : r = R(u, \Omega), \quad (2)$$

with $\mathcal{M}_-$ a regular cavity (Minkowski, or weakly curved ordinary matter satisfying the energy conditions), Σ a timelike worldtube, and $\mathcal{M}_+$ the photon-rocket exterior (Fig. 1).

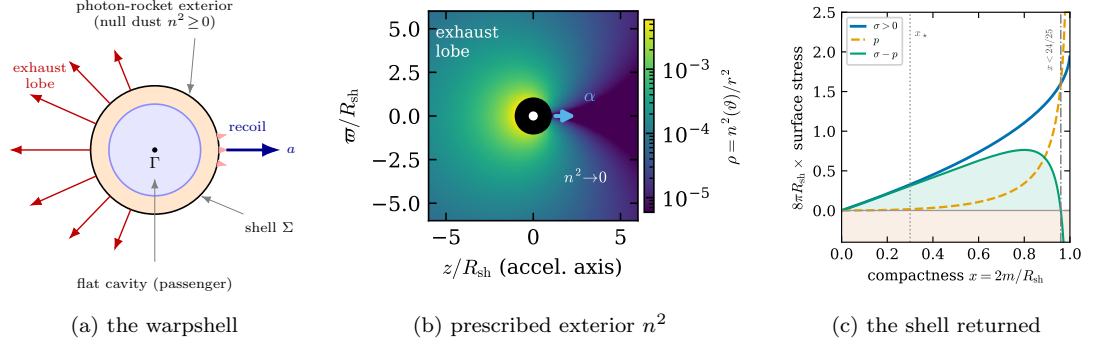


Figure 1. The worldtube-first warpsell: the schematic, the exterior it prescribes, and the shell the field equations return. (a) Schematic cross-section: a flat passenger cavity $\mathcal{M}_-$ is matched across a timelike shell Σ to the exact photon-rocket exterior $\mathcal{M}_+$, whose source is outgoing null dust with $n^2 \geq 0$; the exhaust is a forward-suppressed monopole-plus-dipole pattern whose dipole moment recoils the shell, giving the central worldline Γ a covariant acceleration a paid for by Bondi mass loss $-\dot{m}$, with no exotic matter. The design order is inverted: Γ and the admissible exterior matter are prescribed, and the Einstein equations *return* Σ . (b) The prescribed exterior null-dust density $\rho = n^2(\vartheta)/r^2$ at peak acceleration (a single rear dipole lobe trailing the drive, vanishing at the forward pole, with the cavity ($r < R$) field-free) is the *input*. (c) The shell the field equations then return is the *output*: ordinary matter, with a strictly positive surface energy density $\sigma(x) > 0$ and worst-observer surface dominant-energy margin $\sigma - p > 0$ out to the compactness window $x < 24/25$ ($8\pi R$ units; operating point $x = 0.3$ marked).

4.2 The exterior: an exact photon rocket

In retarded coordinates $(u, r, \zeta, \bar{\zeta})$ the Kinnersley–Robinson–Trautman line element is

$$ds_+^2 = \frac{2r^2}{P^2} d\zeta d\bar{\zeta} - 2 du dr - 2H du^2 \quad (3)$$

$$2H = K - 2r (\ln P)_{,u} - \frac{2m(u)}{r}, \quad K = 2P^2 (\ln P)_{,\zeta\bar{\zeta}}, \quad (4)$$

where $m(u)$ is the Bondi mass and the steering is carried by

$$P(u, \zeta, \bar{\zeta}) = A(u) + B(u) \zeta + \bar{B}(u) \bar{\zeta} + C(u) \zeta \bar{\zeta}, \quad (5)$$

normalized so that $AC - B\bar{B} = \frac{1}{2}$ gives the unit-curvature ($K = 1$) round sphere. The functions $(A, B, C)(u)$ encode the instantaneous Lorentz frame of the source [21, 24]; a time-dependent boost is a genuine, covariant acceleration of the central worldline (an arbitrary $K = 1$ conformal factor not of this boost form does not solve the field equations, App. A).

The stress-energy is pure outgoing null dust,

$$T_{ab}^+ = \frac{n^2}{r^2} \ell_a \ell_b, \quad \ell_a = -(du)_a \text{ (null)} \quad (6)$$

with the Einstein equations fixing the amplitude

$$4\pi n^2(u, \zeta, \bar{\zeta}) = -\dot{m} + 3m (\ln P)_{,u}. \quad (7)$$

Because ℓ_a is null, every contraction of T_{ab}^+ with a causal vector is proportional to n^2 ; the bulk energy conditions therefore reduce to the single positivity $n^2 \geq 0$, proved in Sec. 5.

4.3 The static limit is Schwarzschild

Setting $m = \text{const}$ and (A, B, C) constant ($K = 1$, $(\ln P)_{,u} = 0$), Eqs. (3)–(4) reduce to

$$ds_+^2|_{\text{static}} = -\left(1 - \frac{2m}{r}\right)du^2 - 2du dr + r^2 d\Omega^2, \quad (8)$$

the Schwarzschild metric in retarded Eddington–Finkelstein form, with r the areal radius and $n^2 = 0$ (vacuum). This is the anchor of the existence theorem (Sec. 7).

4.4 The shell and its surface stress

Across Σ the first junction condition $[h_{ab}] = 0$ (continuity of the induced metric) fixes $R(u, \Omega)$, and the Lanczos equation returns the surface stress

$$S_{ab} = -\frac{1}{8\pi}([K_{ab}] - [K]h_{ab}) \quad (9)$$

supplemented (because the null radiation crosses Σ) by the *timelike*-shell flux-source balance laws derived in Sec. 7.2 (the standard Israel relations with an external crossing flux). The cavity interior is vacuum ($T_{ab}^- = 0$), so the jump $[T_{ab}]$ that sources the shell is carried entirely by the exterior radiation. The drive is specified by the pair (Γ, m) : a steering curve and a Bondi mass budget. Einstein’s equations then fix the radiation (7) that carries it, and the junction returns the shell that confines it.

5 Bulk energy conditions of the null-dust exterior

Proposition 3 (Bulk admissibility). *The exterior stress-energy (6) satisfies the null, weak, strong, and dominant energy conditions at every point of $\mathcal{M}_+$ if and only if $n^2 \geq 0$, equivalently*

$$-\dot{m} + 3m(\ln P)_{,u} \geq 0. \quad (10)$$

Proof. $T_{ab}^+ = (n^2/r^2)\ell_a\ell_b$ is algebraically rank-one null (Hawking–Ellis Type II, a defective null eigenvector; the frame-invariant content, the Type-I $\leftrightarrow$ II eigensolver nuance recorded once in App. A). For any causal v^a , $T_{ab}^+v^av^b = (n^2/r^2)(\ell_av^a)^2$, which is ≥ 0 for all v^a iff $n^2 \geq 0$; this is the WEC (and, restricting to null v^a , the NEC). The flux vector $-T_{ab}^+v^b = -(n^2/r^2)(\ell_bv^b)\ell^a$ is proportional to the null ℓ^a ; with $\ell_a = -(du)_a$ a future-directed v^a gives $\ell_bv^b < 0$, so for $n^2 \geq 0$ the flux $-(n^2/r^2)(\ell_bv^b)\ell^a$ is a non-negative multiple of the future-null ℓ^a , hence causal and future-directed: the DEC. The trace vanishes ($\ell_a\ell^a = 0$), so the SEC reduces to the WEC. Conversely $n^2 < 0$ makes $T_{ab}^+\ell^a\ell^b = (n^2/r^2)(\ell_a\ell^a)^2 < 0$ for null ℓ' transverse to ℓ , violating the NEC. $\square$

Proposition 3 reduces the bulk energy conditions to a single scalar, established in all frames by an observer-robust Hawking–Ellis classification [20, 19]. The choice of *pure* outgoing null dust is what keeps this clean: an imperfect-fluid reading of the photon rocket instead carries negative energy density on its trailing side, so that only its leading side satisfies the weak and dominant conditions [51], whereas the rank-one null source (6) is admissible at every point where $n^2 \geq 0$. Unlike Alcubierre-class drives, the exterior *satisfies* the NEC identically; this is the first half of our evasion of the Santiago–Schuster–Visser theorem (Sec. 14).

6 Bondi recoil and the steering control law

6.1 Covariant acceleration and Bondi momentum

The exterior (3) is asymptotically flat at null infinity, with Bondi four-momentum $P_B^\mu(u)$ whose monopole is the Bondi mass $m(u)$ (the rest-frame energy P_B^0 ; the full four-momentum is $P_B^\mu = m v^\mu$) and whose evolution is fixed by the radiated null momentum [52],

$$\frac{dP_B^\mu}{du} = -\oint n^2(u, \Omega) \hat{\ell}^\mu(\Omega) d\Omega, \quad (11)$$

$\hat{\ell}^\mu$ the asymptotic null direction (with n^2 normalized as in Eq. (7), so $\oint n^2 d\Omega = -\dot{m}$; this is the matter channel of the model-independent law (1), whose $4\pi n^2$ inside the $-\frac{1}{4\pi}$ prefactor is the same quantity). The spatial part of Eq. (11) is the recoil: the central worldline accelerates because the radiation carries away anisotropic momentum. This is a coordinate-free statement ($dP_B^\mu/du \neq 0$), so the acceleration in Definition 1(iv) is covariant, answering the coordinate-artifact objection [15, 53]. Equation (11) is the local, dipole form of the conservation law of Sec. 3.

6.2 The axisymmetric control law

Write the steering dipole as $(\ln P)_{,u} = -\alpha(u) \cos \vartheta + (\text{monopole})$, with ϑ measured from the acceleration axis.² With the Kinnersley normalization of Sec. 4 the coefficient α equals the covariant proper-acceleration magnitude, $\sqrt{a_\mu a^\mu} = |\alpha|$, directed along $\vartheta = 0$ (App. A); we set $a = \alpha$ and use the two interchangeably hereafter. Throughout, ϑ is the angle from the proper-acceleration axis *in the momentary rest frame* of Γ at each retarded cut, so (12) and the control law below are per-cut rest-frame statements: at finite rapidity the fixed-frame angular pattern is the aberration of this rest-frame dipole, and “pure dipole” refers to the rest-frame form (App. A). Equation (7) becomes

$$4\pi n^2(u, \vartheta) = -\dot{m} - 3m \alpha(u) \cos \vartheta, \quad (12)$$

peaking at the rear pole $\vartheta = \pi$ (the exhaust, opposite the acceleration) and least at the forward pole (Fig. 1b). Positivity $n^2 \geq 0$ for all $\vartheta \in [0, \pi]$ requires the minimum, attained at the forward pole $\cos \vartheta = +1$, to be non-negative, giving the control law

$$\boxed{-\dot{m} \geq 3m |\alpha|}. \quad (13)$$

For general steering, $-\dot{m} \geq 3m \sup_\Omega [-(\ln P)_{,u}]_+$.

Proposition 4 (Steering positivity). *For any bounded acceleration history $\alpha(u)$ there exists a positive, monotonically non-increasing Bondi mass $m(u) > 0$ satisfying (13) on any finite interval $[u_0, u_f]$; explicitly any $m(u) = m_0 \exp[-3 \int_{u_0}^u |\alpha| du']$ saturating (13) works, and the retained mass fraction is*

$$\frac{m(u_f)}{m_0} = \exp\left[-3 \int_{u_0}^{u_f} |\alpha(u')| du'\right] \quad (14)$$

(so the fraction radiated away is $1 - m(u_f)/m_0$).

Equation (14) is the relativistic-rocket (Tsiolkovsky) budget, in the Bondi-mass-loss form Füzfa [49] used for radiation-powered interstellar travel: acceleration is paid for by Bondi mass loss. It replaces Alcubierre’s negative-energy requirement with a *positive*-energy inequality. The factor 3 (against the ideal collimated photon rocket’s $e^{-\Delta\eta}$) is the price of the mandatory broad Kinnersley dipole exhaust $4\pi n^2 = 3ma(1 - \cos \vartheta)$ at saturation, which delivers only 1/3 of its radiated energy as net momentum (the $\langle \cos^2 \vartheta \rangle = 1/3$ projection). This factor 3 is a property of the news-silent Robinson–Trautman class, whose emission is forced to be a broad dipole, not a fundamental general-relativistic floor: a more collimated exhaust would approach the ideal photon rocket’s $-\dot{m} = m|a|$, at the price of leaving the gravitational-wave-silent class (Secs. 10, 11). The cost is thus severe (order-unity fractional mass loss for order-unity changes in rapidity), but it is positive and finite (Sec. 15).

7 Existence of admissible accelerating shells

The key surface result is that the matched shell can satisfy the *surface* energy conditions while the worldline accelerates. We prove it by deforming a strictly admissible static anchor.

7.1 The static anchor

Match a flat interior to the Schwarzschild exterior (8) across a static shell at areal radius R , with $x := 2m/R \in (0, 1)$. The extrinsic-curvature jump [27] gives the standard surface density and pressure

$$\sigma_0 = \frac{1}{4\pi R} (1 - \sqrt{1-x}), \quad (15)$$

$$p_0 = \frac{1}{8\pi R} \left(\frac{1-x/2}{\sqrt{1-x}} - 1 \right). \quad (16)$$

Both are strictly positive on $(0, 1)$, so the WEC ($\sigma_0 \geq 0, \sigma_0 + p_0 \geq 0$) holds throughout. The DEC requires in addition $\sigma_0 \geq |p_0| = p_0$. Writing $s := \sqrt{1-x}$,

$$8\pi R (\sigma_0 - p_0) s = -\frac{5}{2}s^2 + 3s - \frac{1}{2} = -\frac{1}{2}(5s-1)(s-1), \quad (17)$$

²With ϑ measured from the proper-acceleration axis, the drive accelerates toward the forward pole $\vartheta = 0$ and the exhaust (maximal n^2) trails at the rear pole $\vartheta = \pi$, so $4\pi n^2 = -\dot{m} - 3ma \cos \vartheta$ [Eqs. (7)–(12)], $\oint n^2 d\Omega = -\dot{m}$, and the thrust is $dP_B^z/du = -\oint n^2 \cos \vartheta d\Omega = +ma$. Bulk positivity $n^2 \geq 0$ then binds at the forward pole, where the dipole subtracts, while the surface dominant-energy frontier (Sec. 8) binds at the distinct rear pole, where the effective compactness $x + 2ar$ is deepest.

which is positive iff $1/5 < s < 1$, i.e.

$$\boxed{\text{surface DEC} \iff x = \frac{2m}{R} < \frac{24}{25}}. \quad (18)$$

Lemma 1 (Strictly admissible anchor). *For every $x \in (0, 24/25)$ the static Schwarzschild–Minkowski shell satisfies the surface WEC and DEC strictly. On any compact subinterval $[x_1, x_2] \subset (0, 24/25)$ the worst-case margin $\delta_0 := \min_{[x_1, x_2]}(\sigma_0 - p_0) > 0$ is bounded away from zero.*

The window (18) is wide: a flat-interior shell tolerates compactness almost up to $x = 1$. This $24/25$ window is the flat-interior thin-shell case of the rigorous static thin-shell Buchdahl-type bounds of Andréasson [54] and Horvat–Ilić [55]: it coincides with the $\Omega = 2$ member of the sharp anisotropic ladder $\sup 2m/R = ((2\Omega + 1)^2 - 1)/(2\Omega + 1)^2$ of Andréasson [56], whose $\Omega = 1, 2, 3$ entries are $8/9, 24/25, 48/49$. Here Ω is the constant in that paper’s anisotropic bound $p_r + 2p_t \leq \Omega\rho$, with p_r the radial and p_t the tangential pressure: $\Omega = 1$ is the classic isotropic/Vlasov value ($8/9$), a purely tangential thin shell ($p_r = 0$, surface DEC $p_t \leq \rho$) sits at $\Omega = 2$ through $p_r + 2p_t = 2p_t \leq 2\rho$ (the $24/25$ window), and $\Omega = 3$ gives the most general anisotropic bound $48/49$ [56]. We center the deformation at intermediate compactness, where the anchor margin δ_0 is large. This is precisely the regime that the smooth source-first shells of Ref. [18] could not reach: there the dominant-energy failure lived in the radial gradients of a finite-width transition (Hawking–Ellis Type IV in the tail), whereas the distributional junction concentrates the transition into S_{ab} over a *non-vacuum, positive-radiation* exterior. Section 7.4 makes the reconciliation explicit.

7.2 The junction and flux-source balance laws

Because the null dust (6) streams radially outward, its projections on the timelike Σ (unit outward normal N^a , intrinsic triad $\{e_{\hat{a}}\}$) are

$$T_{NN}^+ = \frac{n^2}{r^2}(\ell_a N^a)^2 \geq 0, \quad T_{N\hat{a}}^+ = \frac{n^2}{r^2}(\ell_b N^b)(\ell_c e_{\hat{a}}^c), \quad (19)$$

whose energy-flux component $T_{N\hat{0}}^+ \neq 0$ (the radially streaming dust carrying no lateral flux, $T_{N\hat{\theta}}^+ = T_{N\hat{\phi}}^+ = 0$), so matter crosses the shell and the static Israel relation must be supplemented by the flux-source balance laws [26, 27] (the standard *timelike*-shell junction with an external crossing flux; *not* the lightlike Barrabès–Israel limit [57, 58], since Σ is timelike)

$$D_b S^b_{\hat{a}} = -[T_{N\hat{a}}], \quad (20)$$

$$S^{ab} \bar{K}_{ab} = -[T_{NN}], \quad \bar{K}_{ab} := \frac{1}{2}(K_{ab}^+ + K_{ab}^-). \quad (21)$$

Physically the shell is a *radiating* layer, the regularized rocket nozzle: the constitutive hypothesis of the construction is that this surface layer supplies the prescribed outgoing null flux at Σ (the radiation Kinnersley’s solution emits from the now-excised worldline singularity) while carrying the admissible S_{ab} , and the recoil (20) is the thrust. Consistency requires that the angular moments of the crossing flux reproduce the bulk budget,

$$\oint T_{N\hat{0}}^+ d\Omega \propto \dot{m}, \quad - \oint T_{N\hat{0}}^+ \cos \vartheta d\Omega \propto m a, \quad (22)$$

where $T_{N\hat{0}}^+ \propto n^2(\vartheta)$ is the normal energy flux (the proportionality absorbing the angle-dependent normal and redshift factors at the shell; the radially streaming null dust carries no lateral component, $T_{N\hat{\theta}}^+ = 0$): i.e. the monopole of the crossing energy flux is the luminosity $-\dot{m}$ and its $\cos \vartheta$ -moment, the axial momentum flux, is the thrust $m a$, reproducing the Bondi budget (11). The moment identities (22) are derived in Appendix B.

7.3 Existence by openness

Introduce the dimensionless deformation parameters

$$\lambda := |a|R, \quad \mu := |\dot{m}|R/m, \quad (23)$$

and expand the surface stress about the anchor,

$$S_{ab} = S_{ab}^{(0)} + \lambda S_{ab}^{(a)} + \mu S_{ab}^{(\dot{m})} + \mathcal{O}(\lambda^2, \mu^2, \lambda\mu), \quad (24)$$

where $S_{ab}^{(a)}$ carries the dipole ($\ell = 1$) steering response (including the induced surface energy flux $S_{\hat{0}\hat{\vartheta}}$ from (20)), and $S_{ab}^{(\hat{m})}$ the monopole ($\ell = 0$) luminosity/tension shift.

Lemma 2 (Darmois solvability at the anchor). *At the static anchor the geometry is exactly Schwarzschild–Minkowski and the shell’s induced 2-metric is the round sphere, so the linearized first-junction (Darmois) map DF_0 acts in its $\ell \geq 2$ shape sector as the round-sphere rigidity operator $\Delta_{S^2} + 2$, whose kernel is the $\ell = 1$ rigid translations. Off that kernel DF_0 is a bounded isomorphism in the $\ell \geq 2$ shape sector and in the $\ell = 0$ monopole block, and it is surjective onto the $\mathcal{O}(a)$ steering source in $\ell = 1$. Exact first-junction solvability for $a > 0$ then reduces to the non-vanishing of the single closed-form scalar $D(x) = \det J(x)/a$, $D(x) = (s^5 - 6s^4 - s^3 + 6s^2 - 4)/(3s^3)$ with $s = \sqrt{1-x}$ (58), which is strictly negative with $|D(x)| \geq \frac{4}{3}$ and monotone in magnitude on $(0, 24/25)$ (60). Hence local Darmois solvability holds at the anchor for every small $a > 0$.*

Proof sketch. The $\ell = 1$ kernel is the rigid translations (Cohn–Vossen/Weyl rigidity [59]); off it the operative bound is the *Schauder* estimate $\|u\|_{C^{2,\alpha}} \leq C_S \|(\Delta_{S^2} + 2)u\|_{C^{0,\alpha}}$ (App. D), which delivers the C^2 shell embedding and hence the continuity of $[K_{ab}]$ and S_{ab} (the cruder L^2 inverse $\|(\Delta_{S^2} + 2)^{-1}\|_{L^2} = \frac{1}{2}$, smallest off-kernel eigenvalue magnitude 2, controls only the spectral gap that sets C_S). The $\ell = 0$ monopole block is invertible in closed form through the redshift Jacobian $\partial_R[\sqrt{f}] = m/(R^2\sqrt{1-x}) \neq 0$. In the $\ell = 1$ steering sector the foliation-tilt row $(\partial[h_{\tau\vartheta}]^{\sin}/\partial w_1 = 1)$ and the closed-form $\cos\vartheta$ redshift row (54) are independent for every $x \in (0, 1)$, spanning the two-dimensional source $([h_{\tau\tau}]^{\cos}, [h_{\tau\vartheta}]^{\sin})$, while the constant part of $[h_{\tau\tau}]$ carries no $\mathcal{O}(a)$ source (a $\cos\vartheta$ shape feeds the $\ell = 0$ jump only through the $\langle \cos^2\vartheta \rangle$ self-coupling, hence at $\mathcal{O}(a^2)$). Exact solvability for $a > 0$ then reduces, by Lyapunov–Schmidt, to the non-vanishing of the constant-direction bifurcation scalar $D(x) = \det J(x)/a$, in closed form (58) with $|D(x)| \geq \frac{4}{3}$ by an elementary factorization (App. D). $\square$

Theorem 2 (Existence of admissible accelerating warps). *Fix $x_* \in (0, 24/25)$ with anchor margin $\delta_0 > 0$, and restrict the steering to $\lambda = |a|R < 1$, so that the accelerated cavity admits a regular Rindler chart whose acceleration horizon does not reach Σ (Herglotz–Noether; see Sec. 15). By Lemma 2 the first junction is solvable for $a > 0$, closing on the closed-form scalar $D(x)$ with $|D| \geq \frac{4}{3}$, and the matched surface stress $S_{ab}(\lambda, \mu)$ is then C^1 in (λ, μ) , uniformly in $\Omega \in S^2$. Then there exist $\epsilon_a, \epsilon_{\hat{m}} > 0$ such that for all $\lambda < \epsilon_a$, $\mu < \epsilon_{\hat{m}}$ satisfying the bulk law (13), the matched warpsell is admissible: its surface stress S_{ab} is Hawking–Ellis Type I and satisfies the surface WEC and DEC strictly. A sufficient explicit condition is*

$$C_a \lambda + C_{\hat{m}} \mu < \delta_0, \quad C_a := \|S_{ab}^{(a)}\|, \quad C_{\hat{m}} := \|S_{ab}^{(\hat{m})}\|. \quad (25)$$

The parameters are not independent: the bulk law (13) forces $\mu = |\hat{m}|R/m \geq 3\lambda$, so the binding case is the minimum-fuel slice $\mu = 3\lambda$, for which (25) reads $(C_a + 3C_{\hat{m}})\lambda < \delta_0$.

Proof sketch. Admissibility is decided by the worst-observer surface dominant-energy margin, the Hawking–Ellis Type-I eigenvalue margin (which for a Type-I stress also signs the WEC),

$$\Phi(\lambda, \mu) := \min_{\Omega \in S^2} (\tilde{\sigma} - \max\{|\tilde{p}_{\hat{\vartheta}}|, |p_{\hat{\varphi}}|\}), \quad (26)$$

built from the Type-I eigenvalues of the $(\hat{0}, \hat{\vartheta})$ stress block, $\tilde{\sigma} = \frac{1}{2}[(\sigma - p_{\hat{\vartheta}}) + \sqrt{\Delta_s}]$ and $\tilde{p}_{\hat{\vartheta}} = \frac{1}{2}[(p_{\hat{\vartheta}} - \sigma) + \sqrt{\Delta_s}]$, with discriminant $\Delta_s = (\sigma + p_{\hat{\vartheta}})^2 - 4q^2 \geq 0$ (the stress is Type I iff $\Delta_s \geq 0$); the recoil flux $q = S_{\hat{0}\hat{\vartheta}}$ enters only through this boosted block and vanishes at the binding rear pole $\vartheta \rightarrow \pi$, where $\Phi = \sigma - p$. The extrinsic curvatures are algebraic-plus-first-derivative functionals of the smooth metric data $m(u)$ and $P(u, \zeta, \bar{\zeta})$, and the balance laws (20)–(21) are linear in the smooth crossing flux; hence $S_{ab}(\lambda, \mu)$ is C^1 near the anchor (no denominator $\sqrt{1-x_*}$ vanishes). The maps $S_{ab} \mapsto \{\sigma, \sigma \pm p_A, \dots\}$ are continuous, and min over the compact S^2 preserves continuity, so Φ is continuous. $\Phi(0, 0) = \delta_0 > 0$ by Lemma 1 (at the anchor all flux terms vanish). Continuity gives an open neighborhood on which $\Phi > \frac{1}{2}\delta_0 > 0$; the bound (25) follows from the linear Lipschitz estimate $\|S_{ab} - S_{ab}^{(0)}\| \leq C_a \lambda + C_{\hat{m}} \mu + \mathcal{O}(\lambda^2, \mu^2, \lambda\mu)$. By Lemma 2, $[h_{ab}] = 0$ is solvable for the interior / R at fixed (λ, μ) , the $\ell = 1$ closure resting on the nonzero closed-form $D(x)$, and the implicit function theorem returns the matched shell (Proposition 9, App. D). The same anchor solvability underlies the dynamical Proposition 6 and the Newton–Kantorovich estimate of App. C. $\square$

Theorem 2 is the existence half of the paper’s claim: *accelerating, positive-energy, energy-condition-respecting warpsells exist*. With Lemma 2 it is *unconditional* at small amplitude, the $\ell = 1$ non-degeneracy carried by the proven bound $|D(x)| \geq \frac{4}{3} > 0$ (60). The one input not in closed form is the *explicit* size of the admissible neighborhood, i.e. whether the rigorous reach extends to the numerical lower-bound frontier. That explicit reach is mapped numerically in App. C; a rigorous version (explicit elliptic constants, conservative; or the interval-arithmetic/embedding route for the sharp frontier) is set out as future work in App. C.

7.4 Reconciliation with the boundary-cost theorem

Reference [18] found *no* admissible smooth shells, the dominant-energy failure localizing at the source–vacuum transition and persisting in the static limit, the closed-form radial-tension obstruction of Prop. 7 below. This does not contradict Theorem 2. The boundary cost is carried by the radial gradients and induced anisotropy of a *finite-width* matter profile dying into *vacuum*; it appears as Type IV stress-energy in the transition tail. The radiative warpsell removes both ingredients: the transition is a measure-zero distributional junction (no finite-width gradient tail), and the exterior is *not vacuum*: it is positive null radiation, so there is no “matter dies into vacuum” curvature spike at the place where momentum must be transported. The static anchor of Sec. 7.1 is the clearest example: the thin Schwarzschild–Minkowski shell satisfies DEC strictly for $x < 24/25$, exactly the regime the smooth scan could not reach. The thin-shell limit does not *hide* the boundary cost; it *replaces* the obstructing geometry by an admissible one, as Theorem 2 establishes.

8 The acceleration–compactness frontier

How fast can a positive-energy shell accelerate before DEC fails? Decompose the deformed surface stress in spherical harmonics about the anchor. The monopole ($\ell = 0$) is the luminosity/tension shift; the dipole ($\ell = 1$) is the steering, which produces a front–back asymmetry in σ, p and an equatorial surface flux $q_{\hat{g}} = \lambda q_1 \sin \vartheta + \mathcal{O}(\lambda^2)$. The flux-inclusive Type-I DEC is $\sigma \geq \sqrt{p^2 + q^2}$; since $q = \mathcal{O}(\lambda)$ vanishes at the anchor, the recoil flux enters this margin only at $\mathcal{O}(q^2) = \mathcal{O}(\lambda^2)$, so the binding angular value to linear order is set by the cos-dipoles alone,

$$\Phi_{\text{DEC}}(\Omega) \geq (\sigma_0 - p_0) - \lambda(|\sigma_1| + |p_1|) + \mathcal{O}(\lambda^2), \quad (27)$$

and $\Phi_{\text{DEC}} \geq 0$ yields a frontier

$$\boxed{a_{\text{max}} R \leq g(x)}, \quad (28)$$

where $g(x)$ is the largest $\lambda = aR$ admitting a *regular* admissible matched shell, i.e. the lesser of the λ at which the observer-robust worst-observer surface DEC margin first vanishes (App. C) and the λ at which the exterior areal foliation degenerates ($f - 2aR \rightarrow 0$, Prop. 5); the latter caps $g(x)$ below the kinematic ceiling $\frac{1}{2}(1 - x)$ *by construction*, since admissibility of a regular shell presupposes a regular foliation. A naive L^1 estimate $g \simeq (\sigma_0 - p_0)/(|\sigma_1| + |p_1|)$ *over-counts* the response (it sums the dipoles as if co-binding, whereas the matched DEC binds at the rear pole) and is superseded by the margin; the recoil flux q_1 peaks at the equator and enters the margin only at $\mathcal{O}(q^2)$, not linearly. Two facts then control g (App. C). The linear surface response is *locally Schwarzschild*, $\sigma_1 = \mathcal{O}(x^2)$, and the recoil flux does not enter the margin at linear order (it appears only as q^2 , whose λ -derivative vanishes at the anchor); the $\mathcal{O}(\lambda)$ maneuver envelope is therefore *wide* and shape-dominated, scaling as $\mathcal{O}(1/x)$ at small compactness. The fully nonlinear multipole matched solve (App. C) closes the junction at finite λ *numerically* and yields a strictly positive Hawking–Ellis Type I worst-observer DEC margin (rear-pole binding) at a computed lower bound $\lambda = \underline{g}(x)$ that the independent observer-robust dipole scan reproduces. The frontier is *shape-limited* at the rear pole; the wide $\mathcal{O}(\lambda)$ envelope $g_{\text{lin}} \sim 1/x$ is the tangent the $\mathcal{O}(\lambda^2)$ margin overtakes. The computed lower bounds $\underline{g}(x)$ (45) (App. C) keep the positive-energy maneuver window broad and only weakly compactness-dependent through the lower-to-middle range, narrowing toward $x \rightarrow 1$. While $g(x)$ has no known elementary closed form (its value is set beyond the low-order steering expansion, App. C), the numerics obey a physically motivated upper envelope. The rear pole carries an effective redshift $|\nabla r|_g^2 = f - 2ar$ (Eq. (47)), so by the equivalence principle its effective compactness is $x_{\text{eff}} = x + 2ar$; treating the *rigid-shape* rear pole as a static shell at x_{eff} reaching the surface-DEC threshold $x_{\text{eff}} = 24/25$ gives

$$g(x) \lesssim g_{\text{env}}(x) := \frac{1}{2} \left(\frac{24}{25} - x \right), \quad (29)$$

respected by the numerical frontier at every x . This envelope is *empirical, not a rigorous inequality*: its static-shell-at- x_{eff} derivation is heuristic, and the first-principles *frozen-shape*

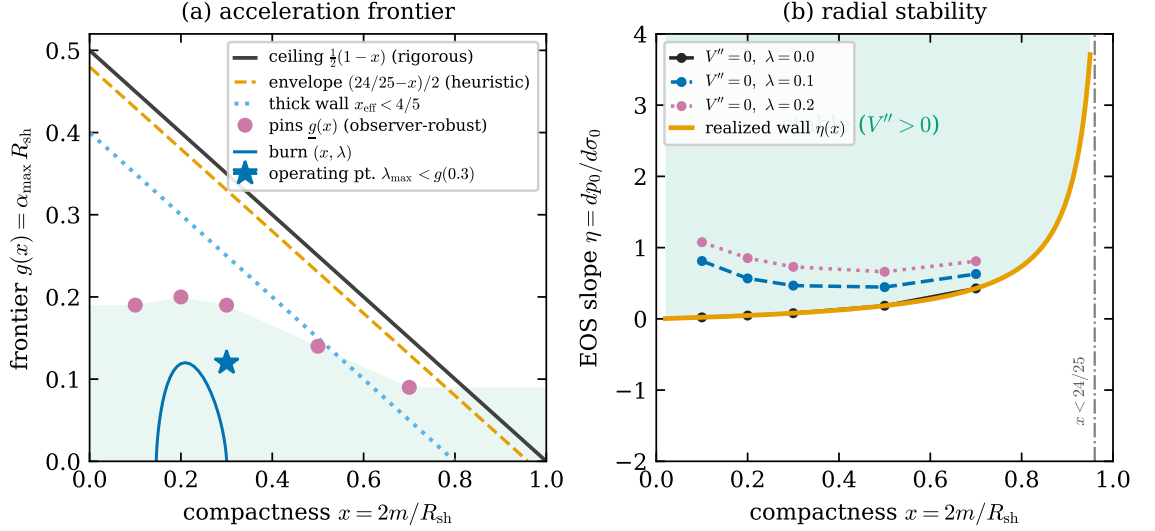


Figure 2. The static anchor’s two admissibility windows versus compactness $x = 2m/R$. (a) The acceleration–compactness frontier $g(x) = a_{\max}R$, read by authority level: the kinematic ceiling $\frac{1}{2}(1-x)$ (Eq. (30), solid) is a rigorous, shape-independent *upper* bound; the computed values $\underline{g}(x)$ are observer-robust *lower bounds* at the sampled compactnesses (the faint shading merely interpolates them as a guide); the envelope $\frac{1}{2}(\frac{24}{25}-x)$ (Eq. (29), dashed) is heuristic; and the thick-wall realizability bound $\lambda < \frac{1}{2}(\frac{4}{5}-x)$ (dotted, $x_{\text{eff}} = x + 2\lambda < 4/5$) is the binding finite-thickness constraint. The saturating burn of Theorem 3 and its operating point $\lambda_{\max} = 0.12 < g(0.3)$ stay strictly inside every window. (b) Poisson–Visser radial stability (App. J): the realized tangential-pressure wall $\eta(x)$ lies *exactly* on the marginal $V'' = 0$ curve (a zero mode), the frozen-background acceleration λ shifting it upward; the surface-DEC window $x < 24/25$ is marked.

rear-pole margin in fact *rises* at low order (positive $\mathcal{O}(\lambda^2)$ coefficients, App. C), so over the operating range the margin is closed by the global $\mathcal{O}(\lambda^2)$ shape response, not by the rigid-shape redshift. The rear pole’s status as the binding angle rests on the observer-robust all-angle margin, not a closed-form angular-dominance proof.

One rigorous, closed-form bound nonetheless survives, shape-independent.

Proposition 5 (Rigorous kinematic ceiling). *The exterior areal foliation is regular at the shell (the areal-radius gradient spacelike) only if the rear pole stays outside its acceleration-induced effective horizon, $f - 2aR > 0$. On the symmetry axis the inverse metric is exactly $g^{rr} = 2H = f \mp 2aR$ (– rear, + front), where coordinate and areal radius coincide (the off-axis form $f + 2ar \cos \vartheta$ is the leading-order/axis expression, deviating by $\mathcal{O}(a^2)$ at the equator); the binding rear-pole value $f - 2aR$ is exact. Here R is the rear-pole areal radius, equal on the symmetry axis to the shell’s areal radius there and entering both $x = 2m/R$ and $\lambda = aR$, so the bound is shape-independent (Eq. (47)):*

$$g(x) = a_{\max}R < \frac{1}{2}(1-x), \quad x_{\text{eff}} = x + 2aR < 1. \quad (30)$$

This is exact and shape-independent (a property of the exterior geometry, not of S_{ab}), strictly tighter than the interior Herglotz–Noether bound $\lambda < 1$ (Sec. 15), and the computed lower bound lies strictly inside it, the gap closing as $x \rightarrow 1$. It is a regularity ceiling (the condition that the rear pole stay outside its acceleration-induced effective horizon) of the specific Kinnersley exterior foliation, not a universal dominant-energy frontier; the operative DEC frontier $g(x)$ binds below it on the numerically tested range (Sec. 8, Eq. (46)).

Equation (30) is the one rigorous, closed-form member of the frontier hierarchy (Fig. 2): the empirical envelope (29) is tighter but heuristic, and $g(x)$ itself is a nonlinear global isometric-embedding constant (App. C). For a *general* accelerating shell (30) holds only as a quasi-local normal-form condition $2m_{\text{ql}}/R + 2aR < 1 + \varepsilon$, the exact areal combination being specific to the Kinnersley foliation.

The inequality (28) is a Buchdahl-type bound for *kinematics in form*: just as $2m/R \leq 8/9$ caps the static compactness of a fluid sphere [56], $a_{\max}R \leq g(x)$ caps the proper acceleration of a positive-energy shell, achieved here with the energy conditions intact everywhere, in contrast to exotic-matter warp drives. Independently of the size of the admissible set, (28) is itself a positive-energy maneuver envelope.

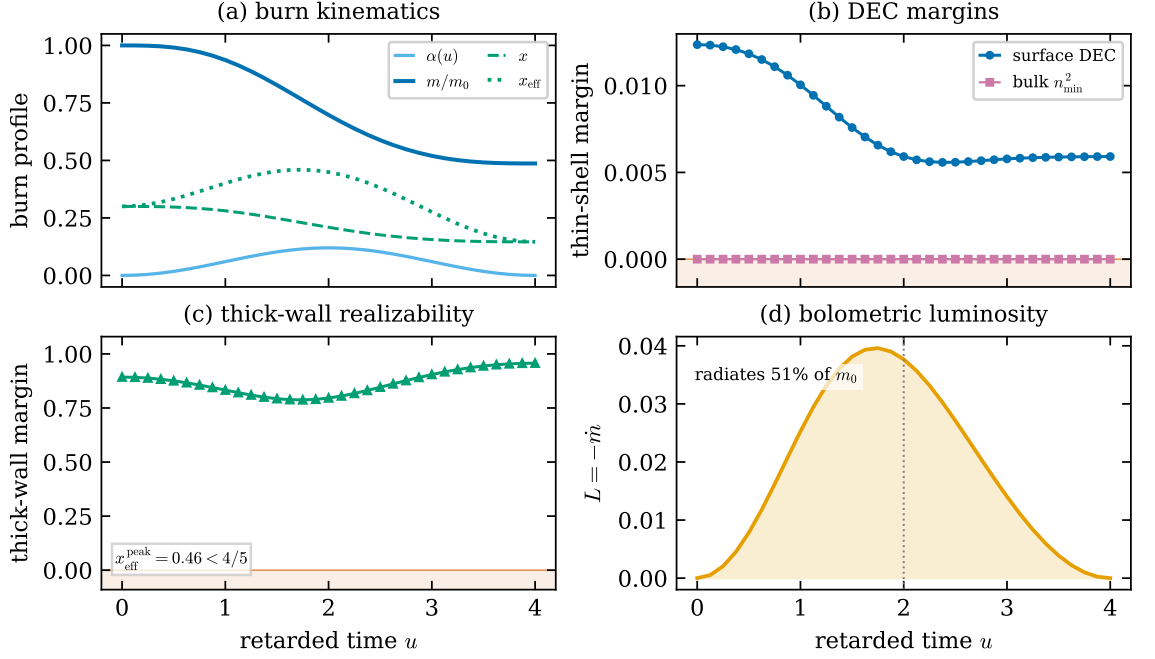


Figure 3. The complete finite-duration maneuver (Theorem 3) for one saturating burn: admissible three ways at every retarded time, at the mandatory cost of a radiated exhaust. (a) the kinematics $a(u)$, $m(u)/m_0$, $x(u) = 2m/R$, and rear-pole effective compactness $x_{\text{eff}} = x + 2\lambda$; (b) the worst-observer thin-shell surface DEC margin (Type I) and the bulk minimum n_{min}^2 , both above the shaded violating half-plane; (c) thick-wall realizability, the static tangential-pressure-wall margin at $x_{\text{eff}}(u)$ staying strictly positive ($\Leftrightarrow x_{\text{eff}} < 4/5$, Sec. 9.3); (d) the bolometric luminosity $L(u) = -\dot{m}$, the burn radiating $\approx 51\%$ of the rest mass to steer. The angular emission pattern $4\pi n^2(\vartheta) = -\dot{m} - 3ma \cos \vartheta$ is the minimum-radiation profile of Fig. 5.

8.1 From the frontier to a complete maneuver

Theorem 2 is local in retarded time and the frontier (28) caps a single instant; assembled along a trajectory they give a *complete finite-duration maneuver*.

Theorem 3 (Admissible finite maneuver, per-instant quasi-static). *Fix $x_\star < 24/25$ and a steering history $a(u)$ supported on $[u_0, u_f]$ with $a(u_0) = a(u_f) = 0$ and nonzero net rapidity $\Delta\eta = \int_{u_0}^{u_f} a \, du \neq 0$ (so the endpoint Bondi velocities differ, $v(u_f) \neq v(u_0)$; in particular any single-signed $a \geq 0$, as in the saturating bump below, while a sign-changing round-trip with $\Delta\eta = 0$ returns to rest and steers no net velocity), the saturated Tsiolkovsky budget $m(u) = m_0 e^{-3 \int_{u_0}^u |a|}$ (14) (so $x(u) = 2m(u)/R$, and $\lambda(u) = |a(u)|R < \underline{g}(x(u))$ throughout, with $\underline{g}$ the numerical lower-bound frontier of Sec. 8. Then at every retarded cut $u \in [u_0, u_f]$ the matched quasi-static warshell is admissible: (i) the bulk obeys $n^2 \geq 0$ for all ϑ (closed form, the control law); (ii) the surface stress is Hawking–Ellis Type I with worst-observer surface DEC margin > 0 ; and (iii) the Rindler horizon never reaches Σ ($\lambda < 1$, closed form); the endpoints are the strictly admissible static anchor. The net rapidity $\Delta\eta \neq 0$ (by hypothesis) changes the Bondi velocity $v^i = P_B^i/P_B^0$ between the stationary (zero-acceleration) endpoints, $\Delta v \neq 0$ (covariantly $dP_B^\mu/du = \dot{m} v^\mu + m a^\mu$, the integrated $\ell = 1$ flux), a genuine velocity change radiated as positive null momentum.*

These hypotheses assemble the per-instant results uniformly in retarded time: the anchor DEC window (Lemma 1), the frontier (28), and the Herglotz–Noether ceiling $\lambda < 1$ (Sec. 15). Clauses (i) and (iii) are closed-form and hold for every admissible amplitude; clause (ii) is established on the numerical lower-bound frontier $\underline{g}$, where the frozen-shape rear-pole margin is in addition closed-form and strictly positive at the operating amplitude (App. C), so the conservative bound $\lambda(u) < \underline{g}(x(u))$ guarantees admissibility throughout. The construction is *per-instant quasi-static*: at each retarded cut the shell is the gauge-invariant finite- λ junction at the instantaneous $(x(u), \lambda(u))$ (App. C), with the dynamical $\mathcal{O}(\mu^2)$ corrections deferred to future work (App. F). Because the rocket radiates, $x(u) = 2m(u)/R$ decreases monotonically from its initial value $x_\star$, so the whole path $(x(u), \lambda(u))$ stays inside the admissible window. Along such a bump profile $a(u)$ the worst-observer surface margin and the bulk minimum stay non-negative at every retarded step, the Bondi balance (11) closes identically, and the endpoints recover the static anchor; a worked instance is collected in App. K (Fig. 3). This closes, at the per-instant quasi-static level, the

surface-DEC question that a static anchor and a perturbative existence theorem leave open: a finite burn that changes the velocity is admissible *throughout*, not only at its stationary endpoints.

8.2 From snapshots to a time-evolved spacetime

Theorem 3 establishes a *sequence* of instantaneous junctions. We now lift it to a statement about the *time-evolved* spacetime. The construction is exact on both sides at every retarded time (the exterior is the Kinnersley photon rocket for the full $m(u)$ and worldline, the interior is exactly flat); matched across a C^2 worldtube ($[h_{ab}] = 0$) it *would be* an exact, time-dependent distributional Einstein solution. Two questions remain: whether the per-cut matches assemble into that C^2 worldtube (Prop. 6), and whether its *exact* (fully retarded) surface stress is then admissible. This requires the dynamical junction to be evaluated on a cut of *constant retarded time* (a constant- t slice mixes retarded times across the shell and is not the physical cut); the explicit retarded-time derivatives ($\dot{a}$, $\dot{m}$) then enter S_{ab} through the time-dependent extrinsic curvature.

Proposition 6 (Conditional dynamical admissibility for slow, fixed-axis, sub-horizon burns). *Fix $x_\star \in (0, 24/25)$. For a slow, fixed-axis (non-rotating, reflection-symmetric), sub-horizon burn ($\inf(f - 2aR) > 0$), the per-cut matched shells of Theorem 2 assemble into a single gauge-fixed C^2 family of matched timelike worldtubes $\Sigma(\lambda, \varepsilon)$, conditional on the constant-retarded-cut compatibility identity (64) (the $\ell \geq 2$ retarded-energy-row consistency, a flat-interior embeddability condition), whose leading $\mathcal{O}(a^2)$ ($\ell = 2$) obstruction is shown to cancel gauge-completely (App. E); the all-orders identity and the rotating-axis case remain open (Sec. 15). The exact retarded surface stress S_{ab} is then continuous in (λ, ε) , where $\lambda = aR$ and ε is the burn rate in a C^2 norm of $(m(u), a(u))$, reducing at $(\lambda, \varepsilon) = (0, 0)$ to the static anchor with strict Hawking–Ellis Type I surface DEC margin $\delta_0 > 0$ (Lemma 1). Hence there exist $\lambda_\star, \varepsilon_\star > 0$ such that for $\lambda < \lambda_\star$, $\varepsilon < \varepsilon_\star$ the matched shell is Type I with strict surface DEC throughout the burn: a time-evolved admissible accelerating spacetime, not a sequence of snapshots.*

This is the openness of the strict (Type-I eigenvalue) DEC margin under a continuous perturbation of an exact admissible solution, the dynamical counterpart of Theorem 2 now carrying the retarded $\dot{m}, \dot{a}$ terms. The per-cut shape match $[h_{AB}] = 0$ is unconditional at small amplitude (the static elliptic problem of Lemma 2 at each cut); it is the compatibility identity (64) that upgrades the admissible snapshots to a single C^2 worldtube. The *fixed-axis* restriction is on the time-evolved *assembly*, not on the steering direction: at small amplitude a matched shell exists per retarded cut for an *arbitrary* acceleration three-vector by $SO(3)$ -equivariance (Cor. 6), and the optimal-control reading of Sec. 11.1 is fully three-dimensional. Its analytic content is the constant-cut elliptic reduction of Lemmas 3–4, closed by a Grönwall bound in u once the time-row solvability (63) holds, with the multipole solve of App. C extending it to finite amplitude; the exterior’s Ricci tensor keeps the rank-one null-dust form $R_{ab} \propto \ell_a \ell_b$ along the burn, and the junction reduces to the static anchor as $a \rightarrow 0$. Two limits fix its reach. First, it is *perturbative*: small acceleration and slow burns, not the saturating amplitude. Second, the leading dynamical term is the outgoing mass-loss flux, which *tightens* the surface DEC margin through the dense wall core ($d(\text{margin})/d\mu < 0$ where the lapse deepening dominates), exactly as the $\mathcal{O}(\mu)$ back-reaction of Sec. 9.2 finds (App. F), so the fully dynamical frontier is somewhat tighter than $g(x)$; the saturating-amplitude, all-angle reach is left open (Sec. 15). The analytic route is a gauge-fixed coercive estimate $\|\delta Y\|_{H^{s+2}} \leq C \|D_Y F \delta Y\|_{H^s}$ for the linearized constant-retarded-cut junction $F := [h_{ab}] = 0$, propagated in u . The one structural question is the principal angular (Douglis–Nirenberg) symbol of the reduced map; we resolve it.

Lemma 3 (Noncharacteristic constant-cut symbol). *After Schur-eliminating the finite-dimensional $\ell \leq 1$ lapse/tilt gauge block (non-degenerate at the anchor, its $\ell = 1$ determinant the closed-form $D(x) \neq 0$ (58) and its $\ell = 0$ Jacobian $\partial_R[\sqrt{f}] \neq 0$) the principal angular symbol of the reduced first-junction map on the shape Y is that of the static infinitesimal-bending operator $\Delta_{S^2} + 2$ (App. D), i.e. $|\xi|_{S^2}^2$, noncharacteristic for $\xi \neq 0$. The retarded data enter only at subordinate angular order: on a cut of constant retarded time the dynamical components $h_{uu} = g_{\mu\nu} \partial_u X^\mu \partial_u X^\nu$ and $h_{uA} = g_{\mu\nu} \partial_u X^\mu \partial_A X^\nu$ carry the retarded derivative Y_u (and $\dot{m}, \dot{a}$) at angular order 0 and ≤ 1 respectively ($\partial_u X$ carries no angular derivative of Y and $\partial_A X$ at most one) while the order-two angular content $\partial_\Omega^2 Y$ (the Δ_{S^2} of the bending reduction) enters only the shape block $[h_{AB}]$, with coefficients that the time dependence of $(m(u), a(u))$ shifts but whose leading angular order it does not raise. Hence the order-two angular principal part is block-diagonal and identical to the static one; Y_u is a genuine transport term in u , not a principal-order coupling.*

The principal symbol alone is not a well-posedness statement; we close the estimate. First, the Schur elimination must survive the burn, not only the anchor.

Lemma 4 (Finite-rate Schur non-degeneracy). *On a constant-retarded cut the $\ell \leq 1$ lapse/tilt block of the dynamical junction is a smooth $\mathcal{O}(\varepsilon)$ perturbation of its static value, so $\det J_{\text{dyn}}(x, \varepsilon) = a D(x) [1 + \mathcal{O}(\varepsilon)]$. Since $|D(x)| \geq \frac{4}{3}$ on $(0, 24/25)$ (60), there is $\varepsilon_0(x_\star) > 0$ with $\det J_{\text{dyn}} \neq 0$ for $\varepsilon < \varepsilon_0$: the Schur elimination of Lemma 3 remains valid at finite burn rate, not only at the anchor.*

By Lemmas 3–4 the reduced constant- u junction is an over-determined elliptic system in the angles whose principal part is the static $\Delta_{S^2} + 2$, with the retarded Y_u of subordinate *differential* order and the $\ell \leq 1$ Schur block invertible throughout the burn. This reduces the persistence to the solvability of the higher-mode retarded time-rows (63) for Y_u : the elliptic gain compensates the single u -derivative (the no-derivative-loss budget, so no Nash–Moser iteration is needed), but the closure requires that those $\ell \geq 2$ rows determine Y_u , which we isolate as condition (63) and establish in App. E for a fixed-axis burn.

Proof of Proposition 6. The proof (the constant-cut elliptic reduction, the fixed-axis time-row solver, the $\mathcal{O}(a^2)$ compatibility computation, and the Grönwall closure) is given in Appendix E. $\square$

The single quantity the argument leaves implicit is the *explicit* size of the ball $(\lambda_\star, \varepsilon_\star)$ as the rear-pole horizon is approached, $\mu_\star = \inf(f - 2aR) \rightarrow 0$ (the saturating-amplitude reach), set by the round-sphere elliptic constants of (53) and mapped numerically in App. C.

9 Finite-thickness wall and back-reaction

9.1 Thickening with admissible matter

The thin shell is the $\ell \rightarrow 0$ limit of a finite-thickness wall. To avoid the smoothing pathology of Ref. [18], the wall must be a *physically modeled* matter layer rather than an arbitrary smoothstep. Two microscopic models satisfy the dominant energy condition by construction. Collisionless (Einstein–Vlasov) matter has stress-energy $T_{ab} = \int f p_a p_b (d^3p/p^0)$ that is DEC for $f \geq 0$ [60] and interpolates a Minkowski cavity to the radiation exterior without a vacuum gap; but a *purely tangential* ($p_r = 0$) Vlasov layer requires its constituents on circular timelike orbits, which exist only outside the photon sphere, $2m(r)/r < 2/3$, so the kinetic realization of the wall below is limited to that window. Anisotropic *elastic* matter [61] carries tangential stress without circular orbits and is the appropriate microscopic model at the higher compactness of the frontier (App. K). This choice is essential, not cosmetic: a *generic* smooth (radial-tension) thickening fails, and the failure is closed-form.

Proposition 7 (Radial-tension thickening violates the dominant energy condition). *A static Kerr–Schild wall $g_{\mu\nu} = \eta_{\mu\nu} + (2M(r)/r)\ell_\mu\ell_\nu$ with smoothstep mass profile $M(r)$ has, from the exact curvature, the anisotropic stress*

$$\rho = \frac{M'(r)}{4\pi r^2}, \quad p_r = -\rho, \quad p_t = -\frac{M''(r)}{8\pi r} \quad (31)$$

The radial block saturates the dominant energy condition ($p_r = -\rho$), while the tangential margin $\rho - |p_t| < 0$ violates it wherever

$$r |M''(r)| > 2 M'(r). \quad (32)$$

The violation is forced at the matter–vacuum edge, independent of thickness or compactness, and for a thin transition $\delta \ll r$ fills the whole wall. The strict obstruction is this static tangential channel; the u -dependent (generalized-Vaidya) graft only saturates it.

Proof. At the cavity edge $r = r_-$ the C^1 match to the flat ($M \equiv 0$) interior gives $M(r_-) = M'(r_-) = 0$, so $\rho(r_-) = 0$; the DEC there requires also $p_t(r_-) = 0$, i.e. $M''(r_-) = 0$. A generic (C^1 , not C^2) profile has $M''(r_-) \neq 0$ and violates the DEC *exactly* at the edge, $\rho - |p_t| = -|M''(r_-)|/8\pi r_- < 0$; a higher-smoothness match with $M''(r_-) = 0$ merely displaces the violation infinitesimally off-edge, where $\rho = M'/4\pi r^2$ is still vanishingly small while M'' has turned on. Either way the source–vacuum boundary cannot satisfy the DEC (the boundary cost, “worst at the cavity edge”), at every compactness; for a thin transition $\delta \ll r$ the violation (32) fills the whole wall ($r|M''|/2M' \sim r/2\delta \gg 1$). The u -dependent (generalized-Vaidya) version $M(u, r) = \bar{m}(u)\chi(r)$ adds the outgoing flux in a *single* slot of the $(\hat{u}, \hat{r})$ block ($T^{\hat{r}}_{\hat{u}} \neq 0$, $T^{\hat{u}}_{\hat{r}} = 0$), making it a *defective* Jordan block with a repeated real eigenvalue and invariant discriminant $\Delta = (\rho_h + p_r)^2 - 4q_h^2 = 0$ (the field equations force $\rho_h + p_r = 2|q_h|$ for a pure outgoing null flux): it is Hawking–Ellis *Type II*, DEC-saturating, not Type IV (a numerical Type-IV label is eigenvalue-splitting noise on the defective block, the Type I $\leftrightarrow$ II artifact of App. A). (This is distinct from the genuine, vorticity-sourced Type IV of a *velocity*-carrying smooth warp shell [18], whose tilted flow circulates energy; the radial-tension wall here is twist-free and merely defective.) $\square$

Corollary 5 (The boundary cost is the $1/\delta^2$ edge law). *For a smoothstep profile $M(r) = m S((r - r_-)/\delta)$ of width δ matched C^1 to the flat cavity ($S(0) = S'(0) = 0$, $S''(0) \neq 0$), the worst-observer surface-DEC margin at the inner edge is, from (31),*

$$(\rho - |p_t|)|_{r \rightarrow r_-^+} = -\frac{m |S''(0)|}{8\pi r_- \delta^2} < 0, \quad (33)$$

diverging as $\delta \rightarrow 0$ at every compactness. The numerically observed boundary cost (Fig. 4, Ref. [18]) is thus the exact $1/\delta^2$ law of the radial-tension edge, an analytic obstruction rather than a scan artifact (App. K).

Proposition 7 with Corollary 5 is the analytic form of the boundary cost of Ref. [18]: the obstruction is the radial tension $p_r = -\rho$ forced by the Kerr–Schild profile, not thickening per se. A *tangential-pressure* wall (radial pressure $p_r \rightarrow 0$ at both edges, the load carried by the tangential pressure fixed by anisotropic hydrostatic equilibrium, $p_t = \frac{r}{2}\rho \Phi'$ with $\Phi' = m/[r(r - 2m)]$) has the worst-observer pointwise margin $\rho - p_t = \rho [1 - m/(2(r - 2m))]$, which is positive (Hawking–Ellis Type I, strict DEC) *across the whole wall* iff the local compactness satisfies $2m(r)/r < 4/5$. Such a wall is source-consistent ($G^a_b/8\pi$ reproduces $(\rho, 0, p_t)$), is strictly DEC-admissible across the wall in the *static* limit for local compactness $2m(r)/r < 4/5$, and recovers the thin-shell anchor (σ_0, p_0) as $\ell \rightarrow 0$. The three admissibility windows nest by matter model, $2/3$ (collisionless, photon-sphere-limited) $< 4/5$ (anisotropic fluid/elastic) $< 24/25$ (distributional thin shell): the limit widens as the model supports the tangential stress under a progressively weaker kinematic constraint, each carrying the wall to higher compactness. The $4/5$ result is *static and local*, whereas an accelerating shell carries an effective rear-pole redshift $x_{\text{eff}} = x + 2ar$ (Sec. 8, Eq. (47)). The static thick wall therefore covers the accelerating frontier only in the lower-compactness window where $x + 2aR < 4/5$. At higher compactness the accelerated rear pole exceeds $4/5$, and the admissible object is the thin shell, whose wider window $24/25$ the accelerated rear pole still respects; an accelerated thick-wall DEC theorem is left to future work. Within that window the distributional shell is the $\ell \rightarrow 0$ limit of admissible matter precisely where the construction operates.

9.2 The radiative wall

We now graft the photon-rocket radiation *through* the wall, making it itself radiative; strict admissibility survives. Model the radiating wall as the two-component stress

$$T_{ab} = T_{ab}^{\text{wall}} + \mu \ell_a \ell_b, \quad T_{ab}^{\text{wall}} = \rho u_a u_b + p_t \gamma_{ab}, \quad p_r = 0, \quad (34)$$

the $p_r = 0$ tangential-pressure wall above plus a *separately conserved* outgoing null-dust flux $\mu \ell_a \ell_b$ ($\mu \geq 0$) streaming through it: a generalized-Vaidya interior $m(u, r) = \bar{m}(u) \chi(r)$ with $\chi(0) = 0$, so the radiation fills the wall with no beam through the center and no matter $\rightarrow$ vacuum edge. For the idealized fixed-background split (34) at constant μ (with $p_r = 0$), the radial Hawking–Ellis block has discriminant

$$\Delta = (\rho + p_r)(\rho + p_r + 4\mu) = \rho(\rho + 4\mu) > 0, \quad (35)$$

so it stays Hawking–Ellis Type I: superposing the radial null dust does *not* analytically re-trigger the Type IV (rotating-energy) failure a heat-flux model would. For the conserved generalized-Vaidya realization $m(u, r) = \bar{m}(u) \chi(r)$ the exact curvature carries a genuine outgoing-null amplitude $n^2 > 0$ streaming through the wall, sourced by $\dot{\bar{m}}$, vanishing at the center, and conserved ($\nabla_a T_{ab} = 0$). That genuine ramp, however, carries $p_r \neq 0$, whose *static* radial-tension channel already violates the DEC at the matter–vacuum edge (Prop. 7); its $(\dot{u}, \hat{r})$ block is the defective Hawking–Ellis Type II (DEC-saturating, $\Delta = 0$) of that proposition, *not* Type IV (App. K), so it serves only to exhibit the through-wall flux, not as the admissible object. Strict DEC admissibility (Type I) across the whole material wall is instead the property of the idealized $p_r = 0$ split (34) above (discriminant (35)) and, with the metric allowed to respond, of the *separately* constructed adiabatic $p_r = 0$ Eddington–Finkelstein wall of App. F; the pure-null exterior *saturates* DEC (Type II, as outgoing null dust must).

We close the back-reaction at the adiabatic level (App. F): realizing the radiating wall as the exact $p_r = 0$ tangential wall in outgoing Eddington–Finkelstein form with a slowly radiating mass $m(u)$ gives a self-consistent Einstein solution ($\nabla_a T_{ab} = 0$), Type I, whose *curvature-derived* worst-observer DEC margin stays strictly positive across the whole frontier regime over the directly tested radiation amplitudes (App. F). With the metric allowed to respond, the outgoing energy *flux* (the quantity the DEC actually constrains) *decreases* the margin through the dense wall core,

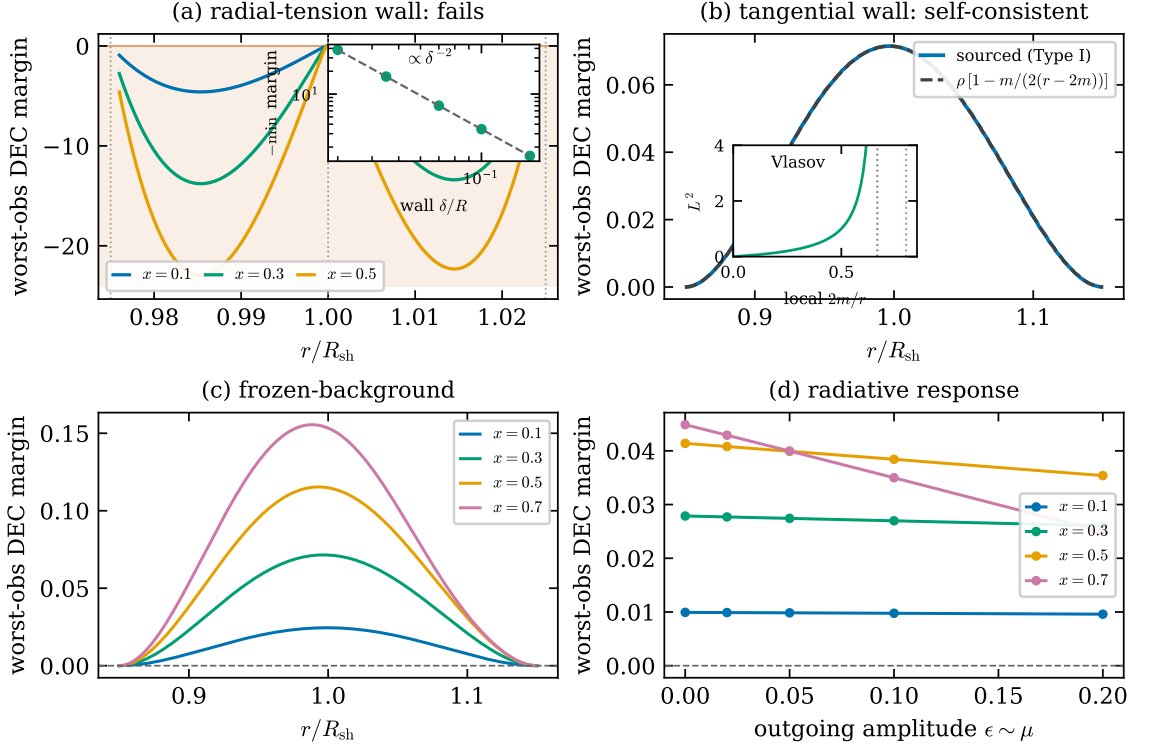


Figure 4. The wall dominant-energy story: the naive thickening fails, the engineered one holds. (a) *fails*: a generic smooth *radial-tension* thickening violates the pointwise DEC: the worst-observer margin across a smoothstep Kerr–Schild wall goes negative (shaded), worst at the cavity→matter inner edge; the inset shows the violation is a boundary cost that diverges as the wall sharpens, $\propto \delta^{-2}$ (Eq. (33)). The engineered *tangential-pressure* wall instead holds three ways. (b) *static* DEC: the metric-sourced margin (Type I) *coincides* with the closed-form single-fluid margin $\rho[1 - m/(2(r - 2m))]$: a genuine equilibrium, not a postulated stress (inset: local compactness against the photon-sphere bound $2/3$ and the elastic bound $4/5$). (c) *frozen positive*: the across-wall margin is strictly positive and grows with x . (d) *back-reaction tightens*: with the metric free to respond, the curvature-derived margin versus the outgoing amplitude $\epsilon \sim \mu$ stays positive but the flux *tightens* the condition (slope < 0), reversing the frozen-background sign while preserving admissibility.

$d(\text{margin})/d\mu = 1 - \frac{r}{2}\rho\partial\Phi'/\partial\mu < 0$ (the analytic sign of Prop. 10, neutralizing only in the $\rho \rightarrow 0$ edge skin), rather than raising it, while leaving it strictly positive; the radiating wall therefore remains admissible (Fig. 4d). A frozen-background calculation gives the opposite, looser sign; that lemma and the controls isolating it are collected in App. F. The residual is the time-dependent $\mathcal{O}(\mu^2)$ term.

The finite-thickness wall stays compatible with the accelerating frontier where $x + 2aR < 4/5$, the thin shell remaining admissible above (Fig. 4). The energy-condition obstruction of Ref. [18] is thereby removed: the radiative wall is strictly admissible, and the generalized-Vaidya geometry exhibits an explicit metric carrying the required through-wall flux. A self-consistent matter realization (DEC by construction) is available: collisionless Einstein–Vlasov matter below the photon sphere ($2m/r < 2/3$) and anisotropic elastic matter [61] at the higher compactness of the frontier; a source-consistent anisotropic equilibrium of the tangential-pressure wall, and the kinetic/elastic boundary, are given in App. K.

9.3 The finite maneuver is thick-wall realizable

The finite-duration maneuver of Theorem 3, established above for the distributional thin shell, is, at the per-instant adiabatic level, the $\ell \rightarrow 0$ limit of an admissible *radiating tangential-pressure thick wall* throughout the burn, *provided* the rear-pole effective compactness stays in the thick-wall window, $x_{\text{eff}}(u) = x(u) + 2\lambda(u) < 4/5$ for all u . This is not implied by the thin-shell frontier $\lambda < g(x)$ at high compactness, but it holds for the worked saturating burn of App. K (whose operating point keeps the rear-pole $x_{\text{eff}} < 4/5$ throughout; the rocket radiates, $x(u)$ falling while $\lambda(u)$ rises). At every retarded step the static tangential-pressure wall at $x_{\text{eff}}(u)$ is then strictly DEC-admissible across its width (Fig. 3c), and the radiating version is curvature-derived Hawking–Ellis Type I with worst-observer DEC margin > 0 and $\nabla_a T_{ab} = 0$ (App. K). The through-wall radiation is modest (well inside the back-reaction window of Sec. 9.2), so the

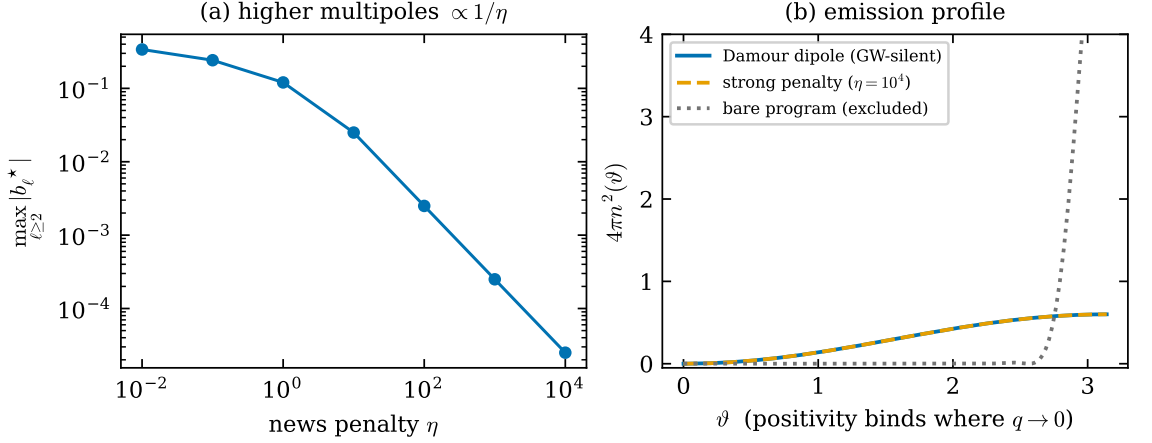


Figure 5. Minimum-radiation steering (Theorem 4). (a) Under a convex surrogate news penalty (App. G; the unconditional dipole optimality is Theorem 4 in the news-silent class, independent of any penalty) the higher multipoles collapse, $\max_{\ell \geq 2} |b_\ell^*| \propto 1/\eta$. (b) The emission profile $4\pi n^2(\vartheta)$: the gravitational-wave-silent Damour dipole (positivity binding where it vanishes at the forward pole), the strong-penalty optimizer converging onto it, and the excluded bare program (rear-pole spike), a lower-fuel singular profile lying outside the silent class.

saturating (minimum-fuel) burn, not merely a gentle one, is thick-wall realizable at the per-instant adiabatic level. The fully anisotropic accelerated thick wall, a dipole-deformed metric carrying both the steering and the through-wall flux, remains future work.

10 Gravitational-wave silence

Whereas the preceding section concerned the wall's *local* matter realization, the gravitational radiation reaching null infinity is fixed by the exterior class alone, and it vanishes. The exterior is left *exactly* Kinnersley by the excision; the Bondi news $\mathcal{N}(u, \Omega)$ of the matched spacetime equals that of the bare photon rocket. Bonnor [23] established that the Kinnersley photon rocket is exactly non-radiative (vanishing Bondi news), and Damour [53] reconciled this with linearized wave-generation theory: for the purely dipolar emission the leading $1/r$ waves from the accelerated mass cancel exactly against those of the momentum-carrying radiation field, so no gravitational radiation reaches null infinity. Hence dipole steering is *gravitational-wave silent*, and the control law (13) suffers no uncontrolled radiative correction. Higher-multipole steering reintroduces a news contribution and an additional energy-loss channel; the minimum-radiation maneuver is therefore the Damour dipole, which we now make precise.

11 Minimum-radiation maneuvers as optimal control

The construction turns warp-drive design into a control problem, and makes the *saturation* clause of Theorem 1 precise: among news-silent maneuvers realizing a prescribed proper-acceleration history $a(u)$ (which, with the budget below, fixes ΔP_B^i), which radiates the least? (Optimizing the acceleration history itself at fixed ΔP_B^i is the free-profile problem of the scope note below.) The *state* is the Bondi four-momentum $P_B^\mu(u)$ (equivalently the worldline kinematics) together with the shell compactness $x(u)$; the *control* is the angular radiation profile, i.e. the mass-loss rate $\dot{m}$ and the multipoles of $(\ln P)_{,u}$; the *dynamics* are Eq. (11); and the *constraints* are the bulk positivity $n^2(u, \Omega) \geq 0$ and the surface admissibility $\Phi(\lambda, \mu) \geq 0$ of Theorem 2 and frontier (28).

A natural objective is the total radiated energy $J = m(u_0) - m(u_f) = \int_{u_0}^{u_f} (-\dot{m}) du$, minimized, at a prescribed proper-acceleration history $a(u)$, over the angular control profiles that realize it (optimizing $a(u)$ itself at a fixed change of Bondi momentum ΔP_B^i is the free-profile problem of the scope note below). Expanding the angular emission in Legendre modes, the conserved momentum couples *only* to the $\ell = 1$ (dipole) mode by S^2 orthogonality, whereas every mode feeds the fuel monopole $-\dot{m}$.

Theorem 4 (Minimum-radiation steering). *Restrict to gravitational-wave-silent maneuvers (vanishing Bondi news, $\mathcal{N} \equiv 0$) within the asymptotically flat Robinson–Trautman pure-radiation class of Sec. 4. These are exactly the dipolar emissions (12): vanishing news forces the conformal factor (5) to $\ell \leq 1$, $P = b(u) + c(u) \cos \vartheta$ in the momentary rest frame, hence the emission (7) to a pure monopole+dipole, for which bulk positivity $n^2 \geq 0$ is equivalent to the control law (13). Among all such news-silent maneuvers realizing a prescribed proper-acceleration history $a(u)$ over $[u_0, u_f]$*

(which, with the budget below, fixes the integrated thrust $\int m a \, du$, hence ΔP_B^i), the radiated energy J is minimized by saturating the control law pointwise, $-\dot{m} = 3m|\alpha|$ (any isotropic mass loss above this minimum raises J without adding thrust), with minimum $J = m_0(1 - e^{-3 \int |\alpha| \, du})$ [the Tsiolkovsky budget (14)]; optimizing the acceleration history itself over all profiles achieving a given ΔP_B^i is the free-profile problem (Theorem 5).

Proof sketch. That the news-silent class is exactly dipolar is classical: the purely dipolar photon rocket has vanishing news (Bonnor [23], with Damour’s [53] linearized reconciliation), and conversely von der Gönna and Kramer [62] proved the Kinnersley rocket is the *unique* no-news solution of the regular *axisymmetric* Robinson–Trautman pure-radiation class, with Cornish [63] removing the axisymmetry restriction (a Robinson–Trautman pure-radiation metric is news-free iff it is a Kinnersley photon rocket for a point mass accelerating *arbitrarily*), so the dipolar restriction holds for a three-dimensional steering history, not only the axisymmetric one. On the dipole class the controls are $(\dot{m}, \alpha)$ and positivity $n^2 \geq 0$ is the affine-in-cos ϑ inequality (13), binding at the forward pole. The control Hamiltonian $\mathcal{H} = (-\dot{m}) + p_\mu \dot{P}_B^\mu - \oint \chi(\Omega) n^2 \, d\Omega$, with Bondi-flux costates p_μ [Eq. (11)] and a Karush–Kuhn–Tucker multiplier field $\chi \geq 0$ enforcing positivity, gives by the Pontryagin maximum principle [64, 65] a constraint-saturating (single active-constraint) optimum: any slack $-\dot{m} > 3m|\alpha|$ could be reduced at fixed $a(u)$, lowering J , so the optimum saturates $n^2 = 0$ at the forward pole, $-\dot{m} = 3m|\alpha|$. Integrating $d \ln m / du = -3|\alpha|$ gives the stated J . The full costate derivation is in App. G. $\square$

Higher multipoles are excluded on two independent grounds (App. G): an $\ell \geq 2$ emission (i) carries energy to infinity without contributing to the $\ell = 1$ momentum (S^2 orthogonality, “energy without momentum”), and (ii) switches on the Bondi news, radiating gravitational waves (Damour). The first effect is the momentum bookkeeping: by S^2 orthogonality only the $\ell = 1$ mode carries net momentum, so a higher multipole radiates energy without adding any net $\ell = 1$ momentum moment of its own. Within the news-silent class the dipole, the lowest momentum-carrying multipole, is therefore the minimum-radiation steering. The optimizer exposes one subtlety: the *bare* program “minimize fuel at fixed thrust subject only to $n^2 \geq 0$ ” is formally optimized not by the dipole but by a singular emission spiked at the *rear* pole; that profile is excluded because it is not a regular pure-radiation exterior and, carrying all multipoles, radiates gravitational waves. The decisive physical input that excludes it is the von der Gönna–Kramer converse: a news-free emission must have $\ell \leq 1$, so the singular spike is simply outside the silent class. The trade-off in the broader (gravitational-wave-allowed) class is governed by the *exact* news energy, not an arbitrary penalty. The radiated gravitational energy is $J_{\text{news}} = \frac{1}{4\pi} \oint |\mathcal{N}|^2 \, d\Omega \geq 0$, and for the Robinson–Trautman class the news $\mathcal{N}$ is, to leading order in the $\ell \geq 2$ amplitudes, a *linear* functional of the $\ell \geq 2$ part of $(\ln P)_{,u}$ (Damour [53]), so J_{news} is a *positive-definite quadratic form*

$$J_{\text{news}} = \sum_{\ell \geq 2} W_\ell |c_\ell|^2 + \mathcal{O}(c^3), \quad W_\ell > 0, \quad (36)$$

in the emission multipoles c_ℓ , vanishing *exactly* (to all orders) on $\ell \leq 1$: the von der Gönna–Kramer characterization [62] that the Kinnersley dipole is the *unique* news-free regular axisymmetric Robinson–Trautman pure-radiation solution. The strict positivity $W_\ell > 0$ is not an assumption but a consequence of news-silence itself.

Lemma 5 (Strict positivity of the news weights). *To leading order the Bondi news is the rotationally covariant linear map $\mathcal{N} = Lc$ of the $\ell \geq 2$ emission multipoles $c = \{c_{\ell m}\}$, with L carrying the spin-2 raising $\eth^2$ of the Robinson–Trautman news in its angular part (Damour [53]). Rotational invariance makes L mode-diagonal, acting on each ℓ -component by the $\eth^2$ weight $\sqrt{(\ell-1)\ell(\ell+1)(\ell+2)}$ (a singular value, since $\eth^2$ raises spin 0 to spin 2), which is nonzero for every $\ell \geq 2$ and vanishes only for $\ell \leq 1$; hence L has trivial kernel on $\ell \geq 2$ (the linearized counterpart of the von der Gönna–Kramer [62]–Cornish [63] characterization $\mathcal{N} \equiv 0 \Leftrightarrow \ell \leq 1$). Therefore $J_{\text{news}} = \frac{1}{4\pi} \oint |\mathcal{N}|^2 \, d\Omega = \langle c, L^\dagger L c \rangle$ is positive-definite on the $\ell \geq 2$ amplitudes, with the single positive eigenvalue $W_\ell = (\ell-1)\ell(\ell+1)(\ell+2) w_\ell > 0$ per multipole (w_ℓ the squared radial Damour weight), giving (36).*

These three properties (non-negativity, convexity, and vanishing precisely on the dipole class (Lemma 5)) are exactly the ones the optimizer uses, so they need not be postulated as a surrogate: J_{news} is strictly positive on any $\ell \geq 2$ admixture, which is why the all-multipole singular spike is forbidden once the gravitational channel is charged at its physical weight. An $\ell \geq 2$ mode radiates this J_{news} *quadratically* in its amplitude while saving fuel only *linearly*: a trade-off whose balance,

for the full (news-charged) objective $J_{\text{fuel}} + J_{\text{news}}$ at fixed thrust, is set by the exact weights W_ℓ against the marginal fuel saving; for a sufficiently weighted gravitational channel the optimum is the dipole (the higher multipoles collapsing as the gravitational channel is weighted, App. G, Fig. 5). This trade-off is qualitative; the *unconditional* statement is Theorem 4 (the gravitational-wave-silent class), where $J_{\text{news}} = 0$ forces $\ell \leq 1$ and the dipole is the only momentum-carrying mode. Outside that class the fuel bound relaxes: a more collimated exhaust lowers the Tsiolkovsky factor toward the pure-rocket value 1, but only by charging the gravitational channel ($J_{\text{news}} > 0$), so the universal 3 is the price of gravitational-wave silence, not a general-relativistic floor.

Theorem 4 fixes a *given* proper-acceleration history $a(u)$ and optimizes only the mass-loss; the *bang-bang* is in that one control. The complementary *free-profile* problem (the least radiation to achieve a prescribed change of velocity) is solved geometrically, and turns warp-maneuver design into geodesy on the velocity space.

11.1 Minimum-radiation steering as a hyperbolic geodesic

The kinematic state of the news-silent rocket is (ρ, v) with $\rho = \log m$ and v^μ the unit worldline four-velocity on the *velocity hyperboloid* $\mathbb{H}^3 = \{v : v \cdot v = -1, v^0 > 0\} \cong SO^+(3, 1)/SO(3)$, so $P_B^\mu = e^\rho v^\mu$. Along any news-silent maneuver the proper acceleration magnitude is the hyperbolic speed, $|a| = \sqrt{g_{\mathbb{H}^3}(\dot{v}, \dot{v})}$, and the saturated control law (13) gives $\dot{\rho} = -3|a|$, so

$$\frac{m_f}{m_0} = e^{-3L_{\mathbb{H}^3}[v]}, \quad L_{\mathbb{H}^3}[v] = \int |a| d\tau = \text{hyperbolic length of } v(\tau). \quad (37)$$

The natural *additive* cost is the log-fuel $C = \log(m_0/m_f) = 3L_{\mathbb{H}^3}$; the radiated energy $J = m_0(1 - e^{-3L_{\mathbb{H}^3}})$ is a monotone transform of it (App. H).

Theorem 5 (Free-profile minimum-radiation maneuver). *Among news-silent maneuvers steering the drive between stationary Bondi frames with four-velocities v_0, v_1 , the minimum log-fuel (equivalently the minimum radiated mass) is*

$$C_{\min} = 3 d_{\mathbb{H}^3}(v_0, v_1) = 3 \operatorname{arccosh}(-v_0 \cdot v_1), \quad \frac{m_f}{m_0} = e^{-3 d_{\mathbb{H}^3}(v_0, v_1)} \quad (38)$$

attained by the hyperbolic geodesic of $\mathbb{H}^3$ (the single one-parameter boost from v_0 to v_1) with the saturated Damour dipole realizing $|a| = |\dot{v}|_{\mathbb{H}^3}$ at each retarded cut. Any non-geodesic velocity path costs strictly more, the boost geodesic being the unique minimizer between fixed endpoints on the Hadamard manifold $\mathbb{H}^3$ (Cartan–Hadamard): a cornered path loses by the triangle inequality, a smooth detour by the uniqueness of the minimizing geodesic.

This is a left-invariant control problem on the Lorentz group $SO^+(3, 1)$, equivalently Riemannian geodesy on the quotient $\mathbb{H}^3 = SO^+(3, 1)/SO(3)$: the cost is the factor-3 hyperbolic length, whose geodesics are the one-parameter boost subgroups, the minimizer between fixed endpoints being unique because $\mathbb{H}^3$ is a complete simply connected manifold of negative curvature (Hadamard/Hopf–Rinow). The Pontryagin maximum principle on the group reproduces (38) by Lie–Poisson reduction [66, 67] (App. H). The content is *not* the optimal-control machinery, which is classical, but that bulk positivity together with news-silence (Damour [53], von der Gönna–Kramer [62]) collapse the warp-steering cost to this single hyperbolic norm with the *universal* constant 3, turning maneuver design into geodesy on the velocity space. The rectilinear Tsiolkovsky budget (14) is the special case $d_{\mathbb{H}^3} = \Delta\eta$ (a boost geodesic through the origin), and “a turn costs strictly more than the collinear figure” (Sec. 15) is the geodesic triangle inequality on $\mathbb{H}^3$, made exact.

The radiation cone and the factor 3. The result rests on the geometry of the radiated-momentum cone. The flux moment $Q^\mu = \oint n^2(\Omega) \hat{\ell}^\mu d\Omega$ obeys $dP_B^\mu/du = -Q^\mu$ (11). For an *arbitrary* positive emission $n^2 \geq 0$, Q^μ ranges over the whole future causal cone, with $Q^0 = -\dot{m} \geq |\vec{Q}| = \text{thrust}$ (equality approached only in the collimated, rear delta-emission limit): the ideal photon-rocket bound $-\dot{m} \geq m|a|$. Restricting to the news-silent $\ell \leq 1$ class $4\pi n^2 = b_0 + \vec{b} \cdot \hat{\Omega}$, positivity is $b_0 \geq |\vec{b}|$, while the moments are $Q^0 = b_0 = -\dot{m}$ and $\vec{Q} = \vec{b}/3$ with $|\vec{Q}| = |\vec{b}|/3 = m|a|$ (the $\ell = 1$ projection $\oint \hat{\Omega}^i \hat{\Omega}^j d\Omega = \frac{4\pi}{3} \delta^{ij}$; $\vec{Q}$ is the radiated momentum, the thrust being the recoil $-\vec{Q} = m\vec{a}$); positivity then reads $-\dot{m} \geq 3m|a|$, the control law (13). The factor 3 is exactly this $\frac{1}{3}$ dipole projection forced by the broad news-silent emission (App. H), the price of gravitational-wave silence rather than a fundamental floor.

Reachability and the moving frontier. Because the boost geodesics reach every point of $\mathbb{H}^3$, every velocity is reachable with positive remaining mass $m_f = m_0 e^{-3\eta} > 0$ (the light speed $\eta \rightarrow \infty$ needs $m_f \rightarrow 0$ and is not reached): the drive is relativistically controllable in velocity. The reachable set in Bondi *four-momentum* is by contrast constrained (the radiated $P_B(u_0) - P_B(u_f)$ must be future causal) so not every future-timelike P_B^f is reachable. The *pure* minimum-radiation cost $3L_{\mathbb{H}^3}$ is geometric (independent of how fast the path is traversed), so its optimum is the *unconstrained* geodesic: one may always burn arbitrarily slowly, and the acceleration frontier $a_{\max} R \leq g(x)$ (28) never binds it. The frontier becomes active once the maneuver is also *time-constrained* (a bounded-duration or minimum-time burn, equivalently a bound on proper acceleration): then $|\dot{v}|_{\mathbb{H}^3} R \leq g(x(u))$ caps the hyperbolic speed, and because the rocket radiates, $x(u) = 2m(u)/R$ decreases so the ceiling $\frac{1}{2}(1 - x(u))$ widens along the burn (the realized $g(x(u))$ following numerically). The minimum-time maneuver is then a *state-constrained* geodesic with constraint-active (maximal-acceleration) arcs $|\dot{v}|_{\mathbb{H}^3} R = g(x(u))$, made precise in Theorem 7 (App. H): the bounded-control trajectory-optimization problem warp design reduces to, with explicit interior/boundary switching and transversality. The structural backbone is the *length reduction* $x(u) = x_0 e^{-3\ell}$, fuel = 3ℓ in hyperbolic arclength ℓ (both bare objectives factor through ℓ , hence single-arc); genuine multi-arc switching requires an objective that breaks this reduction (a dwell/effort penalty), for which the frontier binds early (large x , small g) and releases as the rocket radiates (x falls, g rises).

Steering and swimming as moment map and gauge residual. In this language the no-reactionless-steering law (Sec. 3) is a *moment-map* statement on the Ashtekar–Streubel radiative phase space at $\mathcal{I}^+$ [36]: the Hamiltonian generating the asymptotic-translation (T^4) flow on the radiative data is the *total radiated four-momentum flux* $\int F^\mu du$, while the Bondi four-momentum $P_B^\mu(u)$ is the Dray–Streubel/Wald–Zoupas *cut charge* [31, 32], the two linked by the balance law $dP_B^\mu/du = -F^\mu$ with F^μ future-causal under the DEC. (We reserve “moment map” for the radiated-flux Hamiltonian on the radiative phase space; P_B^μ , being Coulombic, is the cut charge it generates, not itself a functional of the radiative shear/news.) Steering *moves the charge* and must radiate (the no-go). The complementary swimming effect (Prop. 1), a net position change at fixed P_B^μ , is, *by structural analogy* (not a theorem we prove), a holonomy in the position/shape sector of the Shapere–Wilczek/Montgomery type [68, 69], once a Bondi rest frame and a center-of-mass section are fixed; in general relativity that residual is contaminated by supertranslation and memory ambiguities, and (unlike a test body in a fixed curved background) a cyclic deformation of a self-gravitating body generically radiates, so the decomposition is a *balance law plus a frame-dependent residual*, not an exact principal bundle.

Background drift and Finsler geometry. In flat space (and in any static, drift-free background) the cost is the isotropic hyperbolic norm above: Riemannian, not Finsler. A *conformally stationary* exterior (a timelike Killing or homothetic field, as in the de Sitter static patch of Sec. 12, with the drift below the Finsler-breakdown bound $|W| < 1$) introduces a time-independent drift on the velocity space, and the minimum-*time* problem becomes a *Zermelo navigation* problem whose solution is a Randers–Finsler geodesic, the optical-metric correspondence of Gibbons *et al.* [70, 71]: the velocity-space metric acquires a one-form (the drift) and the steering geodesics deform accordingly. (The minimum-*radiation*/log-fuel cost is pure hyperbolic arclength and so is *not* generally Randers under drift (coasting on the drift can connect points at zero fuel) reducing to Randers only in the fixed-speed, fuel- $\propto$ -time case.) This is the natural setting in which the present Riemannian reading generalizes; a general time-dependent drift falls outside ordinary Randers geodesy and is left open.

This is the bridge from general relativity to trajectory optimization: optimal warp maneuvers are the (state-constrained) geodesics of a hyperbolic norm on the velocity space, the energy-budget functional being its factor-3 length.

12 Extension to a positive cosmological constant

Our universe has $\Lambda > 0$, so a physically complete construction should survive in an asymptotically de Sitter background. The *construction* does, almost verbatim, because the photon rocket itself does; the conservation-law *principle*, however, weakens to a supported flux balance whose positive-energy refinement is open (below).

The (anti-)de Sitter photon-rocket exterior. Podolský [72] exhibited the exact Kinnersley rocket in the (anti-)de Sitter universe: the line element (3) is unchanged except that the metric

function acquires the cosmological term,

$$2H = K - 2r(\ln P)_{,u} - \frac{2m(u)}{r} - \frac{\Lambda}{3}r^2, \quad (39)$$

and (this is the decisive point) the source remains *pure outgoing null dust* $T_{ab} = (n^2/r^2)\ell_a\ell_b$ with the *same* amplitude $4\pi n^2 = -\dot{m} - 3m|a|\cos\vartheta$ (12): the cosmological constant enters only the vacuum (cosmological) part of the field equations and does not source the radiation. Three consequences are immediate. (i) The bulk energy conditions of the *matter* still collapse to the single positivity $n^2 \geq 0$ (Prop. 3); $\Lambda > 0$ contributes the usual de Sitter $-\Lambda g_{ab}$ term, which violates only the strong energy condition (as every de Sitter background does) and carries no exotic matter. (ii) The steering control law is *unchanged*, $-\dot{m} \geq 3m|a|$ (13), with the same Tsolkovsky budget (14); the recoil mechanism is identical. (iii) The static limit is the Schwarzschild–de Sitter (Kottler) metric, $f = 1 - 2m/r - \frac{\Lambda}{3}r^2$.

The de Sitter anchor and its surface dominant-energy window. Match a de Sitter static-patch cavity (lapse $f_- = 1 - \frac{\Lambda}{3}r^2$, mass-free) to the Kottler exterior across a static shell at areal radius R , with $x = 2m/R$ and $y = \frac{\Lambda}{3}R^2$. The Lanczos surface stress is closed-form and the worst-observer surface dominant-energy margin is

$$8\pi R(\sigma_0 - p_0) = \frac{\sqrt{1-y}(5x+8y-6) + 2(3-4y)\sqrt{1-x-y}}{2\sqrt{1-y}\sqrt{1-x-y}}, \quad (40)$$

valid on the static-patch domain $0 \leq y < 1$, $x+y < 1$ (both lapses real). For a Type-I surface stress the observer-robust margin is $\min(\sigma_0 - p_0, \sigma_0 + p_0) = \sigma_0 - |p_0|$; Eq. (40) is the $\sigma_0 - p_0$ branch, which binds wherever $p_0 \geq 0$, in particular on the small- y window near x_{crit} used below (the $\sigma_0 + p_0$ branch turns negative only at large y , near the $x+y \rightarrow 1$ edge). It reduces *exactly* to the flat-space window $-\frac{(5s-1)(s-1)}{2s}$, $s = \sqrt{1-x}$ (Eq. (17), surface DEC iff $x < 24/25$) as $y \rightarrow 0$, and shifts the admissible compactness threshold by an explicit amount: implicit differentiation of the numerator N of (40) at $(x, y) = (24/25, 0)$ gives $x_{\text{crit}}(y) = \frac{24}{25} - \frac{4}{5}y + \mathcal{O}(y^2)$, so a positive cosmological constant tightens the window linearly in $y = \frac{\Lambda}{3}R^2$. The flat passenger cavity of the asymptotically flat construction is replaced by a tidally regular de Sitter cavity (the static patch is conformally flat, with curvature set by Λ alone), so the warp feature (an acceleration-protected interior) persists.

The no-go at de Sitter infinity. The conservation-law principle (Sec. 3) requires more care, because in an asymptotically de Sitter spacetime future infinity $\mathcal{I}^+$ is *spacelike* and there is no global timelike Killing field, so the Bondi–Sachs four-momentum is replaced by the asymptotic de Sitter charges of Ashtekar, Bonga, and Kesavan [73]. In that framework the radiative degrees of freedom and their fluxes at $\mathcal{I}^+$ are well defined, and the steering statement becomes a flux balance of the same shape as (1): a confined, positive-energy drive changes its asymptotic de Sitter momentum charge only through the matter (n^2) and gravitational fluxes reaching $\mathcal{I}^+$. Two caveats must be stated. First, the de Sitter charges do not enjoy the unambiguous conservation of the Bondi charges (the absence of a timelike Killing field), so the no-go holds at the level of a flux balance rather than an exact conservation law, and the positive-energy bound replacing Bondi-mass positivity (which closed the negative-mass loophole in the flat case) is *research-level*, not established here. The obstruction is sharp: the static-patch Killing field T^a is timelike only *inside* the cosmological horizon and *null on it*, so the Nester–Witten/Brown–York boundary term degenerates there and cannot be signed by the bulk DEC alone; correspondingly the Kastor–Traschen cosmological-Witten construction [74] finds the *static* time-translation charge *indefinite* (positivity holds only for the conformal-time charge), and DEC-respecting negative-mass Schwarzschild–de Sitter bubbles are known, so DEC together with asymptotic de Sitter structure does not by itself yield a positivity bound. We therefore state the de Sitter steering principle as a supported flux balance rather than a theorem. Second, the cosmological horizon at proper distance $\sim \Lambda^{-1/2}$ bounds the maneuver slab, exactly as the Rindler horizon does in the flat case. Within these limits the explicit construction ((A)dS Kinnersley exterior, $n^2 \geq 0$ matter, control law $-\dot{m} \geq 3m|a|$, and the Kottler anchor with the surface-DEC window (40)) is concrete and positive-energy. For an anchor in the positive-margin domain ($x+y < 1$ with (40) > 0 and $\sigma_0 + p_0 > 0$, automatic since $\sigma_0, p_0 > 0$ on the small- y window) the openness argument of Theorem 2 carries over verbatim (the linearized junction is again the round-sphere rigidity operator, now on the Kottler/de Sitter background), so by continuity admissible accelerating warpphells exist for sufficiently small a with $\Lambda > 0$ as well, the rear-pole ceiling becoming

$\lambda < \frac{1}{2}(1 - x - y)$ on the same Kottler static-patch hypotheses. The energy-budget reading of warp propulsion is therefore not an artifact of asymptotic flatness.

13 Observational signature

The no-reactionless-steering law makes a radiative exhaust mandatory, so a steering maneuver is in principle observable. For a finite burn of Theorem 3 the bolometric luminosity $L(u) = -\dot{m}$ peaks at a sizable fraction of the rest-mass energy per light-crossing time (Fig. 3d; angular profile, Fig. 5). The emission is the photon-rocket dipole $4\pi n^2 = -\dot{m} - 3ma \cos \vartheta$; at minimum-radiation saturation it collapses to a single rear (exhaust) lobe of dipole-to-monopole amplitude unity whose intensity vanishes at the forward pole, so the angular pattern is itself a sharp signature of a saturating drive. The construction fixes this energy-momentum budget and angular pattern but not the exhaust microphysics, so no specific photon spectrum is claimed; the qualitative signature (a transient, beamed, strictly outgoing positive-energy flux) connects to proposals to detect emissions from positive-energy warp bubbles [75] and complements the gravitational-wave burst computed for warp-drive containment failure [76].

The same budget fixes the *cost*: the saturated law integrates to the Tsiolkovsky relation $m_0/m_f = e^{3\phi}$ (Fig. 7, App. K).

14 Comparison with the literature

Having established the construction, its energy budget, its $\Lambda > 0$ extension, and its observational signature, we now place it against the prior no-go and positive-energy literature.

14.1 Invariant warp charges and the classification of the news-silent shell sector

Within its natural family the construction is best placed not by mechanism but by diffeomorphism- and BMS-invariant charges, which we now make precise and use to *classify* it. To a spacetime $(\mathcal{M}, g)$ we attach four bits.

Definition 2 (Warp charges). *(C1) Interior tidal-flatness.* $C_1 = 1$ iff $\mathcal{M}$ contains a nonempty open region $\mathcal{M}_-$ on which $R_{abcd} = 0$ and $T_{ab} = 0$ (a regular Minkowski cavity) whose boundary worldtube $\Sigma = \partial\mathcal{M}_-$ has spacelike cross-sections of strictly positive area $4\pi R^2$, $R > 0$; else $C_1 = 0$. *(C2) Shell admissibility.* $C_2 = 1$ iff the Lanczos surface stress S_{ab} across Σ is Hawking–Ellis Type I and satisfies the surface dominant energy condition at every $\Omega \in S^2$. *(C3) Exterior null matter.* $C_3 = 1$ iff the exterior stress is rank-one null and non-negative, $T_{ab}^+ = \Phi \ell_a \ell_b$ with $\Phi \geq 0$ (Hawking–Ellis Type II with $\Phi \geq 0$; here $\Phi = n^2/r^2$, the vacuum closure $\Phi \equiv 0$ being Schwarzschild). *(C4) Covariant Bondi recoil.* With P_B^μ the Bondi four-momentum, $m_B = \sqrt{-P_B \cdot P_B}$ and $U_B^\mu = P_B^\mu/m_B$ the Bondi rest direction, define the *frame-invariant* recoil

$$R^\mu := (\delta_\nu^\mu + U_B^\mu U_{B\nu}) \frac{dP_B^\nu}{du} = m_B \frac{dU_B^\mu}{du}; \quad (41)$$

$C_4 = 1$ iff $R^\mu \neq 0$. Each of C_1, C_2, C_3 is a diffeomorphism invariant of $(\mathcal{M}, g)$; C_4 is BMS-covariant, unambiguous between stationary Bondi frames and in the canonical shear-free Kinnersley cuts (Cor. 4).

The proper recoil (41) replaces the frame-dependent spatial component dP_B^i/du : a constant-velocity body losing mass isotropically has $dP_B^i/du \neq 0$ in any boosted frame yet does *not* steer, whereas R^μ projects out that longitudinal mass-loss part along U_B and vanishes iff the four-momentum *direction* U_B^μ is constant. For the Kinnersley rocket $P_B^\mu = m(u)v^\mu(u)$ (Cor. 4), so $R^\mu = m a^\mu$ with $|R| = m|a|$ the rest-frame thrust, vanishing iff $a = 0$, and equal to the radiated dipole recoil $-\oint n^2 \cos \vartheta d\Omega$ (App. K).

Proposition 8 (Classification of the news-silent shell sector). *Let $\mathcal{W}$ be the class of spacetimes asymptotically flat at $\mathcal{I}^+$ with peeling and vanishing Bondi news, whose exterior is a regular axisymmetric Robinson–Trautman pure-radiation solution (3), and with $C_1 = C_2 = C_3 = 1$. Within $\mathcal{W}$: (a) the conformal factor is forced to $\ell \leq 1$ (von der Gönna–Kramer [62], Theorem 4), so the emission is the monopole+dipole (12) and $R^\mu = m a^\mu$; hence $C_4 = 1 \iff a \neq 0$; (b) every $C_4 = 1$ member has, up to BMS/Poincaré frame, the Kinnersley photon-rocket exterior family of Sec. 4 (parametrized by $m(u)$ and the acceleration history), admitting a flat-cavity completion through the admissible matched shell of Theorem 2 (perturbatively, with the finite-amplitude completion numerical): the worldtube-first warpsell; (c) the $C_4 = 0$ cell is the constant-velocity face $a \equiv 0$ (exterior $P = \text{const}$, Schwarzschild (8), P_B^μ constant), on which the positive-energy shells of Refs. [8, 9] lie; (d) the bare point photon rocket is the $R \rightarrow 0$ degeneration ($C_1 = 0$, no shell);*

(e) the Alcubierre/Natário class is excluded, its wall stress being observer-robustly Hawking–Ellis Type IV (a complex eigenvalue pair, no rest frame [19]), which fails the rank-one null form $\Phi\ell_a\ell_b$ required by C_3 (independently, it violates the NEC [2]). Thus the single free bit C_4 cleanly separates the accelerating warpsheet from the constant-velocity face.

Proof sketch. (a) News-silence forces $\ell \leq 1$ by von der Gönna–Kramer [62], giving (12); projecting $dP_B^\mu/du = -\oint n^2 \hat{\ell}^\mu d\Omega$ (11) transverse to U_B and using the moments $\oint n^2 d\Omega = -\dot{m}$, $-\oint n^2 \cos\vartheta d\Omega = ma$ gives $R^\mu = m a^\mu$ (App. B). (b) The $\ell = 1$ part of P encodes the instantaneous boost [21, 24]; a non-constant boost is the Kinnersley rocket (App. A), unique up to frame by von der Gönna–Kramer; the matched shell follows from Lemma 2 and Theorem 2. (c)–(d) set $a \equiv 0$ and $R \rightarrow 0$ respectively. (e) C_3 demands the rank-one null (Type II) form; the Alcubierre wall is observer-robustly Type IV [19] and NEC-violating [2], failing C_3 . $\square$

For a *three-dimensional* steering history whose acceleration axis rotates in u there is no global azimuthal Killing vector, so part (a) is stated *per retarded cut*, in the momentary rest frame where the emission is axisymmetric about the instantaneous acceleration axis (App. A). The news-silence characterization itself, however, needs *no* axisymmetry: by Cornish [63] a Robinson–Trautman pure-radiation metric is news-free if and only if it is a Kinnersley photon rocket for an *arbitrarily* (three-dimensionally) accelerating point mass, so the $C_4 = 1$ news-silent cell of $\mathcal{W}$ is the full Kinnersley family even for a rotating acceleration axis; the per-cut framing is only a convenience for the axisymmetric multipole bookkeeping, not a restriction on the steering. What the classification does *not* claim is a sharp dichotomy against a coarse-grained *material* rocket with a finite cabin: a sufficiently idealized hulled rocket shares C_1 – C_2 after coarse-graining and lies *outside* the exact class $\mathcal{W}$ (its exterior is not exact null dust, its cavity only approximately Riemann-flat). The precise statement is therefore a *rocket–warp continuum*: the construction is *mechanically* a rocket (it expends mass as radiation, the passengers feel proper acceleration) and *geometrically* the $C_1 \wedge C_2 \wedge C_3 \wedge C_4$ cell of $\mathcal{W}$: an extended, self-gravitating, exactly tidally flat cavity with covariant Bondi recoil and admissible positive-energy support, the accelerating completion of the positive-energy shell program, distinct from the Alcubierre spatial-expansion class. It is in this precise, invariant sense (exact *within* $\mathcal{W}$, not by a claimed universal separation from all material rockets) that we call it a warp drive.

14.2 Why the Santiago–Schuster–Visser theorem is evaded

Reference [2] proves that *generic warp drives violate the NEC*. Its hypotheses define a warp drive through the shift-vector ansatz with an asymptotically Minkowski (vacuum) exterior, and the violation is a property of the stress-energy that the chosen geometry forces. The radiative warpsheet does not satisfy those hypotheses: we do not prescribe the metric, the exterior is *not vacuum*, and the matter we install (positive null radiation) satisfies the NEC by Proposition 3. The theorem is not contradicted; its premise (a vacuum shift-bubble) simply does not describe our construction. Equally important is *how* positivity is established: a central lesson of Ref. [2] is that earlier “positive-energy” claims inspected only the comoving Eulerian observer, whereas the energy conditions quantify over *all* timelike and null observers; our bulk and shell margins are computed observer-robustly over the boost manifold (App. K), so the binding test is not concealed by a frame choice, the very failure mode the theorem exposes. The only place the NEC could fail is the shell, and Theorem 2 establishes that it does not on an open admissible set.

14.3 Structural soundness beyond the energy conditions

Barzegar, Buchert, and collaborators [16, 15] stress that energy-condition violation is not the only pathology of warp spacetimes: a physical drive must also have a well-defined asymptotic structure and covariant kinematics, and several cataloged constructions fail on those grounds *independently* of any exotic matter. A deeper concern of Refs. [15, 16] is methodological: prescribing a metric and reading off whatever stress-energy the Einstein equations return can manufacture sources with no interpretation as physical matter. The worldtube-first construction is not subject to this objection by design: it does *not* reverse-engineer the stress-energy from a chosen metric but installs interpretable matter region by region (real outgoing null dust outside, vacuum cavity, Hawking–Ellis Type I shell; Sec. 2, Prop. 3), the discipline those references call for. The recent no-go theorems of Ref. [15] and the new realizations of Buchert and Frackowiak [17] (which also report instability in that class, consonant with our stability caveat of Sec. 15) bear on the restricted, metric-first (flow-orthogonal, prescribed-shift) class; the worldtube-first construction, with its interpretable null-dust exterior and a geometrically defined Bondi four-momentum, lies outside their hypotheses. The radiative warpsheet is built to pass these structural tests. Its velocity

and acceleration are Bondi-frame components of the geometrically defined four-momentum $P_B^\mu(u)$ of the exact Kinnersley exterior (Sec. 3, Cor. 4; with $|a|$ the covariant scalar of Definition 1(iv)), not coordinate artifacts. The spacetime is free of closed causal curves and globally hyperbolic on the finite maneuver slab: the Rindler horizon stays off the cavity for $\lambda = aR < 1$, and eternal acceleration is not claimed (Sec. 15). The drive therefore sidesteps the structural no-go catalog, not merely the null-energy one.

14.4 Quantum inequalities do not apply

Ford–Roman–type quantum inequalities [5, 4] bound the magnitude and duration of *negative* energy densities. The radiative warpsell contains none: bulk and (admissible) shell stress-energy are positive. The quantum-inequality no-go that rules out Alcubierre and Krasnikov [77] is therefore inapplicable.

14.5 The acceleration gap

Table 1 positions the construction. Bobrick–Martire and Fuchs *et al.* exhibit positive-energy shells, but only at constant velocity. To our knowledge, the radiative warpsell is the first *subluminal*, *matched flat-cavity* construction to *accelerate* by a covariant, matter-derived recoil, with the *observer-robust dominant* energy condition in bulk and shell and a closed-form steering law, filling the gap those papers identify [8, 9]. The closest membrane-based proposal, Huey [78], also accelerates from an extrinsic-curvature jump, but is *superluminal* and satisfies only the weak condition; the present construction is by contrast *subluminal*, satisfies the full *dominant* condition in bulk and shell, is matched to a compact flat cavity, and prices its acceleration by the closed-form conservation law $-\dot{m} \geq 3m|a|$; four discriminators it does not share. Fuchs *et al.* [9] point to radiation- and momentum-transfer steering (mass shed as focused emission, with ADM/Bondi four-momentum [79] as the governing constraint) as an open direction beyond simple photon rockets; the present work carries out that program, supplying the explicit matched positive-energy metric, the conservation law that prices it, and the dominant-energy-admissible saturating instance. The recent positive-energy constructions of Rodal and Garattini–Zatrimaylov sharpen the same distinction (Sec. 1); the table below and its footnotes record where each sits.

Table 1. Where the radiative momentum warpsell sits. “Bulk NEC” is the matter outside the passenger cavity. “Obs.-robust DEC” is whether the pointwise dominant energy condition is established over *all* timelike and null observers, in bulk *and* shell, the sharpest discriminator from the recent positive-energy constructions. “Accel. (covariant)?” is a genuine, matter-derived change of a covariantly defined state of motion. “FTL?” records only whether a superluminal *coordinate* configuration is permitted, a separate question from physical acceleration; the metric-first bubbles prescribe a coordinate velocity but no covariant acceleration. [†]Rodal reports predominantly positive invariant energy density and global Hawking–Ellis Type I, but the null, weak, and dominant conditions are not globally satisfied (the negative-energy requirement is sharply reduced relative to Alcubierre, not eliminated); [‡]Garattini–Zatrimaylov satisfy the *averaged* conditions in a de Sitter bubble; Huey accelerates from a membrane (extrinsic-curvature jump) but is superluminal and satisfies only the weak condition.

	Bulk NEC	Obs.-robust DEC	Accel. (covariant)?	Mechanism	Energy	FTL?
Alcubierre [1]	violated	no	no	prescribed shift	negative	yes
Natário [6]	violated	no	no	prescribed shift	negative	yes
Van Den Broeck [7]	violated	no	no	prescribed shift	negative	yes
Bobrick–Martire [8]	satisfied	no	no	static shell	positive	no
Lentz [12, 14]	disputed	no	no	plasma soliton	disputed	yes
Fuchs <i>et al.</i> [9]	satisfied	no	no	constant- v shell	positive	no
Rodal [10]	reduced [†]	no	no	constant- v shift	mostly positive	no
Garattini–Zatrimaylov [11]	averaged [‡]	no	no	de Sitter bubble	positive	no
Huey [78]	satisfied	no	yes	membrane jump	positive	yes
This work	satisfied	yes	yes	positive null flux	positive	no

14.6 What is new

The ingredients (photon rockets [21, 23, 53, 24], Israel junctions [26, 57], radiating and thin shells [80, 81], and the Bobrick–Martire taxonomy [8]) are individually classical; the novelty is their synthesis answering an explicitly open problem (the contributions of Sec. 1). Three near neighbors sharpen the distinction. Membrane models that read acceleration off an extrinsic-curvature jump [78] are kinematically related but do not supply an exact, positive-energy radiative driver or a steering law. Füzfa [49] modeled relativistic-rocket interstellar travel with Kinnersley’s solution, but as a bare accelerating point mass, without an excised cavity, a matched passenger shell, a surface energy-condition guarantee, or a kinematic frontier. The thrust here is sourced by physical radiation recoil, which distinguishes the construction from the other classical accelerating exact

solution, the C-metric [82], whose uniform acceleration is supplied by a conical-singularity strut reaching to infinity rather than by emitted radiation.

15 Limitations and scope

We state the boundaries of the claim plainly. *Subluminal.* The construction is causal and subluminal; it is not a faster-than-light proposal. *Not reactionless.* It is a rocket: it expends mass as radiation, with the budget (14). *Felt acceleration.* Passengers feel proper acceleration, not the zero- g free fall of an idealized Alcubierre interior; but in a flat Minkowski cavity the *tidal* tensor vanishes, so there is no differential stretching. This is the warp feature of Definition 1, a tidally protected passenger laboratory, and it locates the construction as the accelerating member of the positive-energy shell family. *Energy cost.* The exact budget (14) is $m_f/m_0 = e^{-3 \int |a| du}$; for a *rectilinear* boost the integrated proper acceleration equals the net rapidity, $\int |a| du = \Delta\eta$, and the cost is severe: $m_f/m_0 = e^{-3\Delta\eta}$, so even mildly relativistic boosts radiate the bulk of the rest mass (the worked saturating burn of App. K sheds about half its rest mass), and approaching the speed of light ($\Delta\eta \rightarrow \infty$) sends $m_f/m_0 \rightarrow 0$. A *steering* history that changes direction has $\int |a| du \geq |\Delta\eta|$ (the integrated rapidity arc exceeds the net rapidity, as boosts do not add as vectors), so a turn costs strictly more than the collinear figure. The cost is steep, but *positive and finite*: no negative energy and no quantum-inequality violation appears at any stage. The barrier is the size of the budget, the sense in which warp propulsion is here an engineering problem. *Local in retarded time.* The solution is a finite maneuver on a u -slab; global, eternal existence is not claimed, and $m(u) > 0$ on the slab by (13) (Prop. 4). *Positivity for the distributional shell.* The Bondi-mass step of Theorem 1 needs only the sign of the junction surface term, which the surface DEC fixes (Lemma 6), and the construction has $m_B = m(u) > 0$ directly. A full distributional positive-Bondi-mass theorem at $\mathcal{I}^+$ is not needed for the bound and is left open; its one missing analytic step is the well-posedness (Fredholm solvability and decay) of the Sen–Witten spinor *transmission* problem on an asymptotically null slice crossing the timelike C^0 shell, the spinor matched across the junction with the surface term of Lemma 6, extending the smooth-slice results [39, 40] as Shibuya [43] extends the spatial-infinity ones. *Existence: unconditional at small amplitude, numerical at finite amplitude.* Theorem 2 is unconditional in the small-amplitude regime: the $\ell \geq 2$ anchor linearization is the round-sphere rigidity operator, the $\ell = 1$ junction is surjective onto the steering source in closed form (Lemma 2), and the exact match rests on the single scalar $D(x)$, closed-form (58) and nonzero on $(0, 24/25)$ (App. D). Its reach to finite amplitude is numerical: the matched-junction solve of App. C returns a small Darboux residual, Type I surface stress, and a strictly positive observer-robust DEC margin, pinning the frontier between a numerical lower bound (with a Newton–Kantorovich estimate) and the rigorous ceiling, $\underline{g}(x) \leq g(x) < \frac{1}{2}(1-x)$ (46). *Dynamical existence: fixed-axis only.* Proposition 6 lifts admissibility to a time-evolved spacetime for slow sub-horizon burns. The constant-retarded-cut junction is elliptic with the static angular symbol and the $\ell \leq 1$ Schur block is non-degenerate (Lemmas 3–4); for a fixed-axis (non-rotating, reflection-symmetric) burn the axial row vanishes by reflection parity, the surviving polar time-rows form a sub-horizon triangular isomorphism (63), and C^2 persistence closes by a Grönwall bound with no Nash–Moser. This is conditional on the constant-cut compatibility identity (64), a flat-interior embeddability condition (not a contracted-Codazzi propagation) whose leading $\mathcal{O}(a^2)$ $\ell = 2$ *obstruction* we show to cancel gauge-completely (the four $\ell = 2$ junction rows independent, the reduced Jacobian non-degenerate, App. E), the identity itself remaining conditional on the all-orders retarded-input step below; the per-cut shape match $[h_{AB}] = 0$ is itself unconditional, being the static elliptic problem at each cut. Two items are left for future work: a self-contained proof of the all-orders identity (64), and the rotating-axis burn, where the magnetic-parity row is sourced and couples all $\ell \geq 2$ modes with no symmetry protection. The leading retarded correction, the outgoing mass-loss flux, *tightens* the surface DEC margin (App. F), so the fully dynamical frontier is somewhat tighter than $g(x)$. *Axisymmetric finite-amplitude shell.* The exterior news-silence holds for an arbitrarily (three-dimensionally) accelerating worldline (Cornish [63]), and at small amplitude the matched *shell* likewise exists per cut for an arbitrary acceleration *axis*: the $\ell = 1$ closure is $SO(3)$ -equivariant and reduces, on the translation representation, to the same nonzero scalar $D(x)$ (Cor. 6). The finite-amplitude results (the higher-order $D(x)$ and the frontier $g(x)$) are computed per retarded cut, in the momentary rest frame where the emission is axisymmetric about the instantaneous axis; this covers any planar (fixed-axis) maneuver in full and a rotating one cut by cut. Assembling the per-cut shells into a single matched worldtube for a rotating axis (a non-axisymmetric Darboux solve in vector/tensor harmonics on S^2) is left to future work. *Thickening: radiative wall realized, self-consistent equilibrium.* A naive *radial-tension* smoothstep wall reproduces the source–vacuum boundary cost, whereas the *tangential-pressure*

wall is strictly DEC-admissible across its whole width for local $2m/r < 4/5$, recovers the thin-shell anchor, is a self-consistent anisotropic equilibrium, and stays admissible once outgoing null radiation streams through it (Sec. 9, App. F). Its $\mathcal{O}(\mu)$ back-reaction is closed at the adiabatic level, but the saturating burn has an order-unity ωR , so the fully time-dependent $\mathcal{O}(\mu^2)$ correction is not parametrically negligible there and remains open. *Linear stability: marginal, conditionally burn-limited.* The realized tangential-pressure wall sits *exactly* on the Poisson–Visser marginal curve $V''(R) = 0$ (App. J): a radial displacement is a neutral zero mode. Under the frozen-background rear-pole redshift the marginal wall e-folds in $\tau_{\text{efold}} = \sqrt{2/|V''|}$, of order a light-crossing time (App. J), so the Tsolkovsky burn outruns the instability only in a restricted high-compactness, brief-rapidity corner. A slightly *stiffer*, still-admissible wall is strictly stable at no cost in the surface DEC margin, which is junction-fixed and independent of the equation-of-state slope, so strict stability and strict dominant energy decouple; this stiffened wall is a nearby model, not the realized one (App. J). The interior cavity is tidally regular ($K^\mu{}_\rho \equiv 0$) for $\lambda < 1$, and the fully-dynamic, flux-coupled stability of the *radiating* shell remains open.

16 Conclusion

We have inverted the warp-drive design problem. Rather than prescribing a metric (and inheriting exotic matter) or a static source (and forgoing acceleration), we prescribe the passenger worldtube, accept positive null radiation as the exterior, and solve for the shell. The exterior is an exact photon rocket whose energy conditions reduce to $n^2 \geq 0$, steering obeys the positive-energy control law $-\dot{m} \geq 3m|a|$, and admissible accelerating shells exist unconditionally at small amplitude about the static Schwarzschild–Minkowski anchor, the $\ell = 1$ matching reducing to the single closed-form scalar $D(x)$. An acceleration–compactness frontier caps the proper acceleration, below the rigorous closed-form ceiling $\frac{1}{2}(1 - x)$.

The organizing result is model-independent: no asymptotically flat, dominant-energy drive with a confined material source can change its velocity without radiating four-momentum to null infinity (Theorem 1). This flux balance is classical Bondi–Sachs; our contribution is to read it as a no-reactionless-steering principle, the general-relativistic counterpart of the relativistic rocket, and to read it directly from the curvature across the Kinnersley rocket, static Schwarzschild, and Vaidya. A confined positive-energy system cannot change its Bondi four-momentum without radiating, so reactionless steering is impossible; the velocity reading is exact between stationary Bondi frames and, for the closed-form Kinnersley rocket, instantaneously (Prop. 1). Our construction is the explicit positive-energy *saturating* instance of this law for an accelerating warp drive, with the minimum-radiation maneuver the gravitational-wave-silent Damour dipole. Acceleration is paid for by positive radiation, not negative energy. Within the compactness window $x + 2aR < 4/5$ the thin shell is realized as the limit of an admissible *radiating* tangential-pressure wall that stays dominant-energy-admissible under its own $\mathcal{O}(\mu)$ back-reaction, and the worked saturating burn (App. K) lies inside this window throughout, so that burn itself, not merely its endpoints, is realized by an admissible radiating wall at the adiabatic level (Secs. 9.2, 9.3).

Several results are fully analytic; we record them as the rigorous results:

- the closed-form control law and the bifurcation scalar $D(x)$ (Lemma 2, App. D);
- the warp-drive identity as a diffeomorphism-invariant charge classification (Prop. 8);
- the radiating-wall back-reaction sign (Theorem 6, Prop. 10);
- the radial-tension thickening obstruction with its $1/\delta^2$ boundary-cost law (Prop. 7, Cor. 5);
- the gravitational-wave silence of the Damour dipole, analytic from the exact Kinnersley news, with a numerical peeling cross-check (App. I);
- maneuver design as a state-constrained geodesic-control theorem on $\mathbb{H}^3$ with universal Tsolkovsky constant 3 (Theorems 7, 5).

The construction extends to $\Lambda > 0$ (Sec. 12).

Four problems remain open, each a sharply posed problem, the scope set out in Sec. 15.

(i) *Two-sided rigorous frontier.* Prove $\underline{g}(x) \leq g(x)$ to complement the already-rigorous ceiling $g(x) < \frac{1}{2}(1 - x)$ (Prop. 5); the gap is a Newton–Kantorovich existence bound at the computed amplitude, requiring the inverse bound at the finite-amplitude solution *and* interval enclosures of the residual and second-variation moduli, by interval arithmetic on the multipole contraction or positive-curvature (Weyl) embedding. (ii) *All-orders compatibility* (64): that the $\ell \geq 2$ retarded energy row $[h_{uu}]$ vanishes identically, equivalently that the exterior 3-metric on Σ isometrically

embeds in the flat interior (a Gauss–Codazzi class-one consistency); its leading $\mathcal{O}(a^2)$ $\ell = 2$ *obstruction* is shown to cancel gauge-completely (App. E), with the $\mathcal{O}(a^3)$ orders and the magnetic-parity (rotating-axis) coupling open. (iii) *Radiating-shell dynamics*, beyond the analytic $\mathcal{O}(\mu)$ wall theorem (Theorem 6): (a) the non-adiabatic $\mathcal{O}(\mu^2)$ surface margin at $\omega R = \mathcal{O}(1)$, by a generalized-Vaidya characteristic interior evolution closing on the contracted Bianchi identity; and (b) the flux-coupled linear stability of the radiating shell, a non-autonomous Poisson–Visser potential in which the e-fold competes with the burn rate. (iv) *Rigorous de Sitter positive-energy bound*. Kastor–Traschen [74] *already* prove positivity of the *conformal-time* charge under the DEC; only the *static-time* charge, against which the steering balance is naturally read, is indefinite. The open object is to recast dP_B^μ/du for the (A)dS rocket in the conformal-charge framework [73] and show the maneuver’s conformal-charge change is sign-definite.

Data availability

The code that reproduces every analytic and numerical result of this paper, with its run configurations and tolerances, is openly available at https://github.com/anindex/world_tube.

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A Photon-rocket conventions and curvature

We use the null tetrad aligned with $\ell_a = -(du)_a$ and the Robinson–Trautman conventions of Refs. [21, 24, 53]. The cleanest route to Eq. (7) is the Kerr–Schild form of the photon rocket on a flat seed,

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2m(u)}{r} \ell_\mu \ell_\nu, \quad (42)$$

with ℓ the retarded outgoing null direction from an accelerated worldline $z^\mu(u)$ (u its proper time) and $r = -(X - z) \cdot \dot{z}$ the retarded distance; an arbitrary conformal factor $P(u, \zeta, \bar{\zeta})$ with $K = 1$ but *not* of genuine boost form does not solve the field equations. For an accelerated worldline with proper acceleration a along a fixed axis ($\vartheta = 0$), evaluated in the momentary rest frame of the cut ($\eta = 0$) (App. K): (i) the Ricci scalar vanishes; (ii) $T_{ab} = (n^2/r^2)\ell_a \ell_b$ is rank-one null to $\sim 10^{-18}$; and (iii) $4\pi n^2 = -\dot{m} - 3ma \cos \vartheta$, a *pure dipole* (in that rest frame) of amplitude exactly $3m|a|$, maximal at the rear (exhaust) pole $\vartheta = \pi$ and vanishing at the forward pole at saturation; at finite rapidity the fixed-frame pattern is its aberration. Positivity for all ϑ is therefore equivalent to the control law (13), and the observer-robust optimizer returns margins ≥ 0 iff $n^2 \geq 0$ (Proposition 3). One nuance: exact null dust is *algebraically* Hawking–Ellis Type II (a defective null eigenvector), but it lies on the Type I $\leftrightarrow$ II eigenvalue boundary, and the numerical eigensolver labels it Type I; the rank-one null structure above is the frame-invariant content.

B Junction algebra and the balance laws

For the static shell, $K_{ab\theta}^\theta = \sqrt{f}/R$ and $K_{ab\tau}^\tau = f'/(2\sqrt{f})$ on each side ($f_- = 1$, $f_+ = 1 - 2m/R$); the Lanczos equation (9) then gives Eqs. (15)–(16) *exactly*, and

$$8\pi R(\sigma_0 - p_0)s = \frac{1}{2}(1-s)(5s-1), \quad s = \sqrt{1-x}, \quad (43)$$

positive iff $1/5 < s < 1$, i.e. the surface-DEC window $x < 24/25$ of Eq. (18). The moment identities (22) follow from the angular structure of n^2 alone: with $4\pi n^2 = -\dot{m} - 3ma \cos \vartheta$,

$$\oint n^2 d\Omega = -\dot{m}, \quad -\oint n^2 \cos \vartheta d\Omega = ma, \quad (44)$$

(the dipole using $\oint \cos^2 \vartheta d\Omega = 4\pi/3$), so the monopole of the crossing flux is the luminosity $-\dot{m}$ and the dipole gives the thrust $dP_B^z/du = -\oint n^2 \cos \vartheta d\Omega = ma$, reproducing the Bondi budget (11). The balance laws (20)–(21) are not extra postulates but the contracted Gauss–Codazzi (projected Bianchi) identities for a timelike hypersurface. With N^a the unit spacelike normal and $\bar{K}_{ab}$ the mean extrinsic curvature, the distributional Einstein equations give $D_b S^b_{\hat{a}} = -[T_{N\hat{a}}]$ (tangential) and $S^{ab} \bar{K}_{ab} = -[T_{NN}]$ (normal): the standard *timelike*-shell relations of Israel [26] (Poisson [27], §3.7–3.8). The surface stress is *not* conserved precisely because the crossing flux $[T_{N\hat{a}}] \neq 0$ sources it. Here the interior is a vacuum cavity ($T_{ab}^- = 0$), so $[T_{N\hat{a}}] = T_{N\hat{a}}^+$ and $[T_{NN}] = T_{NN}^+$: the radiation is *emitted at* Σ (the regularized rocket nozzle), not transmitted through it from the cavity, and the induced surface energy flux $S^{\hat{0}\hat{\theta}}$ balances the lateral recoil. (We use “Darmois” for the first junction condition $[h_{ab}] = 0$, continuity of the induced metric, which fixes $R(u, \Omega)$; the second fundamental form jumps, and the Lanczos S_{ab} carries that jump, so a genuine shell is present.)

Surface dominant energy and the junction term. Step (iii) of Theorem 1 uses one property of the distributional shell: that it adds a non-negative term to the Witten–Nester spinor expression for the Bondi four-momentum. We record it.

Lemma 6 (Surface dominant energy and the junction term). *Let S_{ab} be a Hawking–Ellis Type I surface stress, with surface rest frame u^a , $S_{ab} = \sigma u_a u_b + \sum_A p_A e_{\hat{A}a} e_{\hat{A}b}$ (σ the surface energy density, p_A the principal surface pressures). Then $S_{ab} V^a W^b \geq 0$ for every pair of future-directed causal V^a, W^a if and only if the surface dominant energy condition $\sigma \geq \max_A |p_A|$ holds. Consequently, integrating the Sen–Witten identity across Σ on an asymptotically null slice, the distributional shell contributes the single boundary term $\oint_\Sigma S_{ab} \xi^a J^b dA$ (up to a positive normalization), with ξ^a the asymptotic time translation and $J^a = \bar{\lambda} \gamma^a \lambda$ the Witten–spinor Dirac current (both future-directed causal) which is non-negative whenever Σ is admissible.*

Proof. Write $\gamma_V = -u \cdot V > 0$, $\gamma_W = -u \cdot W > 0$ and the spatial parts $V_\perp^A = e_{\hat{A}} \cdot V$, $W_\perp^A = e_{\hat{A}} \cdot W$; for causal V, W , $-V \cdot V = \gamma_V^2 - |V_\perp|^2 \geq 0$ gives $|V_\perp| \leq \gamma_V$ and likewise for W . Then $S_{ab} V^a W^b = \sigma \gamma_V \gamma_W + \sum_A p_A V_\perp^A W_\perp^A$, and by Cauchy–Schwarz

$|\sum_A p_A V_\perp^A W_\perp^A| \leq (\max_A |p_A|) |V_\perp| |W_\perp| \leq (\max_A |p_A|) \gamma_V \gamma_W$, so $S_{ab} V^a W^b \geq (\sigma - \max_A |p_A|) \gamma_V \gamma_W \geq 0$. Conversely, if $\sigma < |p_B|$ for some B , the future-null pair $V = u + e_{\hat{B}}$, $W = u - \text{sgn}(p_B) e_{\hat{B}}$ gives $S_{ab} V^a W^b = \sigma - |p_B| < 0$. The Sen–Witten boundary integrand is $S_{ab} \xi^a J^b$ with ξ, J future-causal. Since S_{ab} is tangential to the timelike Σ ($S_{ab} N^b = 0$), only the tangential projections $\xi_\parallel, J_\parallel$ enter the contraction, and the projection of a future-causal spacetime vector onto a *timelike* hypersurface is itself future-causal in the induced Lorentzian 3-metric; the bound above, applied to $\xi_\parallel, J_\parallel$, gives the non-negativity. $\square$

The matched shell of Theorem 2 is Type I with strictly positive surface-DEC margin, so its boundary term is strictly positive; the flat cavity, being vacuum, contributes none. This sign is the only property of the distributional junction that the Bondi-mass positivity step of Theorem 1 requires; the spinor existence and decay across the C^0 junction that a full distributional positive-Bondi-mass theorem at $\mathcal{I}^+$ would also establish is left open (Sec. 15).

C The dipole response and the frontier

The dipole coefficients σ_1, p_1, q_1 require solving the Darmon condition $[h_{ab}] = 0$ for the shell shape $R(u, \Omega)$. We emphasize that this is *not* a scalar root-find but a nonlinear isometric-embedding problem on S^2 : the induced 2-metric of the angularly anisotropic exterior must match that of the Rindler-accelerated flat cavity. Linearizing about the static anchor in λ and μ , the steering enters only through the $\ell = 1$ (dipole) sector of $(\ln P)_{,u} = -a \cos \vartheta$, which by S^2 orthogonality decouples from the $\ell = 0$ monopole; the embedding then reduces, at first order, to a linear per-mode system, which is the route by which σ_1, p_1, q_1 are obtained. We carry out this computation through a manifestly gauge-invariant matched junction against the exact Kerr–Schild exterior, agreeing across coordinate gauges to $\sim 10^{-13}$ and evaluated with the observer-robust surface energy-condition framework of Ref. [19]. The outcome is admissible: the matched surface stress is Hawking–Ellis Type I across the entire admissible set (no Type IV transition), and the worst-observer surface DEC margin is strictly positive on an open subinterval of $2m/R < 24/25$, so Theorem 2’s admissible neighborhood is non-empty: admissible *accelerating* shells exist. Two features fix the frontier. First, the exact photon-rocket exterior is *locally Schwarzschild* to $\mathcal{O}(a)$ (the $\mathcal{O}(a)$ content is the traceless null-dust dipole, which does not curve the Weyl tensor until $\mathcal{O}(a^2)$), so the geometric density response is small, $\sigma_1 = \mathcal{O}(x^2)$. Second, the recoil flux does *not* enter the worst-observer margin at linear order: it appears only as q^2 , whose λ -derivative vanishes at the anchor (where $q = 0$), so the linear margin response is built from the cos-dipoles $\sigma_1 - p_1$ alone and, since these are positive, binds at the rear pole $\vartheta \rightarrow \pi$ (where the equatorial flux $q_{\hat{\vartheta}} \propto \sin \vartheta$ also vanishes). Both facts make the *linear* envelope $g_{\text{lin}}(x) = (\sigma_0 - p_0)/(\sigma_1 - p_1)|_{\vartheta=\pi}$ *wide*: with $\sigma_0 - p_0 \sim x$ and $\sigma_1 - p_1 \sim x^2$ it scales as $g_{\text{lin}} \sim \mathcal{O}(1/x)$, generous at small compactness and descending to $\mathcal{O}(10^{-1})$ only as $x \rightarrow 1$. A finite- λ scan with the same observer-robust margin instead closes the DEC at $g \sim \mathcal{O}(10^{-1})$ across the window; because the linear term alone permits larger λ at small x , this finite-amplitude frontier is an $\mathcal{O}(\lambda^2)$ (nonlinear) effect (where the recoil q^2 and the nonlinear shape response re-enter), rather than the $\mathcal{O}(\lambda)$ shape dipole. We also carry out the fully nonlinear matched Darmon solve carrying $\ell \geq 2$ multipoles: a finite- λ least-squares solve of the full $[h_{ab}] = 0$ with the areal shape $R(\vartheta) = R^0(1 + \sum_\ell a_\ell P_\ell)$, the lapse, and the foliation tilt on *both* sides expanded in Legendre/vector-harmonic modes. Under a successively refined multipole expansion the residual *falls geometrically with the multipole order*: with the matching frames held timelike (a physical constraint, since tilting a frame past null gives a spurious degenerate frame and a divergent surface stress), $\max |[h_{ab}]|$ falls to 10^{-7} – 10^{-9} by $L = 6$ for $x \lesssim 0.3$, and to $\sim 10^{-3}$ by $L = 6$ at $x = 0.5, 0.7$ (tightening further with $L = 8$ – 10), once the $\ell \geq 2$ shape and the *interior re-slicing* (lapse+tilt) modes are present; the heuristic envelope (29) is an independent (non-rigorous) comparison the computed values respect, not a proven bound (the rigorous upper bound is the kinematic ceiling, Prop. 5). The dipole-only mismatch of 1–10% is thus a missing-multipole artifact, not an obstruction: the residual falls under multipole refinement, *numerical evidence* that the accelerating photon-rocket exterior is Darmon-matchable to a flat cavity through a thin shell at finite amplitude (the rigorous, unconditional existence statement remains the small-amplitude Theorem 2). On the converged match the surface stress is Hawking–Ellis Type I and the worst-observer DEC margin is strictly *positive* at the computed amplitude $\lambda = \underline{g}(x)$ *across the whole window* $x \in [0.1, 0.7]$, binding at the rear pole and stable under multipole refinement: the converged margin runs $\{1.8, 9.1, 18, 21\} \times 10^{-3}$ at $x = \{0.1, 0.3, 0.5, 0.7\}$ (with $\max |[h_{ab}]| = 1.4 \times 10^{-7}$ at $x = 0.1$ and $\sim 10^{-3}$ at $x = 0.5, 0.7$), gauge-invariant to $\sim 10^{-3}$. The two

computations are complementary. The observer-robust *dipole* scan locates the margin zero at

$$\begin{aligned} \underline{g}(x) &\simeq \{0.19, 0.20, 0.19, 0.14, 0.09\} \\ \text{at } x &= \{0.1, 0.2, 0.3, 0.5, 0.7\} \end{aligned} \quad (45)$$

(to within the $\sim 5\%$ scan resolution; every value respects the envelope (29)); the *converged multipole* solve, evaluated at that same $\lambda = \underline{g}(x)$, finds the worst-observer margin still strictly *positive* (the same $\{1.8, \dots, 21\} \times 10^{-3}$ above), because the higher multipoles only relax the junction. The true margin zero therefore lies at $\lambda \geq \underline{g}(x)$, so $\underline{g}(x)$ is a numerical *lower bound* on the frontier $g(x)$ of (28), and the margin grows as $\underline{g}(x)$ falls with compactness. Together with the rigorous kinematic ceiling (Prop. 5) this *brackets* the frontier,

$$\underline{g}(x) \leq g(x) < \frac{1}{2}(1-x), \quad (46)$$

a numerical lower bound $\underline{g}$ (supported by a Newton–Kantorovich estimate below) and a rigorous closed-form upper bound.

Anchor solvability (proof of Lemma 2). At the anchor the exterior is exactly Schwarzschild and the shell’s induced 2-metric is exactly the round sphere, so the $\mathcal{O}(\lambda)$ corrections to the embedding linearization vanish and DF_0 is the round-sphere normal operator $\Delta_{S^2} + 2$ on the shape (no approximation). Its kernel is the $\ell = 1$ infinitesimal isometries (Cohn–Vossen rigidity [59]); its eigenvalues $2 - \ell(\ell + 1)$ have magnitude ≥ 2 off that kernel, giving $\|(\Delta_{S^2} + 2)^{-1}\| = \frac{1}{2}$. The $\ell = 0$ lapse block is algebraic with non-vanishing Jacobian for $x < 1$ ($\partial_R[\sqrt{f}] = m/(R^2\sqrt{1-x}) \neq 0$), and the $\ell = 1$ steering sector is *surjective* onto its $\mathcal{O}(\lambda)$ source through the closed-form rows (54) (the constant direction being $\mathcal{O}(\lambda^2)$), so by Lyapunov–Schmidt the exact match closes on the single non-vanishing scalar $D(x)$ (55), in the elementary closed form (58) ($D < 0$, $|D| \geq \frac{4}{3}$ on $(0, 24/25)$ by elementary factorization). Hence DF_0 is a bounded isomorphism in its $\ell \geq 2$ and $\ell = 0$ sectors and surjective in $\ell = 1$ off the translation gauge, and the implicit function theorem (in its submersion form for $\ell = 1$) makes Theorem 2 *unconditional* in a (rigorous, non-explicit) open neighborhood of the anchor. What this argument does *not* supply is the *explicit* radius of that neighborhood: a conservative rigorous lower bound would follow from explicit elliptic (Schauder) constants for the inverse bound and an explicit second-variation bound (a quantitative implicit-function estimate), and the sharp frontier $\underline{g}(x) \sim 0.2$ across $x \in [0.1, 0.7]$ (hardest near the effective horizon $f - 2aR \rightarrow 0$ at high x) would follow either from an interval-arithmetic evaluation of the contraction below or from a reduction to the positive-curvature isometric-embedding (Weyl) theory; both are left to future work.

Newton–Kantorovich convergence estimate. A genuine Newton–Kantorovich [83] existence proof at a computed amplitude $\lambda = \underline{g}(x)$ needs the inverse bound at the finite-amplitude approximate solution, not the anchor value $\beta_0 = \frac{1}{2}$ of Lemma 2. Using β_0 as an approximation, the Kantorovich product $h = K\beta_0^2\eta$ (with $\eta = \|F\|_\infty$ the converged residual and $K = \|F''\|$ the second-variation modulus) stays well below the $\frac{1}{2}$ threshold across $x \in [0.1, 0.7]$ ($h \leq 1.3 \times 10^{-5}$ for $x \leq 0.3$, $h \approx 6 \times 10^{-2}$ at $x = 0.5, 0.7$), the geometric decay of η in the multipole order doing the work. This is a convergence estimate, evidence that the multipole iteration contracts toward a genuine solution at that amplitude, not a finite-amplitude existence theorem (which needs the finite-amplitude inverse bound and interval bounds on η, K , left open). Accordingly $\underline{g}(x)$ is a *numerical* lower bound, the only rigorous member of the bracket (46) being the kinematic ceiling (Prop. 5).

The frontier is set by the shape anisotropy $\sigma - p$ at the rear pole (the recoil flux decouples, Sec. 8).

Why the rear pole, in closed form. On the symmetry axis the retarded structure is exactly solvable: for a point a coordinate distance b behind the source the areal radius is $r = b(1 - ba/2)$, the conformal factor $w^2 = 1 - 2ar$, and the squared gradient of the areal radius is

$$|\nabla r|^2 = 1 - \frac{2m}{r} \mp 2ar = f \mp 2ar \quad (47)$$

($-$ at the rear, $+$ at the front), matching the exact Kerr–Schild metric. Equivalently, the exact retarded inverse metric is $g^{rr} = 2H = f + 2ar \cos \vartheta$, which is already $f - 2ar$ at the rear pole ($\vartheta = \pi$) and $f + 2ar$ at the front: the rear pole is natively the deeper effective potential, the

coordinate and areal radii agreeing on the axis, and the effective horizon $f - 2ar \rightarrow 0$ is approached at the rear. The acceleration thus enters as an *effective redshift*: the rear pole carries effective compactness $x + 2ar$ and the front pole $x - 2ar$ (the equivalence-principle redshift of a point at proper distance r behind a frame of proper acceleration a). *This is why the dominant-energy frontier binds at the rear pole*: acceleration deepens the effective potential there. The frontier is set by the surface DEC margin (the extrinsic-curvature *jump* $[K]$ below), not by the bare timelikeness $|\nabla r|^2 > 0$; both happen to bind at the rear for the exact metric, but the operative inequality is the observer-robust margin, evaluated angle-by-angle by the scan. The exterior extrinsic curvatures follow in closed form, $K^{\hat{\tau}}_{\hat{\tau}}^+ = (m/r^2 - a)/\sqrt{f - 2ar}$ and $K^{\hat{\theta}}_{\hat{\theta}}^+ = \sqrt{f - 2ar}/r$, with the flat-Rindler interior giving $K^{\hat{\tau}}_{\hat{\tau}}^- = -a/(1 - ar)$, $K^{\hat{\theta}}_{\hat{\theta}}^- = 1/r$. The margin reduces to $\sigma(\pi) - p(\pi) = -\frac{1}{8\pi}([K^{\hat{\tau}}_{\hat{\tau}}] + 3[K^{\hat{\theta}}_{\hat{\theta}}])$. The frame term $[K^{\hat{\tau}}_{\hat{\tau}}]$ is closed-form and nearly λ -independent; the *constant* $g(x)$ is set by the $\mathcal{O}(\lambda^2)$ response of the shape multipoles in $[K^{\hat{\theta}}_{\hat{\theta}}]$ (with the shape frozen, the low-order margin opens rather than closes), so the mechanism is exact while the constant is the global shape response, a Buchdahl-type bound whose structure is closed-form and whose value $g(x) \sim 0.2$ is numerical. An $\mathcal{O}(\lambda^2)$ expansion makes this sharp: with the shape frozen, both the first- and second-order rear-pole margin coefficients are *positive* (acceleration alone makes the rigid pole *safer* at low order; the full rigid margin closes only near the effective horizon $f - 2aR \rightarrow 0$ at large x), so over the operating range the frontier is forced by the global shape response. Extrapolating the *first-principles* rear-pole margin from the converged multipole solve to its zero under an $\mathcal{O}(\lambda^2)$ model is, moreover, non-predictive: the extrapolated thresholds (lying below the envelope at small x and unbounded above it at larger x) bear no systematic relation to the computed values, direct quantitative evidence that the constant lives beyond $\mathcal{O}(\lambda^2)$, in the nonlinear closure. Fitting the computed values under the envelope yields the semi-closed form $g(x) \simeq \frac{1}{2}(\frac{24}{25} - x)[x/(x_0 + x)]^p$ with $x_0 \approx 3.0$, $p \approx 0.23$ (the envelope times a small- x shape-suppression factor, accurate to $\lesssim 4\%$ across $x \in (0, 0.7]$). The high-compactness fall-off of $g(x)$ is the onset of the effective horizon $f - 2ar \rightarrow 0$.

Route to a rigorous lower bound on the frontier. The frozen-shape rear-pole margin is closed-form through the jumps above, $8\pi R M_{\text{frozen}}(x, \lambda) = -R([K^{\hat{\tau}}_{\hat{\tau}}] + 3[K^{\hat{\theta}}_{\hat{\theta}}])$, and its retarded expansion at fixed compactness is

$$8\pi R M_{\text{frozen}}(x, \lambda) = M_0(x) + c_1(x)\lambda + c_2(x)\lambda^2 + \mathcal{O}(\lambda^3), \quad M_0(x) = -\frac{(5s-1)(s-1)}{2s}, \quad (48)$$

with $s = \sqrt{1-x}$, $M_0 = 8\pi R(\sigma_0 - p_0) > 0$ the anchor margin, and the closed forms $c_1(x) = -1 + 7/s + (5x-6)/(2s^3)$, $c_2(x) = -1 + \frac{5}{2s^3} - \frac{3x}{4s^5}$ (from the exact axis $[K]$ -jumps). Both are strictly positive across the tested operating window $x \leq 0.7$: $c_1 > 0$ follows from Sturm's theorem ($2s^3c_1 = -2s^3 + 9s^2 - 1$ has no root on $s \in [\frac{1}{2}, 1]$, i.e. $x < \frac{3}{4}$), and $c_2 > 0$ up to its lone root $x \simeq 0.7606$ (just past the window, where c_2 turns negative), with an interior maximum before that descent; the two-term expansion itself converges only below the effective-horizon branch point $\lambda < \frac{1}{2}(1-x)$ (Prop. 5), which the computed $\underline{g}(x)$ respect. Explicitly $(c_1, c_2) = (3.16, 1.83), (3.52, 2.72), (3.95, 3.95), (4.17, 3.56)$ at $x = 0.1, 0.3, 0.5, 0.7$: at the computed amplitudes $\lambda = \underline{g}(x) \lesssim 0.2$ the rigid pole is strictly *safer* than the anchor, so the surface-DEC closure at $\underline{g}(x)$ is driven by the *global shape response*, not the frozen-shape redshift (the rigid margin itself closes only near the kinematic ceiling for $x \gtrsim 0.6$). This establishes the frozen expansion in closed form: the rigid pole is safer at low order. The *matched* worst-observer margin Φ , by contrast, binds at the rear pole and *falls* with λ , the matched dipole $\sigma_1 - p_1 > 0$ reducing it. Split it as

$$\Phi(x, \lambda) = M_0(x) - b_1(x)\lambda - \mathcal{D}_{\text{sh}}(x, \lambda), \quad b_1(x) = (\sigma_1 - p_1)|_{\pi}, \quad \mathcal{D}_{\text{sh}}(x, \lambda) \geq 0, \quad (49)$$

with $M_0 = 8\pi R(\sigma_0 - p_0) > 0$ the anchor margin, b_1 the matched linear slope, and $\mathcal{D}_{\text{sh}}$ (distinct from the $\ell = 1$ scalar $D(x)$ of (58)) the nonlinear shape-response reduction. This route to a rigorous lower bound rests on three estimates, of which the first is established here. *Quadratic onset*. The matched linear response is locally Schwarzschild ($\sigma_1 = \mathcal{O}(x^2)$) and the recoil flux enters the margin only as q^2 , whose λ -derivative vanishes at the anchor (this appendix and Sec. 8); hence the matched linear slope is parametrically small, $b_1(x) = \mathcal{O}(x^2)$, the linear envelope $M_0/b_1 \sim 1/x$ is wide, and $\mathcal{D}_{\text{sh}} = \mathcal{O}(\lambda^2)$ carries the binding: the frontier is a genuine $\mathcal{O}(\lambda^2)$ shape-response effect, not a frozen-shape redshift. *Shape-reduction bound*. A uniform nonlinear estimate of the DEC-margin functional under the embedding response, built from the Schauder bound (52) of App. D (entering as C_S^2 , together with the margin-functional and curvature-derivative constants),

would give $\mathcal{D}_{\text{sh}}(x, \lambda) \leq C_{\text{shape}}(x) \lambda^2$ with interval-arithmetic bounds. *Angular dominance.* That the rear pole is the binding angle, $\Phi(x, \lambda) = \min_{\vartheta} \Phi(x, \lambda, \vartheta)$, requires the all-angle estimate $\Phi(\vartheta) - \Phi(\pi) \geq c(x, \lambda) \sin^2 \vartheta$ that the observer-robust scan exhibits numerically but for which no closed form is yet known. *Granting the shape-reduction and angular-dominance estimates*, $\Phi > 0$ whenever $M_0(x) > b_1(x) \lambda + C_{\text{shape}}(x) \lambda^2$, the least positive root of equality being a lower bound $\underline{g}_{\text{rig}}(x)$,

$$\underline{g}_{\text{rig}}(x) = \sup \left\{ \lambda < \frac{1}{2}(1-x) : M_0(x) - b_1(x) \lambda - C_{\text{shape}}(x) \lambda^2 > 0 \right\} > 0, \quad (50)$$

which, with the closed-form kinematic ceiling (Prop. 5), would bracket the frontier between two analytic bounds, $\underline{g}_{\text{rig}}(x) \leq g(x) < \frac{1}{2}(1-x)$, the numerical lower bound $\underline{g}(x) \sim 0.2$ lying in between (and below the ceiling at every x) and the Newton–Kantorovich estimate above placing the matched shell within an explicit ball of a genuine solution at that amplitude. What this establishes outright is the quadratic onset (the frontier is a genuine shape-response effect, not a frozen-shape redshift) and reduces a two-sided rigorous frontier to the single interval-bounded estimate $C_{\text{shape}}(x)$ plus the angular-dominance inequality; the large anchor margin $M_0(x)$ and the analytic $\beta = \frac{1}{2}$ make this feasible, and we set it out as the concrete route.

The binding angle is closed-form at *linear* order, where the matched dipole $\sigma_1 - p_1 > 0$ binds the rear pole. At finite amplitude rear-dominance is a real-analytic inequality that the observer-robust scan exhibits numerically and that is decidable order-by-order (a degree- $2N$ polynomial inequality by Sturm/interval arithmetic, with the Schauder tail bound (53) on the embedding remainder), but not in elementary closed form, since the resummed embedding response is not algebraic in $\cos \vartheta$.

The full multipole-frontier solve is detailed with the numerical implementation (App. K).

D The linearized Darboux map at the anchor: rigidity, Schauder bound, and the dipole block

Lemma 2 asserts that the linearized first-junction map DF_0 at the Schwarzschild–Minkowski anchor is a bounded isomorphism off the $\ell = 1$ translation gauge. We prove it here, addressing the two points the claim turns on: (i) that the shape sector reduces to the round-sphere rigidity operator $\Delta_{S^2} + 2$ in the *Schauder* (not merely L^2) topology that the C^2 shell embedding requires, and (ii) that the dipole ($\ell = 1$) steering source (which lies in the *kernel* of that operator) is absorbed by a non-degenerate finite-dimensional lapse/tilt/shape block, rather than being applied on the kernel.

Setup and harmonic split. The first junction condition $[h_{ab}] = 0$ on the timelike worldtube $\Sigma : r = R(\tau, \Omega)$ requires the induced 3-metric to agree from both sides. Linearizing about the anchor (exterior exactly Schwarzschild, interior exactly flat, Σ the round sphere of areal radius R_0 , $x = 2m/R_0$), the perturbation carries three fields on S^2 : the normal shape displacement $u(\Omega)$ of Σ , the lapse/redshift perturbation (the $\tau\tau$ block), and the foliation tilt (the τA block). Correspondingly $[h_{ab}] = 0$ splits into three sectors, the angular-angular $[h_{AB}]$ (*shape*), the time–time $[h_{\tau\tau}]$ (*redshift*), and the time–angle $[h_{\tau A}]$ (*tilt*), each expanded in spherical harmonics $u = \sum_{\ell} u_{\ell}$.

Shape sector, $\ell \geq 2$: rigidity and the Schauder estimate. A normal displacement u of $S_{R_0}^2$ preserves the first fundamental form to first order iff

$$(\Delta_{S^2} + 2)u = \mathcal{S}[u_{\text{tan}}, \delta g_{AB}^+], \quad (51)$$

the inhomogeneity $\mathcal{S}$ being fixed by the tangential embedding field and the $\mathcal{O}(a)$ angular anisotropy of the exterior induced metric; this is the classical infinitesimal-bending equation for the sphere [59]. Its homogeneous operator is the round-sphere rigidity operator, with kernel the $\ell = 1$ harmonics: a rigid translation $\mathbf{a} \cdot \hat{n}$ is an $\ell = 1$ normal displacement and $(\Delta_{S^2} + 2)Y_1 = 0$ (Cohn–Vossen/Weyl rigidity [59]). Its eigenvalues $2 - \ell(\ell + 1)$ have magnitude ≥ 4 for $\ell \geq 2$ (and 2 for $\ell = 0$), so $\Delta_{S^2} + 2$ is invertible on the $\ell \neq 1$ subspace. The bound that controls the matched shell is, however, *not* the L^2 spectral statement $\|(\Delta_{S^2} + 2)^{-1}\|_{L^2} = \frac{1}{2}$ but a Schauder estimate: as a second-order elliptic operator on the compact manifold S^2 , $\Delta_{S^2} + 2$ restricted to the closed subspace $\ell \neq 1$ (equivalently, on fields satisfying the gauge condition $P_1 u = 0$, with P_1 the L^2 -projection onto the $\ell = 1$ kernel) is an isomorphism $C^{2,\alpha}(S^2) \rightarrow C^{0,\alpha}(S^2)$,

$$\|u\|_{C^{2,\alpha}} \leq C_S \|(\Delta_{S^2} + 2)u\|_{C^{0,\alpha}}, \quad P_1 u = 0, \quad (52)$$

the constant C_S depending on the round metric, the Hölder exponent α , and the spectral gap [59]. The constant is moreover *explicit in structure*. On the pure shape sector $\ell \geq 2$ the gap is

$|2 - \ell(\ell + 1)| \geq 4$ (sharper than the $\frac{1}{2}$ off-kernel bound, which is the $\ell = 0$ value $|\mu_0|^{-1}$), so the L^2 inverse there is $\frac{1}{4}$; the elliptic gain $\|\Delta_{S^2} u\|_{L^2} \leq \frac{3}{2} \|(\Delta_{S^2} + 2)u\|_{L^2}$ (since $\ell(\ell + 1)/|2 - \ell(\ell + 1)| \leq \frac{3}{2}$ for $\ell \geq 2$) gives $\|u\|_{H^2} \leq \sqrt{\frac{1}{16} + \frac{9}{4}} \|(\Delta_{S^2} + 2)u\|_{L^2} = 1.5207 \|(\Delta_{S^2} + 2)u\|_{L^2}$, and absorbing the lower-order term in the closed-manifold Schauder estimate bounds

$$C_S \leq C_0(\alpha) (1 + 5.391 C_{\text{Sob}}), \quad (53)$$

with $C_0(\alpha)$ the round-sphere Schauder constant of $\Delta_{S^2} + 2$ and C_{Sob} the $H^2(S^2) \hookrightarrow C^0(S^2)$ embedding constant (finite since $2 > \frac{n}{2} = 1$); C_S is thus reduced to tabulated round-sphere elliptic data, the single input the rigorous frontier lower bound (50) needs through $C_{\text{shape}}(x) \sim C_S^2$ (through the spectral gap and the H^2 constant). Estimate (52) is what yields the C^2 shell embedding, hence the continuity of $[K_{ab}]$ and of S_{ab} that Theorem 2 uses; it replaces the L^2 inverse bound quoted in Lemma 2.

Dipole sector, $\ell = 1$: the steering source is off the kernel. Because the $\ell = 1$ harmonic lies in $\ker(\Delta_{S^2} + 2)$, the shape sector alone cannot fix the dipole response; the full first-junction map, including lapse and tilt, must be used. For the axisymmetric steering source the three first-junction conditions (the constant and $\cos \vartheta$ parts of $[h_{\tau\tau}]$ and the $\sin \vartheta$ part of $[h_{\tau\vartheta}]$) are affine in the three dipole unknowns (ρ_1, A_{t1}, w_1) (shape dipole, exterior lapse dipole, foliation tilt; the interior lapse dipole is fixed to zero by $[h_{\tau\tau}]^{(0)} = 0$). The linearized map is *block-triangular*:

- the $\sin \vartheta$ part of $[h_{\tau\vartheta}]$ is set by the foliation tilt with unit leading coefficient, $\partial[h_{\tau\vartheta}]^{\sin}/\partial w_1 = 1$, the remaining tilt couplings being $\mathcal{O}(a)$ (vanishing at the anchor): the tilt *decouples*;
- the residual 2×2 block $(\rho_1, A_{t1}) \times \{[h_{\tau\tau}]^{\text{const}}, [h_{\tau\tau}]^{\cos}\}$ carries the redshift Jacobian. Its $\cos \vartheta$ row is closed-form: the areal-shape match responds to the shape and lapse dipoles through

$$\frac{\partial[h_{\tau\tau}]^{\cos}}{\partial \rho_1} = -\frac{x}{1-x}, \quad \frac{\partial[h_{\tau\tau}]^{\cos}}{\partial A_{t1}} = -2\sqrt{1-x}, \quad (54)$$

both nonzero on $x \in (0, 1)$ (and underwritten by $\partial_R[\sqrt{f}] = m/(R_0^2\sqrt{1-x}) \neq 0$, $1/\sqrt{1-x} \neq 0$).

The $\mathcal{O}(a)$ steering source has only two components: the $\cos \vartheta$ part of $[h_{\tau\tau}]$ and the $\sin \vartheta$ part of $[h_{\tau\vartheta}]$. The constant part of $[h_{\tau\tau}]$ carries *no* $\mathcal{O}(a)$ source: a $\cos \vartheta$ shape feeds the $\ell = 0$ jump of the quadratic induced metric only through the $\langle \cos^2 \vartheta \rangle$ self-coupling, so $[h_{\tau\tau}]^{\text{const}} = \mathcal{O}(a^2)$ and its derivatives $[h_{\tau\tau}]^{\text{const}}_{,\rho_1}, [h_{\tau\tau}]^{\text{const}}_{,A_{t1}} = \mathcal{O}(a)$. Because the $\cos$ row (54) and the tilt row ($J_{33} = 1$ to $\sim 10^{-5}$) are independent and nonzero for all $x \in (0, 1)$, the linearized junction is *surjective onto this two-dimensional source*: a matched accelerating shell exists at $\mathcal{O}(a)$ *in closed form, unconditionally*. Killing the residual $\mathcal{O}(a^2)$ constant jump reduces, by Lyapunov–Schmidt on the one-dimensional cokernel, to the non-vanishing of the single scalar bifurcation coefficient

$$D(x) := \frac{\det J(x)}{a} = -2\sqrt{1-x} \frac{[h_{\tau\tau}]^{\text{const}}_{,\rho_1}}{a} + \frac{x}{1-x} \frac{[h_{\tau\tau}]^{\text{const}}_{,A_{t1}}}{a}. \quad (55)$$

The two const-row coefficients are the $\mathcal{O}(a^2)$, $\langle \cos^2 \vartheta \rangle$ self-couplings of the induced-metric jump, and are elementary. Explicitly, the $\tau\tau$ jump is built from the exterior redshift $f(R) = 1 - 2m/R$ and the time normalization A_t ; with the dipole shape $R(\vartheta) = R_0(1 + a\rho_1 \cos \vartheta)$ and the exterior redshift seed $A_t = (1-x)^{-1/2} + aA_{t1} \cos \vartheta$, the only $\ell = 0$ contribution at $\mathcal{O}(a^2)$ is the $\cos^2 \vartheta$ self-coupling of the $\cos \vartheta$ dipole, projected by $\langle \cos^2 \vartheta \rangle = \frac{1}{3}$. The resulting Legendre $\ell = 0$ part of $[h_{\tau\tau}]$ is a one-dimensional angular integral of explicit Schwarzschild quantities,

$$\frac{1}{a} \frac{\partial[h_{\tau\tau}]^{\text{const}}}{\partial \rho_1} = \frac{2-x}{3(1-x)} = \frac{1+s^2}{3s^2}, \quad \frac{1}{a} \frac{\partial[h_{\tau\tau}]^{\text{const}}}{\partial A_{t1}} = \frac{x}{3} - \frac{4x}{3\sqrt{1-x}} - \frac{1}{3} = \frac{4s^2 - s^3 - 4}{3s}, \quad (56)$$

with $s = \sqrt{1-x}$. The reduction producing (56) is the order- a^2 expansion of the matched $\tau\tau$ induced metric, carried out as follows. The shell's unit timelike tangent is the seed time direction ξ (exterior $\xi^+ = A_t \partial_t$, interior $\xi^- = \partial_T$) Gram–Schmidt-orthogonalized against the shell's spatial tangent $\partial_\vartheta \Phi$, so the induced lapse on each side is

$$h_{\tau\tau} = g(\xi, \xi) - \frac{(g(\xi, \partial_\vartheta \Phi))^2}{g(\partial_\vartheta \Phi, \partial_\vartheta \Phi)}, \quad (57)$$

the orthogonalization denominator carrying the $1/(1-x)$ and $1/\sqrt{1-x}$ redshift factors. Expanding both sides of (57) to $\mathcal{O}(a^2)$ with the dipole shape and seed above and the rocket redshift $2H = f + 2ar \cos \vartheta$, the $\mathcal{O}(a)$ parts reproduce the cos-row (54), while the $\ell = 0$ part of the $\mathcal{O}(a^2)$ jump is the $\langle \cos^2 \vartheta \rangle = \frac{1}{3}$ self-coupling of the cos ϑ dipole, with three contributions: the curvature of the exterior redshift ($R_0^2 f''(R_0) = -2x$), the areal/shape-gradient response, and the orthogonalization denominator of (57). Collecting these with the anchor data $f(R_0) = 1 - x$, $R_0 f'(R_0) = x$ gives (56). The order-by-order series, an independent evaluation from the exact Kerr–Schild geometry (App. K), and the elementary sign argument below all return the same closed form. Substituting (56) into (55) collapses it, by elementary algebra, to the closed form

$$D(x) = \frac{s^5 - 6s^4 - s^3 + 6s^2 - 4}{3s^3} = \frac{s^2}{3} - 2s - \frac{1}{3} + \frac{2}{s} - \frac{4}{3s^3}, \quad s := \sqrt{1-x} \quad (58)$$

under the spherical-harmonic ($L^2(S^2)$, $\ell = 0/\ell = 1$) projection onto the cokernel that the $\Delta_{S^2} + 2$ decomposition above uses. Its sign, magnitude, and monotonicity are settled by a single elementary factorization, with no root-isolation step. Writing $D = N(s)/(3s^3)$ with $N(s) := s^5 - 6s^4 - s^3 + 6s^2 - 4$ and $3s^3 > 0$ on $(0, 1]$, the bound and the sign come together in one step:

$$D(x) + \frac{4}{3} = \frac{s^5 - 6s^4 + 3s^3 + 6s^2 - 4}{3s^3} = \frac{(s-1)Q(s)}{3s^3}, \quad Q(s) := s^4 - 5s^3 - 2s^2 + 4s + 4. \quad (59)$$

On $(0, 1)$ one has $s - 1 < 0$ and $3s^3 > 0$, so $D + \frac{4}{3} < 0$ exactly when $Q > 0$. Here $Q(0) = 4$, $Q(1) = 2$, and $Q''(s) = 12s^2 - 30s - 4 < 0$ on $[0, 1]$, so Q' is strictly decreasing; since $Q'(0) = 4 > 0 > Q'(1) = -11$, Q' has a *single* zero in $(0, 1)$, Q is unimodal (increasing then decreasing), and its minimum on $[0, 1]$ is attained at an endpoint, $\min_{[0,1]} Q = \min\{4, 2\} = 2 > 0$. Hence

$$D(x) < -\frac{4}{3} < 0 \quad \text{strictly for all } x \in (0, 24/25) \quad (\text{indeed } (0, 1)), \quad |D(x)| \geq \frac{4}{3}, \quad (60)$$

the bound $\frac{4}{3}$ approached only as $x \rightarrow 0^+$ ($s \rightarrow 1$); the strict sign is a corollary of the bound.

Monotonicity is one further elementary positivity: from $D(s) = \frac{s^2}{3} - 2s - \frac{1}{3} + \frac{2}{s} - \frac{4}{3s^3}$,

$$s^4 \frac{dD}{ds} = \frac{2}{3}s^5 - 2s^4 - 2s^2 + 4 = 2(s^2 + 2)(1 - s^2) + \frac{2}{3}s^5 > 0 \quad \text{on } (0, 1], \quad (61)$$

both summands nonnegative and the second strictly positive, so D increases with s ; as $s = \sqrt{1-x}$ decreases in x and $D < 0$, $|D| = -D$ increases strictly monotonically as x rises from 0 to 24/25. Hence the exact matched accelerating shell exists and is locally unique for every $a > 0$ and $x \in (0, 24/25)$: *Theorem 2 is unconditional at small amplitude.* (The determinant $\det J = a D(x)$ vanishes at the strict anchor only through the a prefactor, the absence of a steering source there, not through D .) The $L^2(S^2)$ projection (58) used here gives the sharp $|D| \geq \frac{4}{3}$; a least-squares-over- ϑ junction instead aliases the $\cos^2 \vartheta$ content and reports a larger $|D|$, but never zero (App. K). Either projection gives $D \neq 0$, so the $\ell = 1$ closure holds. The order-by-order derivation and an independent evaluation from the exact Kerr–Schild geometry both return the closed form (58). The $\ell = 0$ monopole block is the single scalar redshift/areal match with Jacobian $\partial_R[\sqrt{f}] \neq 0$, non-degenerate at the anchor.

Existence by Lyapunov–Schmidt. The harmonic-sector facts above assemble into a genuine nonlinear existence statement, not merely linearized solvability.

Proposition 9 (Local existence of the accelerating shell). *Fix $x_\star \in (0, 24/25)$. Write the axisymmetric first-junction defect as a map*

$$F : C^{2,\alpha}(S^2) \times \mathbb{R}^2 \times \mathbb{R} \longrightarrow C^{0,\alpha}(S^2) \times \mathbb{R}^3, \\ F(Y, (A_{t1}, w_1), a) = ([h_{AB}], P_{\ell=0}[h_{\tau\tau}], P_{\ell=1}^{\cos}[h_{\tau\tau}], [h_{\tau\vartheta}]^{\sin}),$$

in the normal shape displacement Y , the exterior-lapse and tilt dipoles (A_{t1}, w_1) , and the steering amplitude a (the interior lapse fixed by $[h_{\tau\tau}]^{(0)} = 0$); the codomain collects the shape block and the gauge-fixed finite-dimensional redshift/tilt projections that carry the obstruction. In this axisymmetric reduction the symmetric-2-tensor angular defect $[h_{AB}]$ is the single conformal shape-defect scalar in $C^{0,\alpha}(S^2)$ (its trace-free magnetic-parity component absent by axisymmetry); the tensorial S^2 map is the non-axisymmetric case of Cor. 6. Then F is C^1 near the anchor with

$F = 0$ there; its shape differential is $D_Y F(0) = \Delta_{S^2} + 2$, a Fredholm operator of index 0 that is a $C^{2,\alpha} \rightarrow C^{0,\alpha}$ isomorphism off the $\ell = 1$ kernel (Schauder estimate (52)); the $\ell = 0$ redshift block is invertible ($\partial_R[\sqrt{f}] \neq 0$); and the $\ell = 1$ block is surjective onto the $\mathcal{O}(a)$ steering source through the independent tilt row ($\partial[h_{\tau\vartheta}]^{\text{sin}}/\partial w_1 = 1$) and $\cos\vartheta$ redshift row (54). The single residual scalar condition (the $\ell = 0$ constant part of $[h_{\tau\tau}]$, generated at $\mathcal{O}(a^2)$ by the $\langle \cos^2\vartheta \rangle$ self-coupling) has Lyapunov–Schmidt reduced derivative $aD(x_*)$, which vanishes at the anchor only through the prefactor a and is nonzero on each slice $a = \text{const} > 0$ (60). The anchor is thus a bifurcation point; rescaling the residual by a , the implicit function theorem yields on each such slice a locally unique $C^{2,\alpha}$ matched shell with $[h_{ab}] = 0$ exactly, a branch parametrized by small $a > 0$. Its surface stress S_{ab} is C^1 in a , so the worst-observer margin Φ of Theorem 2 satisfies $\Phi = \delta_0 + \mathcal{O}(a, \mu) > 0$ on an open neighborhood of the anchor: the accelerating warpsell exists and is admissible there.

Proof. The C^1 regularity and the block structure of $D_Y F(0)$ are the harmonic split above: (51)–(52) (shape), the redshift Jacobian (monopole), and the block-triangular dipole map with rows (54) (dipole). The $\ell = 1$ kernel of $\Delta_{S^2} + 2$ is absorbed by the finite (A_{t1}, w_1) block, leaving the one-dimensional cokernel (55); its closed form (58) and the sign bound (60) give $D(x_*) \neq 0$. The reduced $\ell = 0$ residual $\phi(w, a)$ has linearization $\partial_w \phi(0, a) = \det J = aD(x_*)$, which vanishes at the anchor only through the prefactor a ; the correct desingularization is to apply the implicit function theorem not to ϕ but to the rescaled residual $\tilde{\phi}(w, a) := \phi(w, a)/a$, which is C^1 across $a = 0$ with $\partial_w \tilde{\phi}(0, 0) = D(x_*) \neq 0$, so the IFT yields a locally unique branch $w(a)$ and closes the match. Continuity of S_{ab} in a then propagates the strict anchor margin δ_0 of Lemma 1. $\square$

This is the precise content of Lemma 2, and it makes Theorem 2 unconditional (perturbatively) for every small $a > 0$.

An arbitrary acceleration axis. The closure above is stated for an axisymmetric steering source, but at small amplitude it holds for an arbitrary acceleration *direction* by rotational covariance of the anchor.

Corollary 6 (Per-cut existence for any acceleration axis). *Fix $x_* \in (0, 24/25)$. At the static Schwarzschild–Minkowski anchor the linearized first-junction map is the round-sphere operator $\Delta_{S^2} + 2$, whose $\ell = 1$ kernel is the three-dimensional space of rigid translations, and the block-triangular dipole map of Prop. 9 (shape, exterior-lapse, and foliation-tilt dipoles) is $SO(3)$ -equivariant, the anchor being rotationally invariant. For a general small steering source (an arbitrary proper-acceleration three-vector a^i in place of a fixed axis) the closure follows by rotational covariance: the bifurcation source rotates with a^i , so for any fixed direction one aligns the axis with $\hat{z}$ and applies the axisymmetric reduction of Prop. 9, whose single residual condition is the rescaled scalar $\det J = aD(x)$ with $D(x) \neq 0$ on $(0, 24/25)$ (60). The reduced bifurcation map sends the three-dimensional $\ell = 1$ translation kernel to the one-dimensional $\ell = 0$ cokernel (the constant part of $[h_{\tau\tau}]$, Prop. 9), so it is not an $\ell = 1$ endomorphism; its $SO(3)$ -covariant form is the invariant $D(x)a^i$ along the acceleration axis, the equivariant $\ell = 1 \rightarrow \ell = 0$ part vanishing at the anchor by Schur and the scalar $D(x)$ entering at the rescaled $\mathcal{O}(a^2)$ order through the $\langle \cos^2\vartheta \rangle$ self-coupling. The $\ell = 1$ closure therefore holds for every direction a^i : at each retarded cut, in the momentary rest frame, a locally unique admissible accelerating shell exists for an arbitrary small three-dimensional steering source, not only an axisymmetric one.*

The finite-amplitude non-axisymmetric solve (where the $\ell \geq 2$ multipole response is no longer a single Legendre tower but a tensorial coupling on S^2) and the assembly of the per-cut shells into a single C^2 worldtube for a *rotating* acceleration axis remain future work (Sec. 15).

E Assembly of the time-evolved spacetime (proof of Proposition 6)

We prove Proposition 6: for a slow, fixed-axis, sub-horizon burn the per-cut admissible snapshots of Sec. 8.2 assemble into a single time-evolved C^2 worldtube, conditional on the compatibility identity (64).

Proof of Proposition 6. After Schur-eliminating the $\ell \leq 1$ lapse/tilt gauge block (non-degenerate throughout the burn by Lemma 4, $\det J_{\text{dyn}} = aD(x)[1 + \mathcal{O}(\varepsilon)] \neq 0$ with $|D| \geq \frac{4}{3}$) the reduced first-junction condition on the normal shape $Y \in C^{2,\alpha}(S^2)$ reads

$$(\Delta_{S^2} + 2)Y = \mathcal{N}(Y, Y_u; \lambda, u) + \sigma(\lambda, u), \quad (62)$$

with steering source $\sigma = \mathcal{O}(\lambda)$ (the dipole row of Lemma 2) and $\mathcal{N}$ the subordinate part: by Lemma 3 the retarded derivative Y_u enters $\mathcal{N}$ at angular *differential* order ≤ 1 and the quadratic

embedding nonlinearity at order ≤ 2 with the *same* leading symbol as the static case, so $\mathcal{N} : C^{2,\alpha} \times C^{1,\alpha} \rightarrow C^{0,\alpha}$ is smooth with $\mathcal{N}(0, 0; 0, u) = 0$, and $(\Delta_{S^2} + 2)^{-1} : C^{0,\alpha}(S^2) \rightarrow C^{2,\alpha}(S^2)$ is bounded off the $\ell = 1$ kernel (the Schauder estimate (53), gaining two angular derivatives). Equation (62) is the *shape* block: it fixes Y given (Y_u, λ, u) , but to propagate it as an evolution one must *solve for* Y_u , and that closure comes not from (62) but from the higher-mode retarded time-rows. Subordinate differential order is *not* lower harmonic degree: a term with one angular derivative of Y_u still carries arbitrary $\ell \geq 2$ content, so after the $\ell \leq 1$ Schur elimination of Lemma 4 the higher-mode projection of the retarded junction

$$R_{\geq 2}(Y, Y_u; \lambda, u) := P_{\ell \geq 2}([h_{uu}], [h_{uA}])_{\text{red}} = 0 \quad (63)$$

is a genuine system of equations in Y_u , not a consequence of (62). The persistence closes *provided* (63) is solvable for Y_u by a bounded, locally Lipschitz map $Y_u = G(Y; \lambda, u) : C^{2,\alpha}(S^2) \rightarrow C^{2,\alpha}(S^2)$ with no loss of angular derivative, the over-determined $[h_{uA}]$ row carrying no cokernel obstruction. Lemma 3 supplies the no-derivative-loss *budget* (the elliptic gain compensates the single Y_u derivative) and for a fixed-axis (non-rotating) burn the solvability itself follows by equivariance. An axisymmetric source about the burn axis makes the entire junction map $SO(2)$ -equivariant, so an admissible matched worldtube may be sought within the axisymmetric invariant subspace (existence, not uniqueness, is at issue): the shape $Y = Y(\vartheta) \in C_{\text{ax}}^{2,\alpha}(S^2)$ (∂_φ -free), the worldtube embedding $X^\mu(u, \vartheta)$ with no azimuthal motion ($\partial_u X^\varphi = \partial_\vartheta X^\varphi = 0$), and the interior slicing $T_- = T(u) + \beta(\vartheta)$ with tilt $B_\vartheta = \partial_\vartheta \beta$. On this subspace the time-angle defect $[h_{uA}]$ is a one-form with $[h_{u\varphi}] \equiv 0$ (the angular metric is diagonal and nothing moves in φ): in the Hodge split of axisymmetric one-forms into the gradient class $\partial_\vartheta(\cdot) d\vartheta$ and the axial class $\propto d\varphi$, only the gradient class is present on both the data and the unknown $B = d\beta$, so the axial (magnetic) sector (the sole candidate cokernel of $[h_{uA}]$) is empty and the gradient row is square. The two surviving rows are lower-triangular. In the constant-retarded graph gauge $r = R + Y$ the $-2 du dr$ term of (3) gives $[h_{uu}] = -2Y_u + \mathcal{O}(\lambda, \varepsilon)$, so $[h_{uu}] = 0$ fixes $Y_u \in C_{\text{ax}}^{2,\alpha}$ *algebraically*, with $\partial_{Y_u}[h_{uu}] = -2$ independently of the burn content (the interior re-slicing $T_- = T(u) + \beta(\vartheta)$ enters the $[h_{uu}]$ jump only at the subordinate $\mathcal{O}(\lambda, \varepsilon)$ already displayed, shifting the value but not this nonzero leading coefficient, which is all the algebraic solvability uses), free of the tilt and carrying no angular derivative of Y_u (hence no loss). The polar row then reads $-(f + 2aR \cos \vartheta) \partial_\vartheta \beta = \Lambda(Y_u, Y; \lambda, u)$, with diagonal coefficient $\partial_{B_\vartheta}[h_{u\vartheta}] = -(f + 2aR \cos \vartheta)$ the interior timelikeness factor (47), bounded below by $\inf(f - 2aR) > 0$ through the sub-horizon regime and degenerating only at the rear pole $\vartheta = \pi$ as $f - 2aR \rightarrow 0$ (Prop. 5), and right side $\Lambda \in C_{\text{ax}}^{1,\alpha}$ the already-fixed Y_u at angular order ≤ 1 . Built from the pole-regular axisymmetric Y_u, Y , the datum Λ is a smooth axisymmetric one-form component and so vanishes at $\vartheta = 0, \pi$; dividing by the nonvanishing factor and integrating the first-order ϑ -equation, the integration restoring the one angular derivative ($\partial_\vartheta \beta \in C^{1,\alpha} \Rightarrow \beta \in C_{\text{ax}}^{2,\alpha}$), gives a β with $\partial_\vartheta \beta \rightarrow 0$ at the poles, unique once fixed by the gauge $\beta(0) = 0$ (an additive interior time shift). The triangular system thus has a unique, bounded, locally Lipschitz solution $(Y_u, B_\vartheta) = \mathcal{G}(Y; \lambda, u)$ on $C_{\text{ax}}^{2,\alpha}(S^2)$ with inverse norm $\lesssim (\inf(f - 2aR))^{-1}$ and no loss of angular derivative on Y_u , its Y_u -component the map G of (70) (here on the axisymmetric subspace). The one-form $B = B_\vartheta d\vartheta$ is closed automatically on the axisymmetric subspace ($dB = \partial_\vartheta B_\varphi - \partial_\varphi B_\vartheta = 0$) and pole-regular, so $B = d\beta$ with β a single-valued interior time tilt and the cavity stays exactly flat. It remains to show that the time-row solver $\mathcal{G}$ is consistent with the *full* first junction $[h_{ab}] = 0$, not only its time rows. Here the structure helps. The angular block carries *no* retarded derivative ($h_{AB} = g_{\mu\nu} \partial_A X^\mu \partial_B X^\nu$ involves only angular tangents) so on each constant- u cut $[h_{AB}] = 0$ is the *same* elliptic shape problem as the static existence theorem at the instantaneous $(m(u), a(u))$: the round-sphere rigidity reduction with the closed-form $\ell = 1$ closure $D(x) \neq 0$ (Lemma 2). The shape match therefore holds at every cut *unconditionally* at small amplitude, with $Y(u, \cdot) = Y_\star(u, \cdot)$ the per-cut matched shape; there is no shape *evolution* to propagate. What the dynamical problem adds is a *compatibility* between the two ways the retarded derivative enters: the burn rate Y_u fixed by the energy row $[h_{uu}] = 0$ (graph gauge $\partial_{Y_u}[h_{uu}] = -2$) must coincide with the u -derivative $\partial_u Y_\star$ of the per-cut elliptic shape:

$$Y_u|_{[h_{uu}]=0} = \partial_u Y_\star. \quad (64)$$

This compatibility is *not* a contracted-Codazzi propagation: the contracted Codazzi balance (20) presupposes the full $[h_{ab}] = 0$ and so cannot establish it. Its precise content is that the $\ell \geq 2$ part of the retarded energy row $[h_{uu}]$ (which carries *no* free junction parameter once the shape, tilt, and $\ell \leq 1$ lapse/time rows are solved) vanishes identically, equivalently that the exterior-induced 3-metric on Σ is isometrically embeddable in the flat interior (a Gauss-Codazzi class-one

consistency). The candidate leading $\mathcal{O}(a^2)$ $\ell = 2$ obstruction is shown (not merely estimated) to cancel. The worry an obstruction would express is that the per-cut shape match fixes the quadrupole ρ_2 to a value *conflicting* with the ρ_2^* the energy row demands. A direct computation removes that worry. Write the shell areal radius on the momentary-rest cut as

$$R(\vartheta) = R_0 [1 + a\rho_1 \cos \vartheta + a^2 \rho_2 P_2(\cos \vartheta)], \quad (65)$$

matched at equal areal radius to the flat interior and the Schwarzschild exterior $f(R) = 1 - 2m/R$. The two angular induced-metric jumps are then

$$[h_{\varphi\varphi}] = R^2 \sin^2 \vartheta - R'^2 \sin^2 \vartheta \equiv 0, \quad [h_{\vartheta\vartheta}] = \left(R^2 + \frac{R'^2}{f(R)} \right) - (R^2 + R'^2) = R'^2 \frac{1 - f(R)}{f(R)}, \quad (66)$$

the shared areal R^2 canceling in each ($R' = dR/d\vartheta$). Since $R' = -R_0 a \rho_1 \sin \vartheta + \mathcal{O}(a^2)$ carries the quadrupole only at $\mathcal{O}(a^2)$, $R'^2 = R_0^2 a^2 \rho_1^2 \sin^2 \vartheta + \mathcal{O}(a^3)$ is ρ_2 -free at $\mathcal{O}(a^2)$, whence

$$[h_{\vartheta\vartheta}]^{(a^2)} = R_0^2 a^2 \rho_1^2 \sin^2 \vartheta \frac{x}{1-x}, \quad \partial_{\rho_2} [h_{\vartheta\vartheta}]^{(a^2)} = \partial_{\rho_2} [h_{\varphi\varphi}] = 0, \quad (67)$$

with $x = 2m/R_0$; the quadrupole first reaches the jump at $\mathcal{O}(a^3)$ (the Kerr–Schild anisotropy, matched against the exact geometry in App. K). The shape sector therefore imposes *no* $\ell = 2$ constraint, so no shape-fixed value can conflict with ρ_2^* ; the quadrupole is left to the retarded energy row, whose static $\ell = 2$ part carries it with nonzero slope $\partial_{\rho_2} [h_{uu}]_{\ell=2} = -x$ and is met at $\rho_2 = \rho_2^*$. This removes the one mechanism by which the leading-order obstruction could survive, and the remaining $\ell = 2$ rows close it completely. Collect the four $\mathcal{O}(a^2)$ $\ell = 2$ first-junction rows in the unknowns $(Y_2, \rho_2, A_{t2}, w_2)$: the shape bending Y_2 , the areal quadrupole ρ_2 , the exterior-lapse quadrupole A_{t2} , and the regular tilt quadrupole w_2 in $\beta = a w_1 P_1 + a^2 w_2 P_2$. The round-sphere rigidity row $(\Delta_{S^2} + 2)Y_2 = \mathcal{S}_2$ fixes the bending (eigenvalue -4 at $\ell = 2$, the ρ_1^2 source $\mathcal{S}_2$ being ρ_2 -blind by the computation above); the polar tilt row $[h_{u\vartheta}]_{\ell=2}$ carries w_2 ; the retarded energy row $[h_{uu}]_{\ell=2}$ carries ρ_2 with slope $-x$ and is *free* of the lapse normalization A_{t2} ; and the seed-normalized lapse row $[h_{\tau\tau}]_{\ell=2}$ (57) *does* carry A_{t2} . In the column order $(Y_2, \rho_2, A_{t2}, w_2)$ the linearized rows are, with $s = \sqrt{1-x}$,

$$J = \begin{pmatrix} -4 & 0 & 0 & 0 \\ 0 & x/s & 0 & -1 \\ 0 & -x & 0 & 0 \\ 0 & -x/s^2 & -2s & 0 \end{pmatrix}, \quad \det J = 8x s = 8x\sqrt{1-x}, \quad (68)$$

read off directly (rows: rigidity, polar tilt, retarded energy, seed lapse). Because the energy row is A_{t2} -free while the lapse row is not, their (ρ_2, A_{t2}) minor is $2x\sqrt{1-x} \neq 0$: the two are independent conditions, not one row in two gauges. Hence the $\ell = 2$ Jacobian is non-degenerate on the whole physical range $x \in (0, 1)$ (the 24/25 window elsewhere is the independent surface-DEC threshold, not a determinant degeneracy). The determinant $\det J$ is built from the row *slopes* alone, so it is independent of the asserted $\mathcal{O}(a^2)$ retarded source: the energy row's ρ_2 -independent piece (the slow-burn graph-gauge retarded input, whose all-orders justification is the open identity (ii) of Sec. 15) shifts the *value* ρ_2^* but neither the non-degeneracy nor the non-obstruction. The unique solution is

$$\rho_2 = \rho_2^* = \frac{2(\rho_1^2 x - 2\rho_1 x - x + 1)}{3x}, \quad \rho_1 = \frac{1}{1 - \sqrt{1-x}}, \quad (69)$$

together with the closed-form (Y_2, A_{t2}, w_2) it determines. The candidate leading $\mathcal{O}(a^2)$ $\ell = 2$ *obstruction* to the compatibility identity (64) therefore *cancels*, gauge-completely, for the slow fixed-axis burn: the bending and the areal, lapse, and tilt quadrupoles are each fixed by a distinct independent row, with no residual $\ell = 2$ obstruction. What the computation settles is the absence of that obstruction and the nondegeneracy of the $\ell = 2$ rows, not the retarded evolution identity (64) in itself, whose graph-gauge retarded input stays conditional on the all-orders step (item (ii) of Sec. 15). Two items remain open and are the frontier on which Proposition 6 stays conditional: the all-orders ($\mathcal{O}(a^3)$ and beyond) energy-row identity, and the rotating-axis assembly. Granting (64) in full (its leading $\mathcal{O}(a^2)$ obstruction shown to cancel above), the per-cut shapes $Y_*(u, \cdot)$ (each the unconditional static elliptic match) and the energy-row burn rate $Y_u = \partial_u Y_*$ are consistent, so the per-cut admissible shells assemble into a single C^2 worldtube solving $[h_{ab}] = 0$ in full, and the contracted Codazzi momentum balance (20) is the consequent stress balance of the matched

solution (the absent magnetic-parity sector leaving it unobstructed). At the burn endpoints $a(u_0) = a(u_f) = 0$ the configuration is the static anchor. The determinant $\det J_{\text{dyn}} = a D(x)$ degenerates there only through the a prefactor, the removable $\mathcal{O}(a)$ bifurcation of Lemma 2: rescaling the $\ell = 1$ residual by a (as at the static anchor) leaves the reduced Jacobian $D(x)$ nonzero, *uniformly* nondegenerate through $a = 0$, so $\mathcal{G}$ and its Lipschitz constant extend continuously across the endpoints, $Y(u) \rightarrow 0$ as $a \rightarrow 0$, and the evolution (70) is uniformly Lipschitz (hence regular) on the *closed* interval $[u_0, u_f]$. This discharges (63) for a fixed-axis burn; the rotating-axis case, where the source charges the axial sector and the magnetic-parity row couples all $\ell \geq 2$ modes with no symmetry protection, is left open (Sec. 15). Thereby (63) holds, its solver $Y_u = G(Y; \lambda, u)$ making the persistence of the shape block (62) the explicit first-order Banach-space evolution

$$Y_u = G(Y; \lambda, u), \quad G : C_{\text{ax}}^{2,\alpha}(S^2) \rightarrow C_{\text{ax}}^{2,\alpha}(S^2) \text{ locally Lipschitz,} \quad (70)$$

uniformly in $u \in [u_0, u_f]$, with no derivative lost and *no* Nash–Moser iteration. Picard–Lindelöf in $C_{\text{ax}}^{2,\alpha}(S^2)$, with the static anchor $Y \equiv 0$ as datum at u_0 (where $a(u_0) = 0$), gives a unique C^1 -in- u solution $Y(u; \lambda, \varepsilon)$ on $[u_0, u_f]$ for λ, ε small; the C^2 burn data $(m(u), a(u))$ bootstrap it to C^2 in u . A Grönwall estimate gives $\|Y(u)\|_{C^{2,\alpha}} \leq C(\lambda + \varepsilon)$ uniformly on $[u_0, u_f]$, with C controlled by the Schauder constant C_S of (53) and the Schur bound $|D| \geq \frac{4}{3}$, and degrading as $\mu_\star^{-p}$ ($\mu_\star = \inf(f - 2aR)$, $p \in \{1, 2\}$) toward the rear-pole horizon. Thus the embedding $X(u, \Omega)$ is C^2 across the worldtube, so $[K_{ab}]$ and the exact retarded surface stress S_{ab} are continuous in $(\lambda, \varepsilon, u)$. At $(\lambda, \varepsilon) = (0, 0)$, S_{ab} is the static anchor with strict Type I surface-DEC margin $\delta_0 > 0$ (Lemma 1); by continuity there are $\lambda_\star, \varepsilon_\star > 0$ with the margin $> \frac{1}{2}\delta_0$ for $\lambda < \lambda_\star$, $\varepsilon < \varepsilon_\star$ throughout the burn. $\square$

F $\mathcal{O}(\mu)$ back-reaction of the radiating wall

On a *fixed* wall background the two-component stress (34) has worst-observer DEC margin $\rho_\star - p_t$ with radial-block eigenvalue $\rho_\star = \frac{1}{2}\rho(1 + \sqrt{1 + 4\mu/\rho})$ and p_t untouched by the radial null dust, so $d\rho_\star/d\mu > 0$: the superposed null dust appears to *raise* the margin (the discriminant $\Delta = \rho(\rho + 4\mu) > 0$ keeping it Hawking–Ellis Type I throughout, never Type IV). We test whether this frozen-background lemma survives the $\mathcal{O}(\mu)$ Einstein back-reaction of the radiation on the geometry, and resolve the sign density-conditionally: the first-variation formula and the lapse response are analytic, the negative *core* sign resting on the profile inequality $\frac{r}{2}\rho \partial\Phi'/\partial\mu > 1$ verified across the wall.

Theorem 6 (Lapse response of the radiating wall). *For the exact tangential-pressure wall ($p_r = 0$, $\Phi' = m(r)/[r(r - 2m(r))]$, $p_t = \frac{r}{2}\rho \Phi'$) carrying an outgoing null-dust amplitude $\mu \geq 0$ (energy density, radial pressure, and flux all $= \mu$ in the static frame), let the metric respond (the Tolman–Oppenheimer–Volkoff lapse with $p_r = \mu$). Then the lapse deepens,*

$$\frac{\partial\Phi'}{\partial\mu} = \frac{4\pi r}{1-x} + (\text{positive mass response}) \geq \frac{4\pi r}{1-x} \geq 4\pi r > 0, \quad (71)$$

uniformly on the local-compactness window $x = 2m/r < 4/5$.

The Type-I structure and the order of the flux effect are elementary. The two-component radial block $[\rho_\star^\mu, \mu_\star^\mu]$ has discriminant $\Delta = \rho(\rho + 4\mu) > 0$ (35), so it stays Hawking–Ellis Type I, with eigen-density $\rho_\star = \frac{1}{2}(\rho + \sqrt{\rho^2 + 4\rho\mu}) = \rho + \mu + \mathcal{O}(\mu^2)$: beyond the bare energy density μ the dust carries diagonally, the flux-coupling correction to $\rho_\star$ is $\mathcal{O}(\mu^2)$. The linear margin response thus has two pieces, the bare $+\mu$ of $\rho_\star$ competing with the *lapse* response of Theorem 6 acting through p_t , which signs as follows.

Proposition 10 (Density-conditional back-reaction sign). *The wall matter keeps $p_r = 0$, so its tangential pressure is set by its own anisotropic equilibrium $p_t = \frac{r}{2}\rho \Phi'$ with only the lapse responding to the added radiation energy; with $\rho_\star = \rho + \mu + \mathcal{O}(\mu^2)$ the worst-observer surface-DEC margin’s first variation is*

$$\left. \frac{\partial(\text{margin})}{\partial\mu} \right|_{\mu=0} = 1 - \frac{r}{2}\rho \frac{\partial\Phi'}{\partial\mu}, \quad (72)$$

with $\partial\Phi'/\partial\mu \geq 4\pi r/(1-x) > 0$ the lapse response of Theorem 6. Outgoing radiation therefore tightens the dominant energy condition ($\partial(\text{margin})/\partial\mu < 0$) wherever the lapse deepening dominates, $\frac{r}{2}\rho \partial\Phi'/\partial\mu > 1$, which holds across the dense wall core; and because $p_t \propto \rho$, the variation returns to $+1$ as $\rho \rightarrow 0$, so the sign reverses only in the vanishing-density edge skin. This

is the correct, density-conditional content of “outgoing radiation tightens the margin”: the frozen-background monotonicity is reversed in the core, with the pure-flux piece (above) a further order down, the precise crossover carrying the nonlocal lapse response rather than a clean local threshold. The first-variation formula (72), the lapse-response positivity $\partial\Phi'/\partial\mu \geq 4\pi r/(1-x) > 0$ (Theorem 6), and the $\mathcal{O}(\mu^2)$ order of the pure-flux piece are established by a direct expansion; the resulting sign $d(\text{margin})/d\mu < 0$ is density-conditional and is read from the exact adiabatic curvature: the margin slope is strongly negative through the dense wall interior, where $\frac{r}{2}\rho\partial\Phi'/\partial\mu > 1$, and neutralizes only in the $\rho \rightarrow 0$ edge skin (Fig. 4d).

Two naive self-consistent closures fail. (i) A generic generalized-Vaidya mass-ramp $M(u, r) = \bar{m}(u)\chi(r)$ carries $p_r \neq 0$, whose static radial-tension channel violates the DEC at the matter–vacuum edge (Prop. 7) as μ grows; its $(\hat{u}, \hat{r})$ block is the defective Hawking–Ellis Type II (DEC-saturating, $\Delta = 0$), not Type IV: the smoothstep pathology again. (ii) Feeding the null dust into a static constraint solve as a radial pressure $p_r = \mu\rho$ forces it into the wall’s hydrostatic equilibrium, which then demands large tangential tension $p_t < 0$ and fails DEC; but the through-wall radiation is a flux (radial momentum balanced by $\dot{m}$), not a static pressure the wall must support. Both controls show the $p_r = 0$ structure must be preserved.

The correct construction rewrites the exact $p_r = 0$ tangential wall $g = \text{diag}(-e^{2\Phi}, e^{2\Lambda}, r^2, r^2 \sin^2 \vartheta)$ in outgoing Eddington–Finkelstein form (the exact transformation $t = u + \int e^{\Lambda-\Phi} dr$),

$$ds^2 = -e^{2\Phi(r)} du^2 - 2e^{\Phi(r)+\Lambda(r)} du dr + r^2 d\Omega^2, \quad (73)$$

and lets it radiate *adiabatically*: a slow modulation $\omega R \ll 1$, $m(u) = m_0(1 - \epsilon \sin \omega u)$, so (Φ, Λ) acquire a slow u -dependence (linear in $m(u) - m_0$, exact to first order in the amplitude ϵ); the u -derivative generates a genuine outgoing null flux ($n^2 > 0$) while the matter stays $p_r = 0$ in its rest frame. Reading the *curvature-derived* stress from the exact curvature, the metric is a self-consistent Einstein solution ($\nabla_a T_{ab} \sim 10^{-16}$, the contracted Bianchi identity) and stays Type I; the worst-observer DEC margin (from the Type-I eigenvalues, robust where the optimizer’s boosted-frame search degenerates on the off-diagonal metric) is strictly positive across the whole frontier regime $x \leq 0.7$ over the directly tested radiation amplitudes (up to $\approx 9\%$ of the local matter density at $x = 0.7$; the quadratic-fit margin zero extrapolates to $\approx 20\%$, beyond the tested grid), and reduces to the static App. K wall as $\epsilon \rightarrow 0$.

The self-consistent calculation *corrects* the sign of the fixed-background lemma: in this adiabatic family, with the metric allowed to respond, *outgoing radiation tightens* the dominant energy condition through the dense wall core ($d(\text{margin})/d\mu < 0$ where the lapse deepening dominates, $\frac{r}{2}\rho\partial\Phi'/\partial\mu > 1$, Prop. 10; the sign neutralizes only in the $\rho \rightarrow 0$ edge skin), because the radiated energy *flux*, not a pressure, is what DEC constrains. The radiating wall therefore remains admissible, but the back-reaction sharpens rather than relaxes the margin in the core (Fig. 4d); the frozen-background monotonicity was an artifact of treating $\mu\ell$ as separately conserved on a fixed geometry. This closes the back-reaction *at adiabatic linear order* for the present construction; the residual (the time-dependent terms beyond the slow-modulation expansion, formally higher order in $(\epsilon, \omega R)$) awaits a fully dynamical evolution. The finite maneuver of Theorem 3, radiating $\approx 51\%$ over ~ 4 light-crossing times, has an effective $\omega R \sim 0.25$, at the edge of the strict-adiabatic $\omega R \ll 1$ regime, so the adiabatic closure is itself quasi-static there and the $\mathcal{O}(\mu^2)$ dynamical terms are not parametrically negligible during a saturating burn.

G Optimal-control derivation and momentum balance

Two computational records support Secs. 11 and 3.

Conservation law (Proposition 2). We evaluate dP_B^μ/du for the exact Kerr–Schild rocket two independent ways: (A) kinematically, from the point-rocket value $P_B^\mu = m(u)v^\mu(u)$ (v^μ the unit worldline velocity), differentiated in u ; and (B) as the radiated flux $\oint n^2 \hat{\ell}^\mu d\Omega$ with n^2 read from the Einstein tensor of the metric on the light cone. Path A never uses the stress-energy and Path B never uses the worldline momentum, yet they cancel, $dP_B^\mu/du + \oint n^2 \hat{\ell}^\mu d\Omega = 0$, to $\sim 10^{-15}$ in both the energy and momentum components: the balance is enforced by the field equations, not imposed.

Minimum-radiation optimum (Theorem 4). Writing $4\pi n^2 = \sum_\ell b_\ell P_\ell(\cos \vartheta)$, the moment identities read $-\dot{m} = b_0$ and $ma = -b_1/3$ (the thrust; $b_1 = -3ma < 0$), and bulk positivity is $\sum_\ell b_\ell P_\ell \geq 0$. We solve the finite-dimensional program over $\{b_\ell\}_{\ell \leq 8}$: (i) restricted to the dipole $\{b_0, b_1\}$, minimizing fuel b_0 at fixed thrust subject to positivity returns $b_0 = |b_1| = 3m|a|$ with the

emission vanishing at the forward pole: the saturated control law, exactly; (ii) on the full mode set with a convex *surrogate* for the radiated news energy, $b_0 + \eta \sum_{\ell \geq 2} \ell(\ell-1)b_\ell^2$ (a positive penalty rising with multipole order, not the exact Damour news functional), the higher multipoles fall as $\max_{\ell \geq 2} |b_\ell| \propto 1/\eta$ while $b_0 \rightarrow 3m|a|$, recovering the dipole; (iii) integrating the saturated control reproduces the Tsiolkovsky budget (14) to 10^{-14} . The bare ($\eta = 0$) program instead drives the emission toward a rear-pole spike ($\langle \cos \vartheta \rangle \rightarrow -1$), the singular profile that Theorem 4 excludes through gravitational-wave silence (Fig. 5).

H The geometric-control reading: hyperbolic norm, factor 3, and reachability

This appendix supplies the derivations behind Sec. 11.1.

The cost is the hyperbolic length. The worldline four-velocity $v^\mu(\tau)$ is a unit timelike curve, $v \cdot v = -1$, hence a curve on the hyperboloid $\mathbb{H}^3$; its tangent $\dot{v}^\mu = dv^\mu/d\tau$ is the proper-acceleration four-vector a^μ , orthogonal to v , with magnitude $|a| = \sqrt{a_\mu a^\mu} = \sqrt{g_{\mathbb{H}^3}(\dot{v}, \dot{v})}$ the hyperbolic speed (the induced metric on $\mathbb{H}^3$ from $\eta_{\mu\nu}$ is the hyperbolic metric of curvature -1). Thus $\int |a| d\tau = L_{\mathbb{H}^3}[v]$ is exactly the hyperbolic arclength. Along a saturated news-silent maneuver $d \log m/du = -3|a|$ (with u the worldline proper time on each cut), so $\log(m_0/m_f) = 3L_{\mathbb{H}^3}[v]$, Eq. (37); minimizing the radiated mass $J = m_0(1 - e^{-3L_{\mathbb{H}^3}})$ is equivalent to minimizing $L_{\mathbb{H}^3}$, whose minimizer between fixed endpoints is the hyperbolic geodesic of length $d_{\mathbb{H}^3}(v_0, v_1) = \text{arccosh}(-v_0 \cdot v_1)$, Eq. (38). The geodesics of $\mathbb{H}^3$ are the one-parameter boost subgroups, so the optimal velocity history is a single boost and the optimal emission is the saturated Damour dipole tracking it; any cornered (multi-direction) path is strictly longer by the triangle inequality on the negatively curved $\mathbb{H}^3$ (numerically, a single boost to rapidity 0.9 costs $3d_{\mathbb{H}^3} = 2.70$, while a sin-modulated detour reaching the same endpoint costs $\approx 3 \times 1.92$). This is a left-invariant control problem on $SO^+(3, 1)$ (Riemannian geodesy on $\mathbb{H}^3 = SO^+(3, 1)/SO(3)$): with $\Lambda(u) \in SO^+(3, 1)$ the boost frame, $\dot{\Lambda} = \Lambda \sum_i u_i K_i$ (K_i the boost generators) and cost $3 \int \|u\| d\tau$, whose Pontryagin/Lie–Poisson extremals are the boost geodesics [66, 67]. Note the *kinematic* (log-fuel) system is left-invariant, whereas the physical energy $\int (-\dot{m}) du = \int 3m\|u\| du$ carries the scalar m and is not literally a left-invariant length; the two share the same minimizers at fixed m_0 because J is monotone in $L_{\mathbb{H}^3}$.

The radiation moment cone and the factor 3. With $\hat{\ell}^\mu = (1, \hat{\Omega})$ the asymptotic null direction and $4\pi n^2 = b_0 + \vec{b} \cdot \hat{\Omega}$ the news-silent ($\ell \leq 1$) emission, the moments $Q^\mu = \oint n^2 \hat{\ell}^\mu d\Omega$ are, using $\oint d\Omega = 4\pi$, $\oint \hat{\Omega} d\Omega = 0$, $\oint \hat{\Omega}^i \hat{\Omega}^j d\Omega = \frac{4\pi}{3} \delta^{ij}$,

$$Q^0 = b_0 = -\dot{m}, \quad \vec{Q} = \frac{\vec{b}}{3}, \quad |\vec{Q}| = \frac{|\vec{b}|}{3} = m|a|, \quad (74)$$

the radiated momentum $\vec{Q}$ pointing along the exhaust and the thrust being the recoil $-\vec{Q} = m\vec{a}$; so $n^2 \geq 0$ for all Ω (i.e. $b_0 \geq |\vec{b}|$) is exactly $-\dot{m} \geq 3m|a|$, the control law (13). Dropping the news-silent restriction, an arbitrary $n^2 \geq 0$ lets Q^μ fill the future causal cone, $Q^0 \geq |\vec{Q}|$, with equality (factor 1) approached only in the collimated rear delta-emission limit; the broad dipole's $\frac{1}{3}$ in $|\vec{Q}| = |\vec{b}|/3$ is the entire origin of the factor 3, a property of the news-silent Robinson–Trautman class, not a general-relativistic floor.

Reachability. In velocity space the geodesics reach all of $\mathbb{H}^3$, so any rapidity $\eta < \infty$ is attained with $m_f = m_0 e^{-3\eta} > 0$; the null boundary $\eta \rightarrow \infty$ needs $m_f \rightarrow 0$ and is excluded. In four-momentum space the constraint is $P_B(u_0) - P_B(u_f) = \int Q du \in \overline{V_+}$ (future causal), a necessary condition that is *not* satisfied by every future-timelike target, so the momentum-space reachable set is a proper sub-cone.

The state-constrained maneuver: Pontryagin structure. The bounded-control problem is posed on the state manifold $\mathbb{H}^3 \times (0, 1)$ with state (v, x) , control $u \in \mathbb{R}^3$ (the rest-frame boost rate, $\|u\|_{\mathbb{H}^3} = |a|$), saturated news-silent dynamics

$$\dot{v} = \left(\sum_i u_i K_i \right) v, \quad \dot{x} = -3x\|u\|_{\mathbb{H}^3}, \quad (75)$$

(K_i the boost generators), the *moving frontier* mixed state–control constraint $C(x, u) = R\|u\|_{\mathbb{H}^3} - g(x) \leq 0$ (taking the frontier g to be C^1 , as the numerical $g(x)$ of Sec. 8

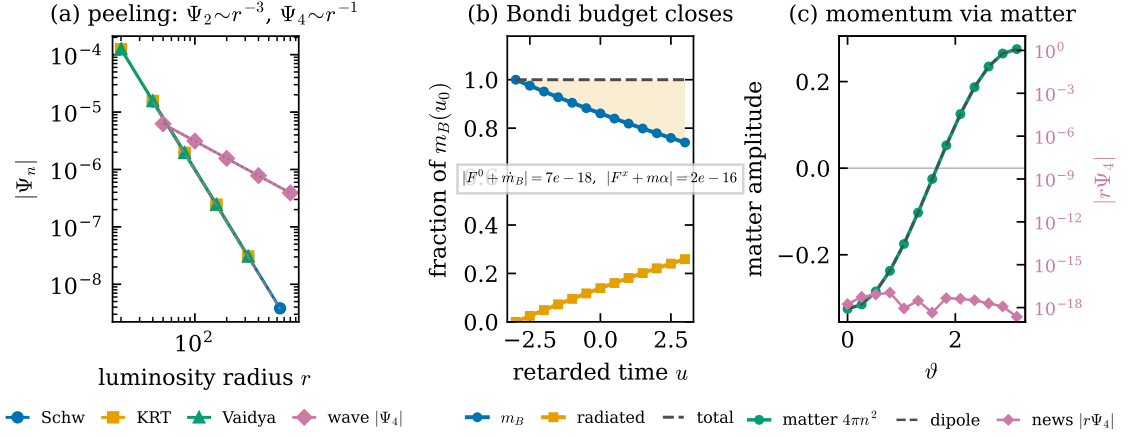


Figure 6. Numerical illustration of Theorem 1: the no-reactionless-steering law as a plot. (a) Newman–Penrose peeling: the Coulombic $\Psi_2 \sim r^{-3}$ on the Kinnersley rocket, Vaidya, and Schwarzschild, and the radiative $\Psi_4 \sim r^{-1}$ on a non-silent linearized-wave control, against reference slopes -3 and -1 . (b) the budget closes: on a saturating maneuver the Bondi mass $m_B(u)$ is strictly positive and monotone, the radiated energy $\int (-\dot{m})du$ accounts for the loss, and their sum is conserved, the curvature-extracted flux matching the mass-loss to the numerical floor ($|F^0 + \dot{m}_B| \sim 7e-18$, $|F^x + m\alpha| \sim 2e-16$). (c) the steering momentum is carried by the $\mathcal{O}(1)$ matter dipole $4\pi n^2(\vartheta)$ while the gravitational news $|r\Psi_4|$ (right, log axis) sits ~ 16 decades below it: momentum via matter, gravitational-wave silent.

supports; the theorem is the optimal-control structure for *any* such frontier), and endpoints $v(u_0) = v_0$, $v(u_f) = v_1$. Because $\partial C/\partial u = Ru/\|u\|_{\mathbb{H}^3} \neq 0$ on the active set, C is a *mixed* constraint of order zero (smoothed to $\frac{1}{2}(R^2\|u\|^2 - g^2)$ at $u = 0$); the decisive identity is the *length reduction* $x(\ell) = x_0 e^{-3\ell}$, fuel $= 3\ell$ in hyperbolic arclength $d\ell = \|u\|_{\mathbb{H}^3} du$.

Theorem 7 (State-constrained multi-burn maneuver). *With the control Hamiltonian $\mathcal{H} = -L + \langle p_v, (\sum_i u_i K_i)v \rangle_{\mathbb{H}^3} - 3xp_x\|u\|_{\mathbb{H}^3} - \nu C(x, u)$ ($\nu \geq 0$ the KKT multiplier), the Pontryagin maximum principle gives the following necessary structure, globally sufficient only for the bare objectives of (ii), where the length reduction collapses the problem to a single arc: (i) the costate is Lie–Poisson, $\dot{P} = \text{ad}_{u \cdot K}^* P$, and on every arc the control aligns with the velocity-costate, so the velocity path is an $\mathbb{H}^3$ geodesic (a single boost direction) between switches; (ii) for the bare objectives $L = 3\|u\|_{\mathbb{H}^3}$ (min log-fuel) and $L = 1$ (min-time) the problem is single-arc: min log-fuel is the unconstrained geodesic, $C_{\min} = 3d_{\mathbb{H}^3}(v_0, v_1)$ (38) (the frontier $\nu \equiv 0$ never binds, the cost being reparametrization-invariant; the radiated mass $m_0(1 - e^{-3d_{\mathbb{H}^3}})$ is its monotone transform, same minimizer), and min-time is the same geodesic traversed at the saturated frontier $\|u\|_{\mathbb{H}^3} = g(x)/R$ a.e. ($\nu > 0$), with $T_{\min} = \int_0^{d_{\mathbb{H}^3}} R/g(x_0 e^{-3\ell}) d\ell$ monotone in path length, so the geodesic minimizes time as well; (iii) genuine multi-arc optima arise only for an objective that breaks the length reduction (e.g. a quadratic dwell/effort penalty), as concatenations of interior geodesic arcs ($\nu = 0$, $\|u\| < g/R$) and boundary maximal-acceleration arcs ($\nu > 0$, $\|u\| = g/R$); the order-zero mixed constraint makes the costates continuous across junctions (no costate jump, no tangency conditions), with $\mathcal{H}$ continuous and complementary slackness $\nu C = 0$; the moving frontier binds early (large x , small g) and releases as the rocket radiates (x falls, g rises), the transversality on $\mathbb{H}^3$ fixing $p_v(u_f)$ from the endpoint.*

This is the precise sense in which warp-maneuver design is bounded-control geodesy on the Lorentz group, whose state-constrained optima include multi-arc interior-geodesic/boundary-acceleration switching. Under a conformally stationary (de Sitter static-patch) drift $|W| < 1$ the *min-time* problem deforms to Zermelo navigation with $\mathbb{H}^3$ -geodesics replaced by Randers–Finsler geodesics (Sec. 11.1), the kinematic ceiling tightening to $\lambda < \frac{1}{2}(1 - x - y)$, $y = \frac{\Lambda}{3}R^2$. The state-constrained version (frontier $|\dot{v}|_{\mathbb{H}^3} R \leq g(x(u))$ with $x(u)$ falling as the rocket radiates) is the multi-burn trajectory-optimization problem of Sec. 11.1.

I Peeling of the asymptotic hypotheses

Theorem 1 assumes Newman–Penrose peeling [28, 29, 30]. For the Kinnersley exterior the hypothesis is analytic: the metric (3) is algebraically special (a Robinson–Trautman pure-radiation spacetime whose repeated principal null direction $\ell = -(du)^\sharp$ is geodesic, shear-free, twist-free, and expanding) so by the generalized Goldberg–Sachs theorem for aligned pure radiation (the Ricci

tensor being null and aligned with ℓ) $\Psi_0 = \Psi_1 = 0$ and the surviving Weyl scalars carry the standard falloff $\Psi_2 \sim r^{-3}$, $\Psi_3 \sim r^{-2}$, $\Psi_4 \sim r^{-1}$ in the asymptotic Bondi tetrad, the Schwarzschild and Vaidya limits being its type-D and pure-monopole specializations. The extraction below illustrates this falloff and isolates the news-silence of the Damour dipole. On an outgoing null cone we build the asymptotic Minkowski null tetrad (ℓ, n, m) of App. A [$\ell = (1, \hat{n})$, $n = \frac{1}{2}(1, -\hat{n})$, $m = (0, e_{\hat{\theta}} + ie_{\hat{\phi}})/\sqrt{2}$] and form the Weyl tensor $C_{abcd} = R_{abcd} - \frac{1}{2}(g_{ac}R_{bd} - g_{ad}R_{bc} - g_{bc}R_{ad} + g_{bd}R_{ac}) + \frac{1}{6}R(g_{ac}g_{bd} - g_{ad}g_{bc})$ from the exact curvature (subtracting the Ricci parts, nonzero on the null-dust cones), then the five scalars $\Psi_0, \dots, \Psi_4$. Sampling $|\Psi_n|$ over luminosity radii and fitting $\log |\Psi_n|$ vs $\log r$ tests $\Psi_n \sim r^{-(5-n)}$ (slope $-(5-n)$). The Coulombic scalar peels as $\Psi_2 \sim r^{-3}$ on all three exact spacetimes: fitted slopes -3.000 (Kinnersley), -3.000 (Vaidya), -2.99 (Schwarzschild), with $|R| \lesssim 10^{-17}$. For the gravitational-wave-silent dipole the radiative Ψ_4 sits at the numerical floor ($\sim 10^{-21}$, reported, not fit); a deliberately *non*-silent linearized outgoing-wave control reproduces the radiative $\Psi_4 \sim r^{-1}$ (slope -1.00), so the near-zero dipole Ψ_4 is genuine silence rather than a coordinate artifact. Finally, on a saturating maneuver $m(u) = m_0 e^{-3 \int |a|}$ the Bondi mass $m_B(u)$ is strictly positive and monotonically decreasing, with the energy closure $\dot{m}_B = -F^0$ holding to $\sim 10^{-17}$ (Fig. 6b).

J Linear stability of the static anchor

We analyze the radial stability of the static Schwarzschild–Minkowski shell by the Poisson–Visser method [80, 81]; the radiating exterior has no autonomous potential, so acceleration enters only as a frozen-background redshift, closed by a timescale estimate. Promoting $R \rightarrow R(\tau)$, the junction reduces to $\dot{R}^2 + V(R) = 0$ with $V(R_0) = V'(R_0) = 0$ (the anchor is an equilibrium of the family $\sigma_0(R), p_0(R)$); stability requires $V''(R_0) > 0$, and V'' is affine in the equation-of-state slope $\eta \equiv dp/d\sigma|_0$. The decisive result is exact: the *realized* tangential-pressure wall has $\eta_{\text{real}}(x) = (dp_0/dR)/(d\sigma_0/dR)$ equal to the marginal value η_{crit} at which $V''(R_0) = 0$, for all x : its equation of state is the tangent to the static equilibrium family, so a radial displacement is a zero mode (neutral stability). Stiffer walls ($\eta > \eta_{\text{crit}}$) sit in a strict potential well, softer walls on a barrier (Fig. 2b); the static anchor stress matches App. B and an independent numerical Israel solve to $\lesssim 5 \times 10^{-3}$. The frozen-background rear-pole redshift $x \rightarrow x + 2\lambda$ raises η_{crit} monotonically with $\lambda = aR$, so under acceleration the (marginal) realized wall slips onto the barrier side, $\eta_{\text{real}} < \eta_{\text{crit}}(\lambda)$. Its growth time follows from the first integral: $\dot{R}^2 = -V$ gives $\ddot{R} = -\frac{1}{2}V'(R)$, so a barrier ($V'' < 0$) e-folds at rate $\sqrt{|V''|/2}$, i.e. $\tau_{\text{efold}} = \sqrt{2/|V''|}$ with *no* factor 2π (the corresponding stable-well oscillation *period* is $2\pi\sqrt{2/|V''|} = 2\pi\tau_{\text{efold}}$, not the barrier e-fold). Evaluated at the realized accelerated wall, the e-fold is short: $\tau_{\text{efold}} \approx \{0.19, 0.36, 0.51, 0.78, 1.04\}$ at $x = \{0.1, 0.2, 0.3, 0.5, 0.7\}$ for $\lambda = 0.1$ (and roughly halved at $\lambda = 0.2$), in units of the light-crossing time. The Tsiolkovsky burn $\tau_{\text{burn}} = \Delta\eta/a$ therefore outruns this static-anchor instability *only in a restricted corner*: at high compactness with a brief rapidity gain (e.g. $x = 0.7$, $\lambda = 0.2$, $\Delta\eta = 0.1$: $\tau_{\text{burn}} \approx 0.5 < \tau_{\text{efold}} \approx 0.6$), whereas for $x \lesssim 0.5$ even the briskest burn has $\tau_{\text{burn}} \gtrsim \tau_{\text{efold}}$, so the marginal realized wall does not by itself guarantee that the burn completes before the (frozen-background) mode develops. Moreover $\tau_{\text{burn}} = \Delta\eta/a$ is the square-pulse *lower bound* on the duration; the $\sin^2$ burn of Theorem 3 ($\langle \sin^2 \rangle = \frac{1}{2}$) lasts $\sim 2\times$ longer, so even the high- x corner is not guaranteed safe without active stabilization. Two facts contain this. First, the realized mode is *neutral* at $\lambda = 0$ (zero growth rate): a static wall drifts only secularly, not exponentially. Second, a slightly *stiffer* wall, $\eta > \eta_{\text{crit}}$, sits in a strict potential well and is linearly stable, and this costs *nothing* in admissibility, because the surface dominant-energy margin $\sigma_0 - p_0$ is junction-fixed (a function of x alone) and *independent* of η , so the stiffer wall carries the identical strictly-positive margin. The required stiffening is moreover causal: $\eta_{\text{crit}}(x) = \{0.021, 0.046, 0.079, 0.184, 0.427\}$ at $x = \{0.1, 0.2, 0.3, 0.5, 0.7\}$ all lie below the subluminal ceiling $\eta = 1$, and at the maneuver rear pole $x_{\text{eff}} \approx 0.46$ both the surface margin and the thick-wall across-wall factor $1 - m/[2(r - 2m)] \approx 0.79$ are bounded away from zero, so a small stiffening to strict stability keeps strict DEC. Strict stability and strict dominant energy are thus decoupled, with no trade-off; the remaining open question is the fully-dynamic, flux-coupled stability of the *radiating* shell. The interior is exactly flat: the tidal eigenvalues vanish, $K^\mu{}_\rho \equiv 0$ for $\lambda \in [0, 1)$, the Rindler lapse $1 - \lambda$ degenerating only as $\lambda \rightarrow 1$ (the Herglotz–Noether bound).

K Numerical results and reproduction

Whether a stress-energy satisfies the dominant energy condition is decided not by the field equations but by a pointwise optimization over all observers, which for an anisotropic, accelerated shell has no closed form; deciding admissibility is therefore an irreducibly computational complement to the analytic development of the main text, carried out observer-robustly [19, 84]

over all timelike and null observers, on the same footing as the boundary-cost null result of Ref. [18]. The bulk (Sec. 5) and the static thin shell (Sec. 7.1) pass; a *generic* smoothstep Kerr–Schild thickening instead fails the pointwise DEC in the radial gradients of the transition, reproducing the boundary cost of Ref. [18] (Fig. 4), so the positive content rests on the distributional thin shell (Lemma 1, Theorem 2) and the carefully modeled tangential-pressure matter of Sec. 9, not an arbitrary profile.

All numerical curvature, energy-condition, and junction computations reported in this paper are carried out with WARPAX [19], a separate observer-robust energy-condition framework (following the approach of WarpFactory [84]), used here solely to study the present theoretical construction numerically; we claim no part of that framework as a contribution of this work. Its configurations and tolerances accompany the implementation referenced in the Data Availability statement.

We implemented a finite-thickness radiative warpsell as a Kerr–Schild metric (42) with a smoothed mass profile $M(r, u) = m(u) S((r - (R - \delta/2))/\delta)$, S a smoothstep (flat cavity inside, $M = 0$; exact photon rocket outside, $M = m(u)$; a regularized radiating wall between) and scanned it with that observer-robust framework. The bulk behaves as designed (flat interior; exact null-dust exterior under steering). However, the *pointwise* dominant energy condition is *violated* in the radial gradients of the wall, worst at the cavity→matter inner edge, already in the static limit $\lambda = 0$ and at small x ; only the exact wall midpoint is marginally admissible (where the Kerr–Schild profile gives $p_r = -\sigma$, saturating DEC). This is precisely the *boundary cost* of Ref. [18], here reproduced in the radiative setting: a *generic smooth thickening does not inherit the admissibility of the distributional thin shell*. Note that the Kerr–Schild radial-mass profile produces radial tension rather than the thin shell’s tangential pressure, so this demonstrates that one natural smooth realization fails, not a universal no-go. The u -dependent (generalized-Vaidya) version of this ramp does *not* produce a rotating-energy Type IV stress: its outgoing-flux block is a defective Hawking–Ellis *Type II*, DEC-saturating like the exterior null dust, and a numerical Type-IV label is the eigenvalue-splitting artifact of that defective null block (App. A). The strict obstruction is thus the *static* tangential-tension channel, not a dynamical Type IV. Section 9 shows the complementary positive result: the *tangential-pressure* wall (the right matter model) is strictly DEC-admissible across its whole width and recovers the anchor for $2m/R < 4/5$.

This wall is moreover a genuine *self-consistent* anisotropic equilibrium, not merely a postulated stress: feeding $(\rho, p_r=0)$ into the numerical constraint solver and reading back the sourced Einstein tensor reproduces the assigned $(\rho, 0, p_t)$ to $\sim 10^{-4}$, the anisotropic (Bowers–Liang) hydrostatic-equilibrium residual vanishes to $\sim 10^{-8}$, the stress is Type I with worst-observer DEC margin > 0 , and the surface reduction recovers (σ_0, p_0) to $\lesssim 1\%$ as $\ell \rightarrow 0$. This removes the “postulated single fluid” caveat at the static level. A collisionless Einstein–Vlasov realization [60] of the $p_r=0$ wall exists only outside the photon sphere ($2m/r < 2/3$, circular timelike orbits); between $2/3$ and the elastic limit $4/5$ the appropriate microscopic matter is anisotropic *elastic* [61], realizing the threshold ordering $2/3 < 4/5 < 24/25$ of Sec. 9 (Fig. 4b) and not constructed here. The radiative graft’s fully Einstein-sourced $\mathcal{O}(\mu)$ back-reaction (radiation streaming *through* the wall) is treated in App. F.

Complete finite-duration maneuver (Theorem 3). We integrate the bump steering $a(u) = a_{\max} \sin^2[\pi(u - u_0)/(u_f - u_0)]$ off the static anchor $x_\star = 0.3$ with the saturated budget $m(u) = m_0 e^{-3 \int |a|}$ and $\lambda_{\max} = 0.12 < g(0.3)$. At every retarded step it evaluates the bulk dust $n^2(\vartheta)$ and the gauge-invariant finite- λ Lanczos surface stress (App. C), establishing $n^2 \geq 0$, a strictly positive worst-observer surface DEC margin (Type I), and $\lambda < 1$ throughout, with the Bondi moment identities $\oint n^2 d\Omega = -\dot{m}$ and $-\oint n^2 \cos \vartheta d\Omega = \dot{m} a$ closing to $\sim 10^{-7}$; the burn radiates $\approx 51\%$ of the rest mass for a rapidity gain 0.24 (Fig. 3), the velocity payoff against an ideal photon rocket being Fig. 7.

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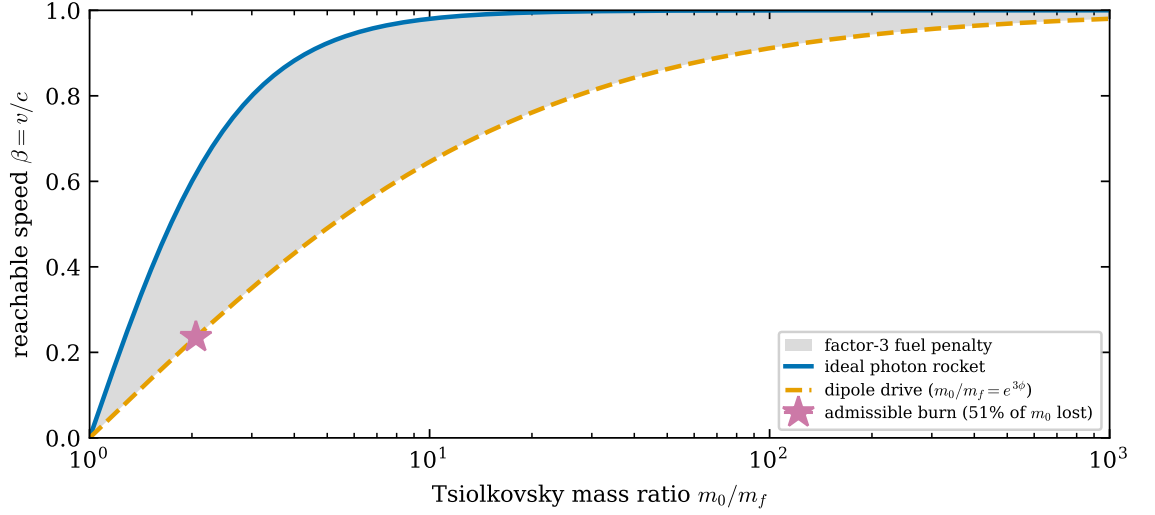


Figure 7. The cost of steering without exotic matter. The saturated control law $-\dot{m} = 3m|a|$ integrates to a Tsiolkovsky budget $m_0/m_f = e^{3\phi}$: reaching rapidity ϕ (speed $\beta = \tanh \phi$) burns the cube of an ideal photon rocket’s mass ratio: the broad-dipole $\langle \cos^2 \vartheta \rangle = \frac{1}{3}$ collimation penalty, not a relativistic floor. The saturating burn of Theorem 3 (star) radiates $\approx 51\%$ of m_0 to reach $\beta \approx 0.24$: a real, paid-for velocity change with strictly positive energy.

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